

Quantitative Economics Notes

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Disclaimer: The notes here are derived from (i) Professor James Duffy's (University of Oxford) notes and lectures from the 2024-2025 Quantitative Economics Course, (ii) Stock and Watson's *Introduction to Econometrics* (scarcely), (iii) Blitzstein and Hwang's *Introduction to Probability* (for the appendix), (iv) Angrist and Pischke's *Mostly Harmless Econometrics*, and literally once (v) Hastie et al's *An Introduction to Statistical Learning* (for the general form of the bias-variance tradeoff). **All and every error is mine** and there **will be errors**. Furthermore, these notes are an interpretation of the primary sources, so there is bound to be a misinterpretation somewhere.

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1 Linear Regression: Mechanics and OLS

Definition 1.1: Linear Causal Model

We assume that the causal model is linear such that our causal model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_m x_m \quad (1.1)$$

where y is the dependent variable, while x_i are the independent variables, and the coefficients β_i can be interpreted as the effect of a change in 1 unit of x_i on y_i , since $\beta_i = \frac{\partial y_i}{\partial x_i}$.

Example.

We might want to assess the causal effects of a worker's characteristics such as age, gender, education on the worker's wage. Hence, we have the following causal model

$$\text{wage}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{gender}_i + \beta_3 \text{edu}_i$$

Remark.

We assume that the model is linear

1.1 Linear Regression with One Regressor

We may be only able to observe y and x_1 , and so we need a simpler model.

Definition 1.2: Single Variable Causal Model

We may simplify (1.1) as the following single variable (linear) causal model

$$y = \beta_0 + \beta_1 x_1 + u \quad (1.2)$$

where $u := \beta_2 x_2 + \cdots + \beta_m x_m$ and $x = x_m$

Motivation:

- Our objective is to estimate the causal effect of x_1 on y , β_1
- We rely on a sample of n observations, denoted $\{Y_i, X_i\}_{i=1}^n$
- Each pair (X_i, Y_i) is regarded as being independently sampled from some joint distribution $F_{X,Y}$ and so are i.i.d (identically and independently distributed)

Since the causal model is assumed to have generated the data, evaluating the causal model at $y = Y_i, x = X_i$ gives

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where u_i captures all the other determinants of Y , which we do not observe.

We call the following equation the ‘causal model’ and write the above equation without the i subscript

$$Y = \beta_0 + \beta_1 X + u \quad (1.3)$$

Remark.

X, Y are treated as generic random variables having the same distribution $F_{X,Y}$ as each of the observations

We want to construct a consistent estimate of β_0 (i.e. estimate converging to population parameter as $n \rightarrow \infty$). So, we need to isolate the effect of X on Y , thereby removing the effects of everything else – hence, we need to assume that u is not systematically related to X .

Assumption 1.3

The following assumptions all express that u is not correlated with X , from weakest to strongest assumption:

Orthogonality (OR):

$$E[u] = 0 \text{ and } E[Xu] = 0$$

or equivalently,

$$E[u] = 0 \text{ and } \text{Cov}(X, u) = 0$$

Mean Independence (MI):

$$E[u] = 0 \text{ and } E[u|X] = E[u]$$

Independence (IN):

$$E[u] = 0 \text{ and } u \perp X$$

Remark.

We assume mean zero in all cases i.e. $E(u) = 0$ and if we want to just estimate β_1 , then we can ignore this assumption.

OR is sufficient to allow (β_0, β_1) to be consistently estimated by linear regression

We have two kinds of questions:

1. Causal Question: Would a counterfactual increase in class size decrease test scores
2. Descriptive Question: Would an actual increase in class size decrease test scores

The causal question is answered by linear regression if OR holds, while the second is always answered by linear regression

Proposition 1.4

IN implies MI implies OR

Proof. We only need to prove the two implications (IN to MI) and (MI to OR) $\square$

Proposition 1.5

OR does not imply MI, MI does not imply IN and OR does not imply IN

Proof. $\square$

1.1.1 Population Linear Regression

Definition 1.6: Best Linear Prediction Problem

To construct the best linear predictor for a random variable Y using another random variable X . In other words, we want to find (b_0, b_1) such that $b_0 + b_1X$ minimises the mean squared prediction error:

$$(b_0, b_1) = \operatorname{argmin} C(b_0, b_1) \quad (1.4)$$

where $C(b_0, b_1)$ is the **mean squared prediction error** defined as

$$C(b_0, b_1) := \mathbb{E}(Y - b_0 - b_1X)^2 \quad (1.5)$$

We may find the solution to the BLP problem via calculus.

Claim

The unique solution to the BLP problem $(b_0, b_1) = (\rho_0, \rho_1)$, where

$$\begin{aligned} \rho_1 &= \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)} \\ \rho_0 &= \mathbb{E}[Y] - \rho_1 \mathbb{E}[X] \end{aligned}$$

Proof for Claim.

We find the FOCs:

$$\begin{aligned} \frac{\partial C}{\partial b_1} &= \frac{\partial}{\partial b_1} \mathbb{E}(Y - b_0 - b_1X)^2 \\ &=_{\text{DCT}} \mathbb{E} \left[\frac{\partial}{\partial b_1} (Y - b_0 - b_1X)^2 \right] \\ &= \mathbb{E}[2(-X)(Y - b_0 - b_1X)] \\ &= -2\mathbb{E}[X(Y - b_0 - b_1X)] \end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_0} &= \frac{\partial}{\partial b_0} \mathbb{E}(Y - b_0 - b_1 X)^2 \\
&=_{\text{DCT}} \mathbb{E} \left[\frac{\partial}{\partial b_0} (Y - b_0 - b_1 X)^2 \right] \\
&= \mathbb{E}[2(-1)(Y - b_0 - b_1 X)] \\
&= -2\mathbb{E}[Y - b_0 - b_1 X]
\end{aligned}$$

So, we have

$$\begin{aligned}
\frac{\partial C}{\partial b_0} &= -2\mathbb{E}[Y - b_0 - b_1 X] = 0 \\
\frac{\partial C}{\partial b_1} &= -2\mathbb{E}[X(Y - b_0 - b_1 X)] = 0
\end{aligned}$$

These are uniquely solved by population regression parameters (ρ_0, ρ_1) where we substitute (b_0, b_1) with (ρ_0, ρ_1) . We then first yield

$$\rho_0 = \mathbb{E}[Y] - \rho_1 \mathbb{E}[X] \quad (1.6)$$

and then substituting this gives

$$\begin{aligned}
0 &= \mathbb{E}[Y - (\mathbb{E}[Y] - \rho_1 \mathbb{E}[X]) - \rho_1 X]X \\
&= \mathbb{E}[(Y - \mathbb{E}[Y]) - \rho_1(X - \mathbb{E}[X])]X \\
&= \mathbb{E}(Y - \mathbb{E}[Y])X - \rho_1 \mathbb{E}(X - \mathbb{E}[X])X
\end{aligned}$$

and since $\mathbb{E}(Y - \mathbb{E}[Y])X = \text{Cov}(X, Y)$, we have

$$\rho_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \quad (1.7)$$

We then define the **population linear regression model** as

$$Y = \rho_0 + \rho_1 X + e$$

where $e := Y - \rho_0 - \rho_1 X$ is the error made by the best linear predictor of Y .

It follows immediately from the proof that e must have

- mean zero: $\mathbb{E}[e] = \mathbb{E}[Y - \rho_0 - \rho_1 X] = 0$
- Orthogonal to X : $\mathbb{E}[Xe] = \mathbb{E}[X(Y - \rho_0 - \rho_1 X)] = 0$ (This is SUPER IMPORTANT TO SEE WHY the error is orthogonal)

which imply that $\text{Cov}(X, e) = 0$ i.e. e and X are uncorrelated. It also follows that for a model $Y = \rho_0 + \rho_1 X + e$ with $\mathbb{E}[e] = 0$ and $\mathbb{E}[Xe] = 0$, then ρ_0, ρ_1 solve the BLP.

The population linear regression model is a decomposition of Y uniquely into a linear function of X and an error e that has mean zero and is uncorrelated with X (satisfying **OR**).

Connection with Causal Model

The population Linear Regression model is **purely descriptive** on how Y could be optimally predicted on the basis of X . The following proposition tells us when the causal model coincides with the linear regression model.

Proposition 1.7

When **OR** holds, the parameters of (β_0, β_1) *must* coincide with the parameters (ρ_0, ρ_1) of the population linear regression model.

Proof. Given the causal model, we can write

$$\begin{aligned}\mathbb{E}(Y - \beta_0 - \beta_1 X) &= \mathbb{E}[u] \\ \mathbb{E}[(Y - \beta_0 - \beta_1 X)X] &= \mathbb{E}[Xu]\end{aligned}$$

which follows by definition of u (since u collects all other determinants) and taking expectation on both sides. Both equations equal 0 since $\mathbb{E}[u] = \mathbb{E}[Xu] = 0$ by **OR**. Then, these moment conditions coincide with the moment conditions of the population linear regression model, hence establishing that (β_0, β_1) coincide with (ρ_0, ρ_1) and hence (b_0, b_1) $\square$

Technical Note: A better way to actually prove this is by contradiction. Assume that $(\rho_0, \rho_1) \neq (\beta_0, \beta_1)$. This means that there are two pairs (ρ_0, ρ_1) and (β_0, β_1) which satisfy the decomposition $Y = \hat{Y} + \epsilon$. But the projection theorem says that for any vector $Y \in \mathbb{R}^n$ (in this case it is just in $\mathbb{R}$), there is a unique decomposition $Y = \hat{Y} + \epsilon$ such that $\hat{Y} \in \text{span}\{1, X\}$ and ϵ is orthogonal to every vector in $\text{span}\{1, X\}$. By assumption, $(\rho_0, \rho_1) \neq (\beta_0, \beta_1)$ and so the decomposition is not unique, a contradiction.

Corollary 1.8

Under **OR**, we have

$$\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \quad \beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X] \quad (1.8)$$

Proof. This follows immediately since (β_0, β_1) coincide with (ρ_0, ρ_1) by **Proposition 1.7** $\square$

1.1.2 Sample Linear Regression: OLS

When we are confined to limited data i.e. a sample, we want to estimate (β_0, β_1) .

Definition 1.9: Sample Linear Regression Problem

We solve the sample linear regression problem by choosing (b_0, b_1) to minimise the sum of squared predicted errors:

$$(\hat{\beta}_0, \hat{\beta}_1) := \operatorname{argmin}_{b_0, b_1} \frac{1}{n} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 \quad (1.9)$$

The estimators $(\hat{\beta}_0, \hat{\beta}_1)$ are termed the ordinary least squares estimators of (β_0, β_1)

Remark.

We can drop $\frac{1}{n}$ in (1.9) but this emphasises the link with expectation

Proposition 1.10

The solution to the Sample Linear Regression Problem is solved by $(\hat{\beta}_0, \hat{\beta}_1)$ i.e.

$$\begin{aligned} \hat{\beta}_1 &:= \frac{\operatorname{cov}(X, Y)}{\operatorname{var}(X)} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ \hat{\beta}_0 &:= \bar{Y} - \hat{\beta}_1 \bar{X} \end{aligned}$$

Proof. See Worksheet 2

□

These are known as the sample analogues of the solutions given to us for the population regression problem. We compute the OLS estimators by regressing Y on X in the sample.

Remark.

We can compute the sample analogues directly as a second way of solving the problem but it really doesn't matter

Sample Decomposition. The OLS estimates $(\hat{\beta}_0, \hat{\beta}_1)$ yield the sample analogue of the decomposition

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i$$

where $\hat{\beta}_0 + \hat{\beta}_1 x$ is the sample regression function, $\hat{Y}_i := \hat{\beta}_0 + \hat{\beta}_1 X_i$ are predicted or fitted values.

The OLS residual $\hat{u}_i := Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$, which can be regarded as an estimator of the error u_i satisfy the following sample analogues:

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0 \quad \frac{1}{n} \sum_{i=1}^n X_i \hat{u}_i = 0$$

and thus,

$$\operatorname{cov}(X, \hat{u}) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(\hat{u}_i - \bar{\hat{u}}) = 0$$

When OR fails

The population regression model

$$Y = \rho_0 + \rho_1 X + e$$

remains well defined and still

- Provides the ‘best linear prediction’ for Y on the basis of X
- decomposes Y into a linear function of X and an error term $e := Y - \rho_0 - \rho_1 X$ that has mean zero and is uncorrelated with X

The OLS estimates $(\hat{\beta}_0, \hat{\beta}_1)$ still can be computed and will provide consistent estimates of the regression parameters (ρ_0, ρ_1) . But the next fact tells us that **we cannot interpret the either the population or OLS regression estimates as the causal effect of X on Y**

Claim

Suppose **OR** fails. Then, (ρ_0, ρ_1) no longer coincide with the parameters (β_0, β_1) of the causal model

Proof for Claim.

(Sketch) We can see this informally. Since **OR** fails, then $\text{Cov}(X, u) \neq 0$. Since $\text{Cov}(X, e) = 0$, then $e \neq u$. Since both $e := Y - \rho_0 - \rho_1 X$ and $u := Y - \beta_0 - \beta_1 X$, then it follows that $(\beta_0, \beta_1) \neq (\rho_0, \rho_1)$. ■

So neither they nor the OLS estimates can be interpreted as the causal effect of X on Y .

Remark.

The estimates (ρ_0, ρ_1) and $(\hat{\beta}_0, \hat{\beta}_1)$ retain a descriptive, but not causal, interpretation. Furthermore, $(\hat{\beta}_0, \hat{\beta}_1)$ still consistently estimates (ρ_0, ρ_1) by construction of the OLS, but NOT (β_0, β_1)

1.1.3 Conditional Expectation and Regression

Definition 1.11: Conditional Expectation Function

For continuous Y with conditional density $f_y(t|X = x)$ at $Y = t$, the Conditional Expectation Function (CEF) is

$$\mathbb{E}[Y|X] = \int t f_y(t|X) dt$$

Remark.

If Y is discrete, then it equals the sum $\sum_t t \mathbb{P}(Y = t|X)$

We have some properties of the CEF (Angrist and Pischke, 2009):

Theorem 1.12: The CEF Decomposition Property

$$Y_i = \mathbb{E}[Y_i|X_i] + \varepsilon_i$$

where (i) ε_i is mean independent of X_i i.e. $\mathbb{E}[\varepsilon_i|X_i] = 0$, and therefore (ii) ε_i is uncorrelated with any function of X_i

Proof. For (i), $\mathbb{E}[\varepsilon_i|X_i] = \mathbb{E}[Y_i - \mathbb{E}[Y_i|X_i]|X_i] = \mathbb{E}[Y_i|X_i] - \mathbb{E}[Y_i|X_i] = 0$. For (ii) Let $h(X_i)$ be any function of X_i . By LIE, $\mathbb{E}[h(X_i)\varepsilon_i] = \mathbb{E}[h(X_i)\mathbb{E}[\varepsilon_i|X_i]]$, and by mean-independence, $\mathbb{E}[\varepsilon_i|X_i] = 0$ $\square$

Theorem 1.13: The CEF Prediction Property

Let $h(X_i)$ be any function of X_i . The CEF solves

$$\mathbb{E}[Y_i|X_i] = \underset{h(X_i)}{\operatorname{argmin}} \mathbb{E}[(Y_i - h(X_i))^2]$$

so the CEF is the minimum mean squared error predictor of Y_i given X_i

Proof. We may write

$$\begin{aligned} (Y_i - h(X_i))^2 &= ((Y_i - \mathbb{E}[Y_i|X_i]) + (\mathbb{E}[Y_i|X_i] - h(X_i)))^2 \\ &= (Y_i - \mathbb{E}[Y_i|X_i])^2 + 2(\mathbb{E}[Y_i|X_i] - h(X_i)) \\ &\quad \times (Y_i - \mathbb{E}[Y_i|X_i]) + (\mathbb{E}[Y_i|X_i] - h(X_i))^2 \end{aligned}$$

The first term doesn't matter because it does not involve $h(X_i)$. The second term can be written as $m(X_i)\varepsilon_i$, where $m(X_i) \equiv 2(\mathbb{E}[Y_i|X_i] - h(X_i))$, and therefore has expectation zero by the CEF decomposition property. The last term is minimised at zero when $h(X_i)$ is the CEF. $\square$

Theorem 1.14: The ANOVA Theorem

$$\operatorname{var}(Y_i) = \operatorname{var}(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[\operatorname{var}(Y_i|X_i)]$$

where $\operatorname{var}(Y_i|X_i)$ is the conditional variance of Y_i given X_i

Proof. The CEF decomposition property implies that the variance of Y_i is the variance of the CEF plus the variance of the residual, $\varepsilon_i := Y_i - \mathbb{E}[Y_i|X_i]$, since ε_i and $\mathbb{E}[Y_i|X_i]$ are uncorrelated. The variance of ε_i is

$$\mathbb{E}[\varepsilon_i^2] = \mathbb{E}[\mathbb{E}[\varepsilon_i^2|X_i]] = \mathbb{E}[\operatorname{var}[Y_i|X_i]]$$

where $\mathbb{E}[\varepsilon_i^2|X_i] = \operatorname{var}[Y_i|X_i]$ because $\varepsilon_i := Y_i - \mathbb{E}[Y_i|X_i]$ $\square$

Theorem 1.15: CEF and Population Regression

Let $\mathcal{X} = (X_1, X_2, \dots, X_k)$ and the population linear regression function be $m(X_1, \dots, X_k) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$. We have the following properties

1. Suppose the CEF is linear. Then the popular regression function is it.
2. The function $m(X_1, \dots, X_k) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$ is the best linear predictor of Y_i given $(X_1, \dots, X_k)$ in the MMSE sense
3. The function $m(X_1, \dots, X_k) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$ provides the MMSE linear approximation to $\mathbb{E}[Y_i|\mathcal{X}]$. That is

$$(\beta_0 + \beta_1 + \dots + \beta_k) = \underset{b}{\operatorname{argmin}} \mathbb{E}[(\mathbb{E}[Y_i|\mathcal{X}] - m(X_1, \dots, X_k))^2]$$

Proof. For proof, see Angrist and Pischke's Mostly Harmless Econometrics (2009) $\square$

Remark.

Crucially, (1) tells us that the population linear regression function is a special case of the CEF (i.e. when it is linear). (2) and (3) tells us that if the CEF is not linear (and so (1) no longer applies), within the class of linear functions, the population linear regression is still the best predictor of Y_i given $(X_1, \dots, X_k)$ in the minimum mean squared error sense (or provides the best linear approximation of the CEF).

1.1.4 Misc

OR vs MI. Stock and Watson assume **MI** which is a stronger assumption than **OR**.

MI entails a unnatural restriction for the marginal distribution of X if $\mathbb{E}[u|X] \neq 0$, but we still want $\operatorname{cov}(X, u) = 0$.

Proposition 1.16

Assume **MI**. Then, $\mathbb{E}[Y|X]$ gives us the best linear predictor of Y using X alone.

Proof. Using the properties of conditional expectation,

$$\begin{aligned} \mathbb{E}[Y|X] &= \mathbb{E}[\beta_0 + \beta_1 X + u|X] \\ &= \beta_0 + \beta_1 \mathbb{E}[X|X] + \mathbb{E}[u|X] \\ &= \beta_0 + \beta_1 X \end{aligned}$$

We know that $\mathbb{E}[Y|X]$ is by definition the best non-linear predictor of Y using X alone. But since it is linear, it must also be the best linear predictor of Y , and therefore coincides with the regression function. $\square$

Remark.

So, assuming **MI** implies that $E[Y|X]$ is linear in X

1.2 Multiple Linear Regression

Definition 1.17: Best Linear Prediction Problem for k variables

The BLP is to construct the best linear predictor for a random variable Y using k random variables $(X_1, X_2, \dots, X_k)$. So, we want to find $(b_0, b_1, \dots, b_k)$ such that $b_0 + b_1X_1 + \dots + b_kX_k$ minimises the mean squared prediction error:

$$(b_0, b_1, \dots, b_k) = \operatorname{argmin} C(b_0, b_1, \dots, b_k)$$

where $C(b_0, b_1, \dots, b_k)$ is the **mean squared prediction error** defined as

$$C(b_0, b_1, \dots, b_k) := \mathbb{E}(Y - b_0 - b_1X_1 - \dots - b_kX_k)^2$$

If we solve the above in an analogous way to the single variable case, we get a system of $k + 1$ first order conditions for $k + 1$ variables. This is computationally inefficient, so we will establish the following result.

Theorem 1.18: Frisch-Waugh-Lovell Theorem

Consider a population linear regression of Y on $(X_1, \dots, X_k)$. Then, the general solution for any coefficient for each regressor is given by

$$\rho_i = \frac{\operatorname{cov}(Y, \tilde{X}_i)}{\operatorname{var}(\tilde{X}_i)}$$

where $\tilde{X}_i$ denotes the error from the regression of X_i on each of the other $k - 1$ regressors i.e.

$$\tilde{X}_i := X_i - (\pi_0 + \pi_1X_1 + \dots + \pi_{i-1}X_{i-1} + \pi_{i+1}X_{i+1} + \dots + \pi_kX_k)$$

Proof. Let $\mathcal{X} = (X_1, \dots, X_k)$ Consider a population linear regression model

$$Y = \rho_0 + \rho_1X_1 + \dots + \rho_kX_k + \varepsilon \tag{1}$$

Then, we consider a population linear regression of X_i on X_{-i} (which denotes all other variables excluding X_i) i.e.

$$X_i = \pi_0 + \sum_{j \neq i} \pi_j X_j + \tilde{X}_i \tag{2}$$

which decomposes X_i into a linear function of X_{-i} and an error $\tilde{X}_i$ that satisfies (i) $\mathbb{E}[\tilde{X}_i] = 0$ and (ii) $\mathbb{E}[X_j \tilde{X}_i] = 0$ for $j \neq i$ i.e. $\operatorname{cov}(X_j, \tilde{X}_i) = 0$.

If we substitute (2) into (1), we get

$$\begin{aligned}
Y &= \rho_0 + \rho_1 X_1 + \cdots + \cdots + \rho_k X_k + \varepsilon \\
&= \rho_0 + \sum_{j \neq i} \rho_j X_j + \rho_i \left(\pi_0 + \sum_{j \neq i} \pi_j X_j + \tilde{X}_i \right) + \varepsilon \\
&= (\rho_0 + \rho_i \pi_0) + (\rho_1 + \rho_i \pi_1) X_1 + \cdots + \rho_i \tilde{X}_i + \cdots + (\rho_k + \rho_i \pi_k) X_k + \varepsilon \\
&= (\rho_0 + \rho_i \pi_0) + \sum_{j \neq i} (\rho_j + \rho_i \pi_j) X_j + \rho_i \tilde{X}_i + \varepsilon \\
&= \gamma_0 + \sum_{j \neq i} \gamma_j X_j + \rho_i \tilde{X}_i + \varepsilon
\end{aligned}$$

where $\gamma_j = \rho_j + \rho_i \pi_j$. Then, consider $\text{cov}(Y, \tilde{X}_i)$:

$$\begin{aligned}
\text{cov}(Y, \tilde{X}_i) &= \text{cov}(\gamma_0 + \cdots + \rho_i \tilde{X}_i + \cdots + \varepsilon, \tilde{X}_i) \\
&= \text{cov}(\gamma_0, \tilde{X}_i) + \rho_i \text{cov}(\tilde{X}_i, \tilde{X}_i) + \sum_{j \neq i} \gamma_j \text{cov}(X_j, \tilde{X}_i) + \text{cov}(\varepsilon, \tilde{X}_i)
\end{aligned}$$

Now, notice that (i) $\text{cov}(\gamma_0, \tilde{X}_i) = 0$ obviously, (ii) $\text{cov}(\tilde{X}_i, \tilde{X}_i) = \text{var}(\tilde{X}_i)$, (iii) $\gamma_j \text{cov}(X_j, \tilde{X}_i) = 0$ for each $j \neq i$ by construction in (2). Then,

$$\begin{aligned}
\text{cov}(\varepsilon, \tilde{X}_i) &= \text{cov}(\varepsilon, X_i - \pi_0 - \sum_{j \neq i} \pi_j X_j) \\
&= \text{cov}(\varepsilon, X_i) - \text{cov}(\varepsilon, \pi_0) - \sum_{j \neq i} \pi_j \text{cov}(\varepsilon, X_j) \\
&= 0
\end{aligned}$$

This is because by construction of (1) where the error term ε is uncorrelated with any of X_j i.e. $\text{cov}(X_j, \varepsilon) = 0$. Hence we have

$$\text{cov}(Y, \tilde{X}_i) = \rho_i \text{var}(\tilde{X}_i)$$

□

Remark.

In the QE notes, the causal model is used directly, but I use the population regression model to maintain that the causal and population model can still come apart and that FWL is a purely **descriptive** theorem

Theorem 1.19: Sample FWL Theorem

Consider a sample linear regression of Y on $(X_1, \dots, X_k)$. Then, the general solution for any coefficient for each regressor is given by

$$\hat{\beta}_i = \frac{\text{cov}(Y, \tilde{X}_i)}{\text{var}(\tilde{X}_i)}$$

where $\tilde{X}_i$ denotes the error from the regression of X_i on each of the other $k - 1$ regressors i.e.

$$\tilde{X}_i := X_i - (\hat{\pi}_0 + \hat{\pi}_1 X_1 + \dots + \hat{\pi}_{i-1} X_{i-1} + \hat{\pi}_{i+1} X_{i+1} + \dots + \hat{\pi}_k X_k)$$

Proof. I'm pretty sure this was in a tutorial sheet □

Connection with the Causal Model

Claim

Let $(\beta_0, \beta_1, \dots, \beta_k)$ be the coefficients of the causal model, and $(\rho_0, \rho_1, \dots, \rho_k)$ be the coefficients of the population regression model. Then, if **OR** fails, $(\rho_0, \rho_1, \dots, \rho_k)$ no longer coincides with $(\beta_0, \beta_1, \dots, \beta_k)$

Proof for Claim.

Consider the causal model

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + u \tag{3}$$

Then, we may adapt the proof of the FWL Theorem (Theorem 1.18) until we reach

$$\begin{aligned} \text{cov}(\varepsilon, \tilde{X}_i) &= \text{cov}(u, X_i - \pi_0 - \sum_{j \neq i} \pi_j X_j) \\ &= \text{cov}(u, X_i) - \text{cov}(u, \pi_0) - \sum_{j \neq i} \pi_j \text{cov}(u, X_j) \\ &\neq 0 \end{aligned}$$

This follows because if **OR** fails, then for at least one X_j , $\text{cov}(u, X_j) \neq 0$. Then, we have

$$\text{cov}(Y, \tilde{X}_i) = \beta_i \text{var}(\tilde{X}_i) + \sum_j^k \text{cov}(u, X_j)$$

which implies

$$\beta_i = \frac{\text{cov}(Y, \tilde{X}_i)}{\text{var}(\tilde{X}_i)} - \frac{1}{\text{var}(\tilde{X}_i)} \sum_j^k \text{cov}(u, X_j)$$

Remark.

- As per previously, $(\hat{\beta}_0, \dots, \hat{\beta}_k)$ still consistently estimates the coefficients of the population regression model and both still retain a descriptive but not causal interpretation.

Notice that if $\sum_j^k \text{cov}(u, X_j) > 0$, then ρ_i overestimates β_i (i.e. positive bias). And conversely, if $\sum_j^k \text{cov}(u, X_j) < 0$, then ρ_i underestimates β_i (negative bias). For instance, suppose you are trying to figure out the effects of education and wealth on future income. Clearly, innate ability is positively correlated with both as well as the dependent future income, but you exclude it. Then, you overestimate the effect in your population regression since some of the increase in education also carries an increase in innate ability (since you do not hold it constant/control for it).

The interpretation of ρ_i depends on other variables in the model.

Interpretation: We may interpret the coefficient ρ_i (or $\hat{\beta}_i$ since that consistently estimates the population coefficient) as the **average** change in Y_i associated with a unit change in X_i holding every other variable constant OR on average, a unit increase in X_i is associated with a ρ_i increase in Y holding each of X_j for all $j \neq i$ constant.

If **OR** is satisfied, then ρ_i is the average causal effect of a unit increase of X_i on Y holding everything else constant.

Interpretation of the FWL Theorem

What the FWL Theorem implies is that the regression coefficient may be obtained equivalently via

- By a linear regression of Y on $X_1, \dots, X_k$
- Or via a two stage procedure
 1. Regress X_i on all other regressors, which yields the error $\tilde{X}_i$
 2. Regress Y on $\tilde{X}_i$

Proxy Variables and Descriptive Interpretations

Often, we do not have data on all major determinants of Y and we may use proxy variables in place of these. For instance, suppose we want to try to measure the effect of income transfers on how much one eats. Data on maintenance calories (as a control) is hard to get, so we may proxy this with height.

In short, we do not observe X_j but observe another variable W that (i) is a good predictor of X_j and (ii) does not itself appear in the causal model and is uncorrelated with u i.e. $\beta_W = 0$.

WARNING: We cannot give a causal interpretation to proxy variables. Never.

2 Linear Regression: Statistical Inference

In this section, we now use the notation

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}$$

where X_{Ii} is the I -th variable for the i -th observation.

2.1 Regression Diagnostics

Consider the OLS regression

$$Y_i = (\hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \cdots + \hat{\beta}_k X_{ki}) + \hat{u}_i = \hat{Y}_i + \hat{u}_i$$

The basic intuition is that the predicted value $\hat{Y}_i$ is the systematic part of the model and also a residual $\hat{u}_i$ where

$$\sum_{i=1}^n X_{Ii} \hat{u}_i = 0$$

for all $I \in \{0, \dots, k\}$ and also

$$\begin{aligned} \sum_{i=1}^n \hat{Y}_i \hat{u}_i &= \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \cdots + \hat{\beta}_k X_{ki}) \hat{u}_i \\ &= \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n X_{1i} \hat{u}_i + \cdots + \hat{\beta}_k \sum_{i=1}^n X_{ki} \hat{u}_i = 0 \end{aligned}$$

Therefore, $\hat{Y}_i$ is regarded as the part of Y_i explained by the regressors while the residual $\hat{u}_i$ is left unexplained by the regression model. We want to measure the adequacy of this regression model

2.1.1 Perfect Multicollinearity

Definition 2.1: Perfect Multicollinearity

Consider regressors $X_1, \dots, X_k$. Then, if X_1 is linearly dependent on $X_2, \dots, X_k$ i.e. may be expressed as a linear combination

$$X_1 = \alpha_1 + \alpha_2 X_2 + \cdots + \alpha_k X_k$$

, we say that $X_1, \dots, X_k$ are perfectly multicollinear. This is equivalent to saying that $X_1, \dots, X_k$ are mutually exclusive and collectively exhaustive in the case of dummy variables

Remark.

The intuition of this is that in a regression model, holding $X_2, \dots, X_k$ constant instantly determines X_1 (follows from the linear dependence) and so it is impossible to identify the effect of X_1 holding $X_2, \dots, X_k$ constant

Claim

Suppose $X_1, \dots, X_k$ are perfectly multicollinear. Then the regression model is not well defined

Proof for Claim.

Consider a regression Y on $X_1, \dots, X_k$. Then, by the FWL Theorem,

$$\rho_i = \frac{\text{cov}(Y, \tilde{X}_i)}{\text{var}(\tilde{X}_i)}$$

We focus on the denominator term. We know that $\tilde{X}_i$ is the error term of regressing X_i on all other models. Hence

$$\tilde{X}_i = X_i - (\pi_0 + \sum_{j \neq i} \pi_j X_j)$$

But notice that since $(\pi_0, \dots, \pi_k)$ is the solution to the problem

$$\min (X_i - (b_0 + \sum_{j \neq i} b_j X_j))^2$$

and WLOG that X_i may be expressed as a linear combination of all the regressors (if strict subset then set the coefficients of those other regressors to zero) and so

$$X_i = \lambda_0 + \sum_{j \neq i} \lambda_j X_j$$

Then,

$$0 = X_i - \lambda_0 + \sum_{j \neq i} \lambda_j X_j$$

so the coefficients of the minimisation problem and linear combination must coincide i.e. $(\lambda_0, \lambda_1, \dots, \lambda_k) = (\pi_0, \pi_1, \dots, \pi_k)$. Hence, $\tilde{X}_i = 0$, and we conclude that

$$\text{var}(\tilde{X}_i) = 0$$

so $\rho_i := \frac{\text{cov}(Y, \tilde{X}_i)}{\text{var}(\tilde{X}_i)}$ is undefined

Remark.

It is not that the population regression coefficients do not coincide with the causal model, but is a logical problem with the population (or sample) regression itself.

Solution: We remove one of the variables and our interpretation takes that missing variable as the base case i.e. ‘the average change associated with working 1 extra hour **instead of sleeping** and the worker’s productivity’

2.1.2 Standard Errors and R^2

Definition 2.2: Standard Error of Regression

Let $\hat{u}_i$ be the residual of the population regression model. Then, we estimate σ_u^2 by computing the sample variance of the residuals as

$$s_{\hat{u}}^2 := \frac{1}{n - k - 1} \sum_{i=1}^n \hat{u}_i^2$$

where k is the number of regressors. The Standard Error of Regression (SER) is simply $s_{\hat{u}}$

Definition 2.3: Sums of Squares Concepts

We have the following sum of squares concepts

- Total Sum of Squares (TSS) $:= \sum_{i=1}^n (Y_i - \bar{Y})^2$
- Explained Sum of Squares (ESS) $:= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$
- Sum of Squared Residuals (SSR) $:= \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (\hat{Y}_i - Y_i)^2$

Proposition 2.4

For any linear regression model, the following identity holds:

$$\text{TSS} = \text{ESS} + \text{SSR} \quad (4)$$

Proof. Consider

$$\begin{aligned} \text{TSS} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n [(\hat{Y}_i + \hat{u}_i) - \bar{Y}]^2 \\ &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n \hat{u}_i^2 - 2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{u}_i \\ &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n \hat{u}_i^2 \\ &= \text{ESS} + \text{SSR} \end{aligned}$$

since

$$\sum_{i=1}^n (\hat{Y}_i - \bar{Y}) \hat{u}_i = \sum_{i=1}^n \hat{Y}_i \hat{u}_i - \bar{Y} \sum_{i=1}^n \hat{u}_i = 0$$

□

Remark.

So basically, the regression model decomposes the TSS (the total variability of Y_i around the sample mean) into the ESS (variability of Y_i explained by the regressors) and the SSR (the variability of Y_i not accounted for by the model)

Definition 2.5: R^2 value

The R^2 is defined as the fraction of the variability of Y_i that is explained by the model (the proportion of variation in Y_i ‘attributable’ to variation in $X_{1i}, \dots, X_{ki}$). Formally,

$$R^2 := \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}} \quad (5)$$

The standard error of regression can be written equivalently as

$$s_{\hat{u}} = \left(\frac{\text{SSR}}{n - k - 1} \right)^{1/2} \quad (6)$$

Proposition 2.6

R^2 is a unitless measure of fit and is invariant with any rescaling of Y_i and/or X_{Ii} . Furthermore, $R^2 \in [0, 1]$

Proof. I think the single-variable case is in a past year paper, but it can be shown we tedious algebra $\square$

Remark.

So are the t -statistics

What is wrong with using R^2 ?

Since OLS chooses $\hat{\beta}_I$ to minimise the SSR, the SSR will not increase when variables are added to the regression and will generally decrease. So, if we judge in terms of SSR alone, a model with more variables will always appear to have a better fit to the data.

So, R^2 will almost always increase (and never decrease) when regressors are added, even if these are irrelevant (with population regression coefficients of zero).

So, we want to reduce the tendency of R^2 to increase when irrelevant variables are added.

Definition 2.7: Adjusted R^2

Denote the adjusted R^2 as $\bar{R}^2$. We define it as follows:

$$\bar{R}^2 := 1 - \frac{n-1}{n-k-1} \frac{\text{SSR}}{\text{TSS}} = 1 - \frac{\text{SSR}/(n-k-1)}{\text{TSS}/(n-1)} = 1 - \frac{s_{\hat{u}}^2}{s_y^2}$$

In effect, we are standardising both SSR and TSS so that we make each term an unbiased estimator of σ_u^2 and $\text{var}(Y_i)$ respectively. So, while adding additional regressors decrease SSR, it also increases k which makes the term smaller – so the effect on $\bar{R}^2$ is indeterminate.

2.2 Inference on Regression Parameters

The model to be estimated is

$$Y_i = \beta_0 + \beta_1 + X_{1i} + \dots + \beta_k X_{ki} + u_i$$

We want to use the OLS estimates to draw inferences about the parameters of the model i.e. we want to estimate a fixed quantity β_1 using the OLS estimator $\hat{\beta}_1$ that is a function of our randomly sampled data on Y_i and X_{Ii} and hence is a random variable with a probability distribution (the sampling distribution of $\hat{\beta}_1$)

2.2.1 Large Sample Properties of OLS

Assumption 2.8

We make the following assumptions:

A1 (Orthogonality/Mean-independence): u_i satisfies either

- **OR:** $\mathbb{E}[u_i] = 0$ and $\text{cov}(u_i, X_{Ii}) = 0$ for $I \in \{1, \dots, k\}$
- **MI:** $\mathbb{E}[u_i | X_{1i}, \dots, X_{ki}] = 0$

A2 (Sampling distribution): $\{(Y_i, X_{1i}, \dots, X_{ki})\}_i^n$ are independently and identically distributed i.e. i.i.d.

A3 (Finite Fourth Moments): $\mathbb{E}[Y_i^4] < \infty$ and $\mathbb{E}[X_{Ii}^4] < \infty$ for $I \in \{1, \dots, k\}$ (large outliers are unlikely)

A4 (Full Rank/No Perfect Multicollinearity): No perfect multicollinearity among any subset of $(X_{1i}, \dots, X_{ki})$

With these assumptions, the following theorems hold (proof is not needed).

Theorem 2.9: Consistency

$\hat{\beta}_I$ consistently estimates β_I for $I \in \{0, \dots, k\}$ i.e.

$$\hat{\beta}_I \xrightarrow{p} \beta_I, \quad \forall I \in \{0, 1, \dots, k\}$$

Remark.

Follows by Law of Large Numbers

Theorem 2.10: Asymptotic Normality

Let n denote the sample size. Then, as $n \rightarrow \infty$,

$$n^{1/2}(\hat{\beta}_I - \beta_I) \xrightarrow{d} \mathcal{N}(0, \sigma_{\beta_I}^2), \quad \forall I \in \{0, 1, \dots, k\}$$

Remark.

Follows by the Central Limit Theorem

In other words, the distribution of the random variable $n^{1/2}(\hat{\beta}_I - \beta_I)$ is well approximated by $\mathcal{N}(0, \sigma_{\beta_I}^2)$ in large samples i.e. it is the limiting distribution of $n^{1/2}(\hat{\beta}_I - \beta_I)$

Corollary 2.11

We may write the asymptotic distribution of $\hat{\beta}_I$ as

$$\hat{\beta}_I \overset{a}{\sim} \mathcal{N}(\beta_I, \frac{1}{n}\omega_{\beta_I}^2) \quad (7)$$

where $\sigma_{\beta_I}^2 = \frac{1}{n}\omega_{\beta_I}^2$

Remark.

We may call $\omega_{\beta_I}^2$ the asymptotic variance of $n^{1/2}(\hat{\beta}_I - \beta_I)$

Hence, in large samples, the asymptotic standard deviation of $\hat{\beta}_I$ is i.e.

$$\text{sd}(\hat{\beta}_I) \simeq \frac{1}{n^{1/2}}\omega_{\beta_I}$$

and if we let $\hat{\omega}_{\beta_I}$ denote an estimator of ω_{β_I} (when we don't know the true value), we obtain

$$\text{se}(\hat{\beta}_I) := \frac{1}{n^{1/2}}\hat{\omega}_{\beta_I}$$

which is the standard error of $\hat{\beta}_I$ (i.e. the estimate of the standard deviation of $\hat{\beta}_I$).

Asymptotic variance of OLS and Homoskedasticity

Here, the goal is to obtain an expression for the asymptotic variance ω_{β_I} of $\hat{\beta}_I$ so that we may

1. assess the factors affecting the precision of $\hat{\beta}_I$
2. understand how to estimate ω_{β_I} which is necessary to compute $\text{se}(\hat{\beta}_I)$

We examine the expressions of asymptotic variance $\omega_{\beta_I}^2$

1. **Single regressor:** For a model $Y_i = \beta_0 + \beta_1 X_i + u_i$, we have

$$\omega_{\beta_I}^2 = \frac{\mathbb{E}[(X_i - \mathbb{E}[X_i])^2 u_i^2]}{[\mathbb{E}(X_i - \mathbb{E}[X_i])^2]^2} \quad (8)$$

2. **Multiple regressor:** $Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + u_i$, we have

$$\omega_{\beta_I}^2 = \frac{\mathbb{E}[(\tilde{X}_{Ii} - \mathbb{E}[\tilde{X}_{Ii}])^2 u_i^2]}{[\mathbb{E}(\tilde{X}_{Ii} - \mathbb{E}(\tilde{X}_{Ii}))^2]^2} \quad (9)$$

with $\tilde{X}_{Ii}$ being the error in a population regression of

$$X_{Ii} = \pi_0 + \sum_{j \neq i} \pi_j X_{ji} + \tilde{X}_{Ii}$$

where $\mathbb{E}[\tilde{X}_{Ii}] = 0$ and $\mathbb{E}[\tilde{X}_{Ii} X_{ji}] = 0$ for $\forall j \neq i$.

Given that $\mathbb{E}[\tilde{X}_{Ii}] = 0$ and $\mathbb{E}[\tilde{X}_{Ii} u_i] = 0$, we can rewrite the above as

$$\omega_{\beta_I}^2 = \frac{\mathbb{E}[\tilde{X}_{Ii}^2 u_i^2]}{[\mathbb{E}(\tilde{X}_{Ii}^2)]^2} = \frac{\text{var}(\tilde{X}_{Ii} u_i)}{[\text{var}(\tilde{X}_{Ii})]^2} \quad (10)$$

where the first assumption holds by construction and the second holds because u_i is uncorrelated with all regressors X_{ji} for all $j \in \{1, \dots, k\}$ by construction (we express $\tilde{X}_{Ii}$ in terms of error of X_{Ii} on all other regressors)

Definition 2.12: Homoskedasticity

Let $\mathcal{X}_i$ denote $(X_{1i}, \dots, X_{ki})$. We say that homoskedasticity holds just when

$$\mathbb{E}[u_i | \mathcal{X}_i] = 0 \quad \text{and} \quad \mathbb{E}[u_i^2 | \mathcal{X}_i] = \mathbb{E}[u_i^2] \quad (11)$$

i.e. when **MI** holds and u_i^2 is mean independent of the regressors.

Proposition 2.13

When homoskedasticity holds, then

$$\text{var}(u_i | \mathcal{X}_i) = \text{var}(u_i) = \sigma_u^2$$

i.e. that the conditional variance of u_i does not depend on $\mathcal{X}_i$

Proof. Write

$$\begin{aligned} \text{var}(u_i | \mathcal{X}_i) &= \mathbb{E}[(u_i - \mathbb{E}[u_i | \mathcal{X}_i])^2 | \mathcal{X}_i] \\ &= \mathbb{E}[u_i^2 | \mathcal{X}_i] \\ &= \mathbb{E}(u_i^2) \\ &= \text{var}(u_i) \\ &= \sigma_u^2 \end{aligned}$$

with the first equality being the definition of conditional variance. □

Claim: Standard Error under Homoskedasticity

Under homoskedasticity,

$$\omega_{\beta_I}^2 = \frac{\sigma_u^2}{\text{var}(\tilde{X}_{Ii})}$$

Proof for Claim.

Under homoskedasticity, we can write

$$\begin{aligned}\mathbb{E}[\tilde{X}_{Ii}^2 u_i^2] &= \mathbb{E}[\mathbb{E}[\tilde{X}_{Ii}^2 u_i^2 | \mathcal{X}_i]] \\ &= \mathbb{E}[\tilde{X}_{Ii}^2 \mathbb{E}[u_i^2 | \mathcal{X}_i]] \\ &= \sigma_u^2 \mathbb{E}[\tilde{X}_{Ii}^2] \\ &= \sigma_u^2 \text{var}(\tilde{X}_{Ii})\end{aligned}$$

with the second last equality following by homoskedasticity. Then, we simplify

$$\omega_{\beta_I}^2 = \frac{\mathbb{E}[(\tilde{X}_{Ii} - \mathbb{E}[\tilde{X}_{Ii}])^2 u_i^2]}{[\mathbb{E}(\tilde{X}_{Ii} - \mathbb{E}[\tilde{X}_{Ii}])^2]^2}$$

to

$$\omega_{\beta_I}^2 = \frac{\sigma_u^2}{\text{var}(\tilde{X}_{Ii})}$$

since $\mathbb{E}[\tilde{X}_{Ii}] = 0$ by definition ■

Interpretation: This implies that $\hat{\beta}_I$ delivers a more precise estimate of β_I i.e. smaller SE is when

- the error has a smaller variance i.e. σ_u^2 is smaller
- the variability of $\tilde{X}_{Ii}$, the part of X_{1i} not explained by the other regressors, is larger

IMPORTANT: when we try to find the standard deviation or standard error of $\hat{\beta}_I$, we rescale ω_{β_I} by $1/n^{1/2}$ i.e.

$$\text{sd}(\hat{\beta}_I) \simeq \frac{1}{n^{1/2}} \omega_{\beta_I} \quad \text{and} \quad \text{se}(\hat{\beta}_I) := \frac{1}{n^{1/2}} \hat{\omega}_{\beta_I}$$

So, if we're willing to assume homoskedasticity, then we can compute $\text{se}(\hat{\beta}_I)$ as

$$\text{se}(\hat{\beta}_I) = \frac{1}{n^{1/2}} \left[\frac{s_{\hat{u}}^2}{\text{var}(\hat{\tilde{X}}_I)} \right]^{1/2} = \frac{1}{n^{1/2}} \cdot \frac{s_{\hat{u}}}{\text{sd}(\hat{\tilde{X}}_I)}$$

where $s_{\hat{u}}^2 = \sum_{i=1}^n u_i^2$ i.e. the estimate of the error variance and $\hat{\tilde{X}}_I$ which denotes the residual of the OLS regression of X_I on all other regressors.

Definition 2.14: Heteroskedastic consistent standard errors

Suppose homoskedasticity does not hold i.e. we assume heteroskedasticity. Then, our standard errors can be computed as

$$se(\hat{\beta}_I) = \frac{1}{n^{1/2}} \left[\frac{\text{var}(\tilde{X}\hat{u})}{[\text{var}(\tilde{X})]^2} \right]^{1/2} = \frac{1}{n^{1/2}} \frac{\hat{sd}(\tilde{X}\hat{u})}{[\text{var}(\tilde{X})]} \quad (12)$$

Basically, OLS underestimates SE under heteroskedasticity, and so inflates t -value and increases Type I error.

2.2.2 Inference on Regression Coefficients: Testing

We have the following definitions for error

Definition 2.15: Error Types

We have the following errors:

1. Type I error rate: $P_{H_0}(\text{reject } H_0)$ i.e. the probability of rejecting the null when it is true (also the significance level of the test)
2. Type II error rate: $P_{\beta_1 \neq b}(\text{accept } H_0)$ i.e. the probability of accepting the null when it is false; which varies depending on the value of β_1

We also have a new concept called the power of tests

Definition 2.16: Power of tests

The power of a test (at $\beta_1 \neq b$) is the rate at which false nulls are detected

$$P_{\beta_1 \neq b}(\text{reject } H_0) = 1 - P_{\beta_1 \neq b}(\text{accept } H_0)$$

which is 1 minus the type II error rate

Single Hypothesis

We have both two sided t tests and one sided t tests

- **Two-sided t tests:** Consider the following null hypotheses and two-sided alternative

$$H_0 : \beta_1 = b$$

$$H_1 : \beta_1 \neq b$$

where b is the hypothesised value of the parameter β_1 .

Claim: Asymptotics of t statistic

Under the H_0 , the t -statistic is asymptotically $\mathcal{N}(0, 1)$

Proof for Claim.

We may write

$$t(b) := \frac{\hat{\beta}_1 - b}{\text{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\hat{\omega}_{\beta_1}/n^{1/2}} = \frac{n^{1/2}(\hat{\beta}_1 - \beta_1)}{\hat{\omega}_{\beta_1}} \xrightarrow{d} \frac{1}{\omega_{\beta_1}} \mathcal{N}(0, \omega_{\beta_1}^2) \sim \mathcal{N}(0, 1)$$

In contrast, under the alternative where $\beta_1 \neq b$, we have

$$t(b) = \frac{\hat{\beta}_1 - b}{\text{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\text{se}(\hat{\beta}_1)} + \frac{n^{1/2}(\beta_1 - b)}{\hat{\omega}_{\beta_1}} = t(\beta_1) + \frac{n^{1/2}(\beta_1 - b)}{\hat{\omega}_{\beta_1}}$$

where $t(\beta_1) \xrightarrow{d} \mathcal{N}(0, 1)$ and $n^{1/2}(\beta_1 - b)/\hat{\omega}_{\beta_1} \rightarrow \pm\infty$, based on whether $b > \beta_1$ or otherwise. Hence, when the null is false, we should expect to see large values of $|t(b)|$. Furthermore, this means that the power of the test reduces (we false accept H_0) if the distribution is near.

So we have the decision rule:

Definition 2.17: Decision Rule for two-sided tests

Reject H_0 if $|t(b)| > c$ where c is the critical value at a α significance level

We choose c to deliver a desired ‘false rejection rate’ (significance level). So in general,

$$c_\alpha = -\Phi^{-1}(\alpha/2)$$

so if $\alpha = 0.05$, then $c_\alpha = 1.96$

- **One sided t tests.** This is not much different except we have

$$H_0 : \beta_1 = b$$

$$H_1 : \beta_1 > b$$

These sorts of tests makes sense only if there is an a priori reason to know that $\beta_1 \geq b$ must be true. It continues to hold that $t(b) \xrightarrow{d} \mathcal{N}(0, 1)$ under the null. Under the alternative, $t(b) \rightarrow \infty$ since we know that $\beta_1 > b$. So, our decision rule is the same, just without the modulus i.e.

Definition 2.18: Decision Rule for one-sided tests

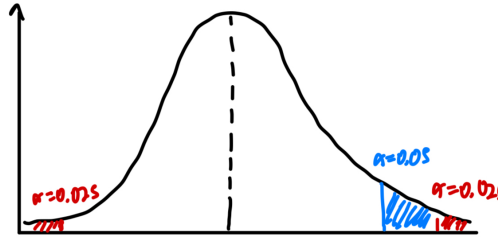
Reject H_0 if $t(b) > c$ where c is the critical value at a α significance level

The important thing to note is that the critical values differ from the two-sided case

since to deliver a test with significance level α , we choose $c = c_\alpha$ such that

$$\alpha = P_{H_0}(\text{reject } H_0) \simeq P(\mathcal{N}(0, 1) > c_\alpha) = 1 - \Phi(c_\alpha) = \Phi(-c_\alpha)$$

So, when $\alpha = 0.05$, the critical value is $c_\alpha = 1.64$ which is smaller than the critical value of 1.96 for the two-sided test. Diagrammatically, this makes sense based on the normal distribution.



Hence, one-sided tests are more powerful than two-sided tests at detecting departures from the nominated direction.

Table 1: Z critical values (standard normal distribution)

Significance Level α	One-sided Test	Two-sided Test
0.10	1.28	1.64
0.05	1.64	1.96
0.01	2.33	2.58

The alternative to critical values are p -values which summarise the weight of evidence against the null. Suppose that the test statistic $t(b)$ computed on the basis of the sample has the value t^{act} . The basic questions are

- What is the probability, under the null, of obtaining a value of the test-statistic that is ‘at least as unfavourable to the null’ as t^{act} is?
- What is the smallest significance level for which we would reject the null, on the basis of this sample? (So as α falls, we become less likely to reject the null)

We may calculate the asymptotic p -value as

$$p := P_{H_0}(|t(b)| > |t^{\text{act}}|) \simeq P(\mathcal{N}(0, 1) > |t^{\text{act}}|) = 2\Phi(-|t^{\text{act}}|)$$

We reject H_0 if

$$\alpha > 2\Phi(-|t^{\text{act}}|) = c_\alpha$$

But since the t test is consistent, even if the null is only slightly violated (i.e. true value of β_1 is very small but nonzero) when the sample size is sufficiently large. So, we may prefer a range of values for β_1 that are consistent with the observed data. This range tells us something about the likely magnitude of β_1 and not just whether β_1 is statistically distinguishable from zero.

Definition 2.19: Confidence Intervals

A confidence interval at a level of significance α is the set of all values of $\hat{\beta}_1$ for which the null hypothesis $H_0 : \beta_1 = b$ is accepted. Formally, the interval is constructed as

$$\mathcal{C} := \{b \in \mathbb{R} | H_0 : \beta_1 = b \text{ is accepted}\} = \{b \in \mathbb{R} | |t(b)| \leq c_\alpha\}$$

To see that this is an interval, notice that we can write

$$\begin{aligned}\mathcal{C} &= \left\{ b \in \mathbb{R} \mid \left| \frac{\hat{\beta}_1 - b}{\text{se}(\hat{\beta}_1)} \right| \leq c_\alpha \right\} \\ &= \{b \in \mathbb{R} \mid -c_\alpha \text{se}(\hat{\beta}_1) \leq \hat{\beta}_1 - b \leq c_\alpha \text{se}(\hat{\beta}_1)\} \\ &= [\hat{\beta}_1 - c_\alpha \text{se}(\hat{\beta}_1), \hat{\beta}_1 + c_\alpha \text{se}(\hat{\beta}_1)]\end{aligned}$$

Remark.

The formal definition of the confidence interval is as for (2.19). The form that we can derive is contingent on the structure of the t statistic

Interpretation: We say that in approximately $(1 - \alpha) \times 100\%$ of all possible samples, the confidence interval will contain the true value of β_1 . We do not talk about the probabilistic behaviour of β_1 since it is a fixed quantity.

The confidence interval has what we may call the *coverage probability* of $\mathcal{C}$ at b . For any fixed $b \in \mathbb{R}$, we have

$$P_{\beta_1=b}(b \in \mathcal{C}) = P_{\beta_1=b}(|t(b)| \leq c_\alpha) \simeq P(|\mathcal{N}(0,1)| \leq c_\alpha) = 1 - \alpha$$

which holds in large samples.

Multiple Hypotheses

Consider a regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

with $\mathbb{E}[u_i | X_{1i}, X_{2i}, X_{3i}] = 0$. Suppose we wanted to test whether each of the regressors have a statistically significant effect on the dependent variable. We may do the single variable test 3 times so the decision rule becomes

$$\text{Reject } H_0 \text{ if } |t_{n,1}(0)| > c \text{ or } |t_{n,2}(0)| > c \text{ or } |t_{n,3}(0)| > c$$

But there is a more efficient way to do this using the F test of multiple restrictions.

F test of multiple restrictions

Suppose that the model is

$$Y_i = \beta_0 + \sum_{j=1}^k \beta_j X_{ji} + u_i$$

where $\mathbb{E}[u_i|\mathcal{X}_i] = 0$ and we want to test $q < k$ restrictions. Our hypotheses are

$$\begin{aligned} H_0 : \beta_1 = \beta_2 = \dots = \beta_q = 0 \\ H_1 : \beta_I \neq 0 \text{ for at least one } I \in \{1, \dots, q\} \end{aligned}$$

Motivation: The idea behind the (homoskedastic) F test is to consider how the fit of the model (in terms of SSR or R^2) deteriorates as a consequence of imposing the null. So, we estimate a restricted model in which we force H_0 to be true and an ‘unrestricted model’ in which the parameters are allowed to freely vary. So in other words, we have two models

1. Restricted model: regress Y_i on $(X_{q+1,i}, \dots, X_{ki})$ i.e. fit

$$Y_i = \beta_0 + \sum_{j=q+1}^k \beta_j X_{ji} + u_i$$

which yields the residuals $\hat{u}_{rs,i}$ and $\text{SSR}_{rs} = \sum_{i=1}^n \hat{u}_{rs,i}^2$

2. Unrestricted model: regress Y_i on $(X_{1i}, \dots, X_{ki})$, yielding residuals $\hat{u}_{un,i}$ and $\text{SSR}_{un} = \sum_{i=1}^n \hat{u}_{un,i}^2$

Then, we have

Definition 2.20: F statistic

We define the F statistic as

$$F := \frac{(\text{SSR}_{rs} - \text{SSR}_{un})/q}{\text{SSR}_{un}/(n - k - 1)} \xrightarrow{d} \frac{1}{q} \chi_q^2 = F_{q,\infty}$$

where q is the number of restrictions, k is the number of regressors, and n is the sample size

We may state the F statistic in terms of the R^2 of the restricted and unrestricted models. Since we have

$$1 - R_m^2 = \frac{\text{SSR}_m}{\text{TSS}}, \quad m \in \{\text{un}, \text{rs}\}$$

we have

$$F = \frac{(R_{un}^2 - R_{rs}^2)/q}{(1 - R_{un}^2)/(n - k - 1)}$$

One special case of the testing problem is when we have a null that no variables are relevant i.e. $q = k$. This simplifies the test such that

- $k = q$ since all k coefficients are restricted
- $\text{SSR}_{rs} = \text{TSS}$ and $R_{rs}^2 = 0$ since the restricted model is $Y_i = \beta_0 + u_i$

We may generalise the approach to more linear restrictions. For instance, we may have the hypotheses

$$\begin{aligned} H_0 : \beta_1 = b \text{ and } \beta_2 + \beta_3 = c \\ H_1 : \beta_1 \neq b \text{ and } \beta_2 + \beta_3 \neq c \end{aligned}$$

where c and b are known constants. Our restricted model will just make the substitutions $\beta_1 = b$ and $\beta_2 = c - \beta_3$.

Power of Tests

In general, we would like a more powerful test than not, since we do not want to not reject false nulls. But the problem is that there is a trade-off between the power and size (critical value) of any test.

Proposition 2.21

There is a trade-off between the power and size of any test. Choosing a smaller critical value c would make it more likely to reject the null when it is false i.e. decrease the type II error rate so the test is more powerful but it will also increase the type I error rate since it is more likely to reject a true null

Intuition: The critical value c and the level of significance are inversely related (when $\alpha = 0.05$, then $c = 1.96$ and when $\alpha = 0.01$, then $c = 2.58$). It follows that when I make c smaller, since if H_0 is false, then $|t(b)|$ will be larger, so it is easier to reject false nulls when they are actually false (i.e. we do not accept H_0 when it is false), minimising type II error rate. But also, this increases type I error rate because decreasing c implies that the critical region is larger and so it is easier for $|t(b)|$ to be rejected even when H_0 is true, hence increase type I error rate.

The following example is instructive

Definition 2.22: Consistency of Tests

For any fixed $\beta_1 \neq b$, then if

$$P_{\beta_1 \neq b}(\text{reject } H_0) = P_{\beta_1 \neq b}(|t(b)| > c_\alpha) \rightarrow 1$$

the test t is said to be consistent

In other words, we are able to detect any failure of the null with probability approaching 1 in large samples. This is because the test statistic diverges to $\pm\infty$ and so eventually exceeds any fixed threshold c_α

2.3 Model Specification

Definition 2.23: Linearity in parameters

A regression model is said to be linear in parameters just when it can be written as a linear combination of its coefficients (parameters) i.e.

$$Y_i = \beta_0 + \beta_1 f_1(X_{1i}, \dots, X_{ki}) + \dots + \beta_m f_m(X_{1i}, \dots, X_{ki})$$

Remark.

There is a difference between a *linear* model and a model that is *linear in parameters*

Example.

The logistic regression model is NOT linear in parameters i.e.

$$P(Y_i = 1|x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

Example.

The following regression models are linear in parameters

- linear-log model:

$$Y_i = \beta_0 + \beta_1 \log X_i$$

- log-linear model:

$$\log Y_i = \beta_0 + \beta_1 X_i$$

- log-log model:

$$\log Y_i = \beta_0 + \beta_1 \log X_i$$

The intuition is clear: insofar as you are able to rewrite some variables, say $X_{1i}X_{2i}$ as W_i , without interfering with the coefficient β_i , then the model is linear in parameters.

Our orthogonality conditions change with respect to the variables. Generally, we use the functions $f_1, \dots, f_k$. Consider a model $Y_i = \beta_0 + \beta_1 X_{1i}X_{2i} + u_i$. Then, our **OR** condition is

$$\mathbb{E}[u_i] = \mathbb{E}[X_{1i}X_{2i}u_i] = 0$$

This holds analogously for all models which are linear in parameters

Remark.

Be careful when using the log transformation because variables that have log cannot take the value 0 since $\log 0$ is undefined. For instance, imagine an RCT to measure the income elasticity of food consumption using income transfers. One may be tempted to use the log of the income transfers but this fails because there are some observations with 0 income transfers.

The fundamental point is that there is no substantive difference between models which are linear in parameters. In other words, we may rewrite **OR** or **MI** in terms of $f_j(X_{1i}, \dots, X_{ki})$ written as W_j as sufficient conditions for the population regression model parameters to coincide with the causal model.

Marginal effects

In a linear model such as

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

we may interpret

$$\frac{\partial Y_i}{\partial X_i} = \beta_1$$

as the causal effect of X on Y . This is because if X_i is increased by an amount ΔX_i , then Y changes by

$$\Delta Y_i = \beta_1 \Delta X_i$$

If we normalise $\Delta X_i = 1$, then we get the natural interpretation of ‘a unit increase in X is causally associated with a β_1 change in Y ’ where $\beta_1 = \Delta Y$.

The problem is that it is not so easy to interpret the coefficients (β_1, β_2) in a nonlinear model like

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i$$

Furthermore, we may not speak unambiguously of ‘the’ causal effect of X and Y since X affects Y both via β_1 and β_2 . In particular,

$$\begin{aligned} \Delta Y &= [\beta_0 + \beta_1(X + \Delta X) + \beta_2(X + \Delta X)^2] - [\beta_0 + \beta_1 X + \beta_2 X^2] \\ &= \beta_1 \Delta X + \beta_2 [2X \Delta X + (\Delta X)^2] \end{aligned}$$

So, there is a multiplicity of causal effects for each of X and ΔX .

Definition 2.24: Marginal Effects

Consider a model that is linear in parameters

$$Y = \beta_0 + \beta_1 f_1(X_1, \dots, X_k) + \dots + \beta_m f_m(X_1, \dots, X_k)$$

Then, the marginal effect of X on Y is

$$\frac{\partial Y}{\partial X_j} = \beta_1 \frac{\partial f_1}{\partial X_j} + \dots + \beta_m \frac{\partial f_m}{\partial X_j}$$

Given the underlying concept of a derivative, this gives a reasonable approximation to the resulting change in Y for a small change in X_j and has the advantage of not requiring an explicit choice of ΔX . It will often still be in terms of X_j .

Estimation of marginal effects: to give an estimate of the marginal effects, we replace the unknown values of $(\beta_1, \dots, \beta_m)$ with their OLS estimates $(\hat{\beta}_1, \dots, \hat{\beta}_m)$ and choose a value of X to evaluate the marginal effect equation. We may commonly choose the sample mean or the median of X . Letting $\bar{x}_j$ denote this, we then have

$$\hat{Y}'(\bar{x}_j) = \hat{\beta}_1 \frac{\partial f_1}{\partial X_j} \Big|_{X_j = \bar{x}_j} + \dots + \hat{\beta}_m \frac{\partial f_m}{\partial X_j} \Big|_{X_j = \bar{x}_j}$$

which provides a consistent estimate the population marginal change (and the causal model if **OR** is satisfied) evaluated at $\bar{x}$

Statistical Tests for Marginal effects

We may perform statistical inference on the value of $Y'(x)$ (for some x). Consider for a model $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + u$ we have

$$\begin{aligned} H_0 : Y'(x) &= b \\ H_1 : Y'(x) &\neq b \end{aligned}$$

for a chosen b . Since the null can be equivalently expressed as

$$H_0 : \beta_1 + 2\beta_2 x - b = 0$$

we may apply the framework of the F tests for linear restrictions.

Alternatively, a test of H_0 can be carried out by a kind of t test. The key insight is that since $\hat{\beta}_I$ s are asymptotically normal, so too is $\hat{Y}'(x)$. It's variance depends on that of $\hat{\beta}_I$ s since

$$\text{var}(\hat{Y}'(x)) = \text{var}(\hat{\beta}_1 + 2\hat{\beta}_2 x) = \text{var}(\hat{\beta}_1) + 4x\text{cov}(\hat{\beta}_1, \hat{\beta}_2) + 4x^2\text{var}(\hat{\beta}_2)$$

Since we can estimate the standard deviation of $\hat{\beta}_I$, we can use the SE to estimate that. The only challenge is to estimate $\text{cov}(\hat{\beta}_1, \hat{\beta}_2)$ which is beyond the scope.

Hence, we may construct a standard error $\text{se}[\hat{Y}'(x)]$ that is an estimate of $\text{sd}[\hat{Y}'(x)]$ and which can be used to construct t statistics and confidence intervals analogously as per individual coefficients. So, under the null, we have

$$t(b) = \frac{\hat{Y}'(x) - b}{\text{se}[\hat{Y}'(x)]} \xrightarrow{d} \mathcal{N}(0, 1)$$

and a $1 - \alpha$ confidence interval for $\hat{Y}'(x)$ can be constructed mechanically as

$$\mathcal{C} = \{\hat{Y}'(x) - c_\alpha \text{se}[\hat{Y}'(x)], \hat{Y}'(x) + c_\alpha \text{se}[\hat{Y}'(x)]\}$$

3 Endogeneity

The basic issue is that exogeneity of a regressor X_I i.e. when $\text{cov}(X_I, u) = 0$ often does not hold. We say that a regressor X_I is *endogenous* when $\text{cov}(X_I, u) \neq 0$ and generally that there is an endogeneity problem when $\text{cov}(X_I, u) \neq 0$ for at least one $I \in \{1, \dots, k\}$. So, we want to find out cases in which endogeneity arises systematically.

3.1 Omitted Variables

The general case. Consider a long regression and a short regression:

- Long regression: $Y = \beta_0 + \sum_{I=1}^{k-1} \beta_I X_I + \beta_k X_k + u$
- Short regression: $Y = \beta_0 + \sum_{I=1}^{k-1} \beta_I X_I + v$

In the long regression, we assume that it is the causal model and that **OR** holds so that $\mathbb{E}[u] = 0$ and $\text{cov}(X_I, u) = 0$ for all $I \in \{1, k\}$. The upshot of OVB is that ignoring one variable will lead to bias and so there will be at least one I such that $\text{cov}(X_I, v) \neq 0$ in the short regression.

Proposition 3.1

Consider the population regression model for the short regression

$$Y = \beta_0 + \sum_{I=1}^{k-1} \gamma_I X_I + e$$

Then, $\hat{\gamma}_j \xrightarrow{P} \gamma_j = \beta_j + \beta_k \pi_j$. In other words, the population regression coefficient no longer coincides with the causal model

Proof. We consider the population regression model for the short regression

$$Y = \beta_0 + \sum_{I=1}^{k-1} \gamma_I X_I + e$$

Then, we know that from the FWL theorem,

$$\gamma_j = \frac{\text{cov}(Y, \tilde{X}_j)}{\text{var}(\tilde{X}_j)}$$

Then, using the properties of covariance, we substitute the causal model (long regression) in

$$\begin{aligned} \text{cov}(Y, \tilde{X}_j) &= \text{cov}\left(\beta_0 + \sum_{I=1}^k \beta_I X_I + u, \tilde{X}_j\right) \\ &= \sum_{I \neq j} \beta_I \text{cov}(X_I, \tilde{X}_j) + \beta_j \text{cov}(X_j, \tilde{X}_j) + \text{cov}(u, \tilde{X}_j) \\ &=_{(1)} \beta_j \text{cov}(\alpha_0 + \sum_{I \neq j} \alpha_I X_I + \tilde{X}_j, \tilde{X}_j) + \beta_k \text{cov}(X_k, \tilde{X}_j) \\ &= \beta_j \text{cov}(\tilde{X}_j, \tilde{X}_j) + \beta_k \text{cov}(X_k, \tilde{X}_j) \end{aligned}$$

So, we take

$$\gamma_j = \frac{\text{cov}(Y, \tilde{X}_j)}{\text{var}(\tilde{X}_j)} = \beta_j + \beta_k \frac{\text{cov}(X_k, \tilde{X}_j)}{\text{var}(\tilde{X}_j)} = \beta_j + \beta_k \pi_j$$

where $\frac{\text{cov}(X_k, \tilde{X}_j)}{\text{var}(\tilde{X}_j)} = \pi_j$ i.e. the coefficient of the population regression of X_k on X_j . So, $\gamma_j = \frac{\text{cov}(Y, \tilde{X}_j)}{\text{var}(\tilde{X}_j)}$. (1) follows from substituting the regression of X_j on all the other variables, and $\text{cov}(u, \tilde{X}_j) = 0$ because **OR** is supposed to hold in the long regression, ex hypothesi. Since $\hat{\gamma}_j$ is consistent for γ_j , this completes the proof $\square$

Remark.

It is easy to see this for the two variable case as the long regression, that $\hat{\gamma}_1 \xrightarrow{P} \gamma_1 = \beta_1 + \beta_2\pi_1$

Utility of the OVB

1. We can ignore the OVB if it is likely to be small. We can therefore still give $\hat{\gamma}_j$ a causal interpretation. This may be because we think that X_k is likely to have a small effect on Y so β_k is very small or is unlikely to be correlated with X_j i.e. π_j is small. Then we may ignore the bias – particularly if it is smaller than the sampling variability of $\hat{\gamma}_j$
2. We also want to know the direction of the bias i.e. if the omitted variable X_k has a positive effect on Y and is positively correlated with X_j , then $\beta_k\pi_j > 0$ and $\hat{\gamma}_j$ can be interpreted as estimating an upper bound of the causal effect of X_j on Y

Proxying for Omitted Variables. Sometimes the missing variable occurs due to unobservability or unmeasurability i.e. ‘outside learning opportunities’ and its effect on students scores. Consider a model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$$

and X_2 is unobservable. Estimating β_1 via a regression of Y on X_1 alone would be subject to OVB. Suppose further that we observe W_1, W_2 that are plausibly correlated with X_2 and assume for simplicity that $\text{cov}(W_1, u) = \text{cov}(W_2, u) = 0$. Then, consider a hypothetical population regression $X_2 = \delta_0 + \delta_1 W_1 + \delta_2 W_2 + e$, where e is uncorrelated with (W_1, W_2) by construction. Substituting this we get

$$\begin{aligned} Y &= \beta_0 + \beta_1 + \beta_2(\delta_0 + \delta_1 W_1 + \delta_2 W_2 + e) + u \\ &= (\beta_0 + \beta_2\delta_0) + \beta_1 X_1 + \beta_2\delta_1 W_1 + \beta_2\delta_2 W_2 + (\beta_2 e + u) \\ &=: \gamma_0 + \beta_1 X_1 + \gamma_2 W_1 + \gamma_3 W_2 + \eta \end{aligned}$$

Since our goal is to estimate β_1 , the condition for the preceding to coincide with a population linear regression is $\mathbb{E}[\eta] = 0$

$$\text{cov}(W_1, \eta) = \beta_2 \text{cov}(W_1, e) = \text{cov}(W_1, u) = 0$$

and analogously with W_2 in place of W_1 . Since **OR** was assumed to hold for the causal model,

$$\text{cov}(X_1, \eta) = \beta_2 \text{cov}(X_1, e) + \text{cov}(X_1, u) = \beta_2 \text{cov}(X_1, e)$$

So, for the regression $Y = \gamma_0 + \beta_1 X_1 + \gamma_2 W_1 + \gamma_3 W_2 + \eta$ to coincide with the causal model, we need either

1. $\beta_2 = 0$, so that X_2 is irrelevant
2. $\text{cov}(X_1, e) = 0$, which expresses that W_1 and W_2 control for the correlation between X_1 and X_2 i.e. that W_1 and W_2 predict X_2 well enough that the part of X_2 not predicted by them is uncorrelated with X_1

Definition 3.2: Valid Proxies

We say that (W_1, W_2) are valid proxies just when the second condition is satisfied

This is in the sense that β_1 is recovered by a population linear regression of Y on (X_1, W_1, W_2)

Remark.

We CANNOT give causal estimates to proxy variables, only those that are regarded as *causal* determinants of Y

However, suppose we have a model

$$Y = \beta_0 + \beta_1 X + \beta_2 W + u$$

and turns out that $\text{cov}(W, u) = \mathbb{E}[u] = 0$ but $\text{cov}(X, u) \neq 0$. It follows that we may not consistently estimate β_1 with the OLS estimator, but what about β_2 ?

We claim that unless $\text{cov}(X, W) = 0$, then we may not consistently estimate β_2 . In other words, either our OLS estimate is inconsistent or $\text{cov}(X, W) = 0$. This is equivalent to the claim, if our OLS estimate is consistent, then $\text{cov}(X, W) = 0$. Consider the population regression

$$Y = \gamma_0 + \gamma_1 X + \gamma_2 W + e$$

By the FWT theorem, we have

$$\gamma_2 = \frac{\text{cov}(Y, \tilde{W})}{\text{var}(\tilde{W})}$$

Considering the numerator and defining $\tilde{W} = W - (\pi_0 + \pi_1 X)$, we yield

$$\begin{aligned} \text{cov}(Y, \tilde{W}) &= \text{cov}(\beta_0 + \beta_1 X + \beta_2 W + u, \tilde{W}) \\ &=_{(1)} \beta_2 \text{cov}(W, \tilde{W}) + \text{cov}(u, \tilde{W}) \\ &=_{(2)} \beta_2 \text{var}(\tilde{W}) + \text{cov}(u, W - \pi_0 - \pi_1 X) \\ &= \beta_2 \text{var}(\tilde{W}) - \pi_1 \text{cov}(u, X) \end{aligned}$$

where (1) follows since $\text{cov}(X, \tilde{W}) = 0$ by construction, (2) follows since $\text{cov}(W, \tilde{W}) = \text{cov}(\pi_0 + \pi_1 X + \tilde{W}, \tilde{W}) = \text{var}(\tilde{W})$ by construction. Then,

$$\gamma_2 = \frac{\text{cov}(Y, \tilde{W})}{\text{var}(\tilde{W})} = \beta_2 - \pi_1 \cdot \frac{\text{cov}(u, X)}{\text{var}(\tilde{X})} = \beta_2$$

ex hypothesi. Since we know that generally that $\text{cov}(u, X) \neq 0$, it must be that $\pi_1 = \frac{\text{cov}(X, W)}{\text{var}(X)} = 0$ which implies that $\text{cov}(X, W) = 0$.

3.2 Measurement Errors

Two kinds of measurement error

- Misreporting of survey responses
- Observed variables are imperfectly related to the underlying variable i.e. ‘total years of

education' is only imperfectly related to an individual's actual educational attainment

The basic goal is we want to find the conditions for the regression model to coincide with the causal model, for a regression model

$$Y = \beta_0 + \beta_1 X + u$$

for which **OR** holds i.e. $\mathbb{E}[u] = \text{cov}(X, u) = 0$

Measurement error in dependent variable. Instead of observing Y we observe

$$Y^* := Y + e_Y$$

Then, estimating β_1 via a regression of Y^* on X yields

$$Y^* - e_Y = \beta_0 + \beta_1 X + u \implies Y^* = \beta_0 + \beta_1 X + (u + e_Y)$$

Hence, for β_1 to coincide with the coefficient on X in a population regression of Y^* on X , we need only that

$$0 = \text{cov}(X, u + e_Y) = \text{cov}(X, u) + \text{cov}(X, e_Y) = \text{cov}(X, e_Y)$$

where the last equality holds by **OR**. Hence, so long as the measurement error in Y is not correlated with X , an OLS regression of Y^* on X would consistently estimate β_1 .

The only impact is that the error $u + e_Y$ may have a larger variance if e_Y and u are independent since $\text{var}(e_Y + u) = \text{var}(u) + \text{var}(e_Y)$.

Measurement error in independent variable. Suppose that X is measured with error i.e. we observe

$$X^* = X + e_X$$

instead of X .

Proposition 3.3

Suppose that e_X is uncorrelated with X and Y i.e.

$$\text{cov}(X, e_X) = \text{cov}(Y, e_X) = 0$$

which implies that e_X is uncorrelated with $u := Y - \beta_0 - \beta_1 X$. Then β_1 would not be consistently estimated by an OLS regression of Y on X^*

Proof. Substituting $X^* = X + e_X$ (or $X = X^* - e_X$) into the causal model yields

$$\begin{aligned} Y &= \beta_0 + \beta_1(X^* - e_X) + u \\ &= \beta_0 + \beta_1 X^* + (u - \beta_1 e_X) \\ &=: \beta_0 + \beta_1 X^* + \varepsilon \end{aligned}$$

Since $\text{cov}(X^*, e_X) = \text{var}(e_X)$, it follows that

$$\begin{aligned}\text{cov}(X^*, \varepsilon) &= \text{cov}(X^*, u - \beta_1 e_X) \\ &= \text{cov}(X^*, u) - \beta_1 \text{cov}(X^*, e_X) \\ &= -\beta_1 \text{cov}(X^*, e_X) \neq 0\end{aligned}$$

So, X^* is endogenous in the causal model and β_1 would not be consistently estimated by an OLS regression as $\text{cov}(X^*, \varepsilon) \neq 0$. $\square$

Proposition 3.4

Consider the coefficient of the population linear regression of Y on X^* , β_1^* . Then,

$$\beta_1^* = \beta_1 \frac{\text{var}(X)}{\text{var}(X) + \text{var}(e_X)}$$

and $\hat{\beta}_1^*$ will consistently estimate this

Proof. We know that

$$\beta_1^* = \frac{\text{cov}(Y, X^*)}{\text{var}(X^*)}$$

Considering the numerator, we have

$$\begin{aligned}\text{cov}(Y, X^*) &= \text{cov}(\beta_0 + \beta_1 X^* + \varepsilon, X^*) \\ &= \beta_1 \text{cov}(X^*, X^*) + \text{cov}(\varepsilon, X^*) \\ &= \beta_1 \text{var}(X^*) + \text{cov}(u - \beta_1 e_X, X + e_X) \\ &= \beta_1 [\text{var}(X^*) - \text{var}(e_X)]\end{aligned}$$

Then, we yield

$$\begin{aligned}\beta_1^* &= \beta_1 \frac{\text{var}(X^*) - \text{var}(e_X)}{\text{var}(X^*)} \\ &= \beta_1 \left(1 - \frac{\text{var}(e_X)}{\text{var}(X) + \text{var}(e_X)} \right) = \beta_1 \frac{\text{var}(X)}{\text{var}(X) + \text{var}(e_X)}\end{aligned}$$

$\square$

Remark.

Since the term multiplying β_1 must lie between 0 and 1, the effect of the measurement error is to shrink β_1^* towards 0

Since an OLS regression of Y on X^* will consistently estimate β_1^* rather than β_1 we call the above result the least squares attenuation bias formula with the extent of that attenuation being proportional to the extent of the measurement error in X .

Qualitative Generalisation: Suppose there were multiple regressors. If only one of these is measured with error, then

- the OLS estimate of the coefficient on the mismeasured variable will suffer from an attenuation bias analogous to that of the above
- the OLS estimate of all other coefficients will be biased, but possibly in different directions (depending on the directions of the correlations between the regressors)

3.3 Simultaneity

This happens when one of the regressors in a model partly depends on the dependent variable. Two variables may be regarded as ‘jointly’ or ‘simultaneously’ determined rather than one being determined as a function of the other.

The prime example is for estimation of demand and supply equations. We have

$$Q^d = \beta_0 + \beta_1 P + \beta_2 X + u$$

and

$$Q^s = \gamma_0 + \gamma_1 P + v$$

Clearly, given the market clearing condition, $Q = Q^s = Q^d$, these three equations jointly determine the equilibrium price and quantity.

In the causal model,

$$Q = \beta_0 + \beta_1 P + \beta_2 X + u$$

Notice that because P and Q are simultaneously determined by market clearing, we cannot say that $\text{cov}(P, u) = 0$ i.e. **OR** fails. Suppose there is an increased preference for the commodity among consumers i.e. u increases. This shifts the demand curve upwards causing a movement along the supply curve until equilibrium is re-established at a higher quantity Q and also at a higher price P . Intuitively, $\text{cov}(P, u) \neq 0$.

Consider the market clearing condition

$$\begin{aligned} \gamma_0 + \gamma_1 P + v &= \beta_0 + \beta_1 P + \beta_2 X + u \\ \implies (\gamma_1 - \beta_1)P &= (\beta_0 - \gamma_0) + \beta_2 X + (u - v) \\ \implies P &= \frac{\beta_0 - \gamma_0}{\gamma_1 - \beta_1} + \frac{\beta_2}{\gamma_1 - \beta_1} X + \frac{u - v}{\gamma_1 - \beta_1} \end{aligned}$$

so that even if $\text{cov}(u, v) = 0$ we have

$$\text{cov}(P, u) = \frac{\text{var}(u)}{\gamma_1 - \beta_1} \neq 0$$

3.4 Randomised Control Trials

In RCTs, we call the value of X the value of the treatment, and when D is a binary variable, we call $D = 1$ the treatment group and $D = 0$ the control group. RCTs imply that there is random assignment.

Definition 3.5: Random Assignment

We say that the treatment X is randomly assigned just when

$$X \perp\!\!\!\perp u$$

In particular, when we have a set of controls $\mathcal{W} = (W_1, W_2, \dots, W_k)$ that is supposed to control for every other determinant of Y that is not determined by X , we may say that the treatment is conditionally independent of u .

Definition 3.6: Conditional Independence Assumption

The CIA implies that for a treatment X and controls $\mathcal{W} = (W_1, W_2, \dots, W_k)$, we have

$$X|\mathcal{W} \perp\!\!\!\perp u$$

We may model the assigned treatment X as a function of these covariates

$$X = \gamma_0 + \gamma_1 W_1 + \dots + \gamma_k W_k + \tilde{X}$$

and $\text{cov}(\tilde{X}, W_I) = 0$ for all $I \in \{1, \dots, k\}$. Then, our regression model includes W_I i.e.

$$Y = \beta_0 + \beta_1 X + \gamma_1 W_1 + \dots + \gamma_k W_k + u$$

Proposition 3.7

The coefficient on X in a population linear regression of Y on X and $(W_1, \dots, W_k)$, assuming CIA, is given by

$$\frac{\text{cov}(Y, \tilde{X})}{\text{var}(\tilde{X})} = \frac{\beta_1 \text{var}(\tilde{X}) + \text{cov}(u, \tilde{X})}{\text{var}(\tilde{X})} = \beta_1$$

Proof. We know that by the FWL, the coefficient of X is given by $\text{cov}(Y, \tilde{X})/\text{var}(\tilde{X})$. Then, we have

$$\begin{aligned} \text{cov}(Y, \tilde{X}) &= \text{cov}(\beta_0 + \beta_1 X + \gamma_1 W_1 + \dots + \gamma_k W_k + u, \tilde{X}) \\ &= \beta_1 \text{cov}(X, \tilde{X}) + \text{cov}(u, \tilde{X}) \\ &= \beta_1 \text{var}(\tilde{X}) + \text{cov}(u, \tilde{X}) \end{aligned}$$

The second and third equality follows since $\text{cov}(\tilde{X}, W_I) = 0$ for all $I \in \{1, \dots, k\}$ and $X = \gamma_0 + \gamma_1 W_1 + \dots + \gamma_k W_k + \tilde{X}$. Furthermore, $\text{cov}(u, \tilde{X}) = 0$ since $\tilde{X}$ is the part of X that is randomly assigned so it is orthogonal to both W_I 's and to any other determinants of Y , including u . This yields the result in the coefficient $\square$

Bad Controls. Even if X or D is unconditionally randomly assigned, adding controls can improve the precision of the estimate of the effect of the treatment. However, we should only add the control to the regression when it is a **pre-treatment characteristic** i.e. it can

be measured prior to the assignment of the treatment and they should ideally be plausibly relevant to the determination of Y .

If we violate this condition, then OLS does not consistently estimate the effect of the treatment.

Example.

Suppose we have random assignment such that

$$\begin{aligned}Y^{\text{end}} &= \beta_0 + \beta_1 D + u \\Y^{\text{mid}} &= \gamma_0 + \gamma_1 D + v\end{aligned}$$

where the first is the end of year test score and the second is the mid year test score and D is the treatment of small class size, and under random assignment $\text{cov}(D, u) = \text{cov}(D, v) = 0$. Notice that β_1 and γ_1 may be consistently estimated by OLS since **OR** holds. Suppose we control for Y^{mid} i.e.

$$Y^{\text{end}} = \delta_0 + \delta_1 D + \delta_2 Y^{\text{mid}} + \varepsilon$$

Then, substituting yields

$$\begin{aligned}Y^{\text{end}} &= \delta_0 + \delta_1 D + \delta_2(\gamma_0 + \gamma_1 D + v) + \varepsilon \\&= \delta_0 + (\delta_1 + \delta_2 \gamma_1) D + (\delta_2 v + \varepsilon)\end{aligned}$$

with $\beta_0 = \delta_0, \beta_1 = \delta_1 + \delta_2 \gamma_1$ and $u = \delta_2 v + \varepsilon$. But notice that since ε and v depend on (unobserved student ability), we have **OR** failing since

$$\text{cov}(Y^{\text{mid}}, \varepsilon) = \text{cov}(\gamma_0 + \gamma_1 D + v, \varepsilon) = \text{cov}(v, \varepsilon) \neq 0$$

Thus Y^{mid} is endogenous and OLS will be inconsistent for both δ_1 and δ_2

Bad controls arise when outcomes are included as controls in a model.

Remark.

It may still be possible to estimate its parameters even with bad controls (it still makes sense as a causal model) – we can use instrumental variables to do this

3.4.1 OLS with a binary regressor

In this set up, we have a binary regressor $D \in \{0, 1\}$ indicating whether the observation i received the treatment. We focus on the mechanics of regression first, before connecting it with the Neyman-Rubin Causal Model.

The RCT model be given by

$$Y = \beta_0 + \beta_1 D + u$$

where we may consistently estimate β_1 by an OLS regression of Y on D .

Theorem 3.8: Difference of Means estimator

Suppose we estimate ρ_1 via an OLS regression of Y on D . Then,

$$\rho_1 = \frac{1}{n_1} \sum_{\{i|D_i=1\}} Y_i - \frac{1}{n_0} \sum_{\{i|D_i=0\}} Y_i$$

where n_0 denotes the number of untreated units and $n_1 = n - n_0$ the number of treated units

Proof. Notice that $\mathbb{E}[Y|D]$ is the CEF function and thus minimises the mean squared error prediction of Y given D . Then we may write

$$\begin{aligned} \mathbb{E}[Y|D] &= \begin{cases} \mathbb{E}[Y|D=0] & \text{if } D=0 \\ \mathbb{E}[Y|D=1] & \text{if } D=1 \end{cases} \\ &= \mathbb{E}[Y|D=0] + \{\mathbb{E}[Y|D=1] - \mathbb{E}[Y|D=0]\} \cdot D \\ &=: \rho_0 + \rho_1 D \end{aligned}$$

Since $\mathbb{E}[Y|D]$ is linear, it is the best linear predictor of Y and hence must coincide with the population linear regression of Y on D . Hence,

$$\frac{\text{cov}(Y, D)}{\text{var}(D)} = \rho_1 = \mathbb{E}[Y|D=1] - \mathbb{E}[Y|D=0] = \frac{1}{n_1} \sum_{\{i|D_i=1\}} Y_i - \frac{1}{n_0} \sum_{\{i|D_i=0\}} Y_i$$

□

Remark.

This holds simply because D is a binary variable

Corollary 3.9

The sample analogue of **Theorem 3.7** is

$$\begin{aligned} \frac{\text{cov}(Y, D)}{\text{var}(D)} &= \hat{\mathbb{E}}[Y|D=1] - \hat{\mathbb{E}}[Y|D=0] \\ &= \frac{1}{n_1} \sum_{\{i|D_i=1\}} Y_i - \frac{1}{n_0} \sum_{\{i|D_i=0\}} Y_i \end{aligned}$$

3.4.2 Heterogenous causal effects

It should be noted that the causal effect of X differs from individual to individual because of

- different X_i 's i.e. observed heterogeneity
- different u_i 's i.e. unobserved heterogeneity

Remark.

- We already can accommodate heterogeneous effects via non-linear models or through
- interaction effects but these are all observed heterogeneity

Proposition 3.10

Suppose $X_i \perp\!\!\!\perp (\beta_{1i}, u_i)$ i.e. X_i is assigned such that it is independent of unobserved individual characteristics. Suppose also that $\mathbb{E}[u_i] = 0$. Then, $\mathbb{E}[\beta_{1i}]$ is the coefficient on X in the population regression of Y on X

Proof. We can write

$$\begin{aligned}\mathbb{E}[Y_i|X_i] &= \mathbb{E}[\beta_0 + \beta_{1i}X_i + u_i|X_i] \\ &= \beta_0 + \mathbb{E}[\beta_{1i}|X_i]X_i + \mathbb{E}[u_i|X_i] \\ &= \beta_0 + \mathbb{E}[\beta_{1i}]X_i\end{aligned}$$

Clearly, $\mathbb{E}[Y_i|X_i]$ is linear – and since it is the best nonlinear predictor of Y_i given X_i , it is also the best linear predictor of Y_i given X_i , so it must coincide with a population linear regression of Y on X . Hence, $\mathbb{E}[\beta_{1i}]$ is the coefficient of X in this population regression $\square$

Remark.

The interpretation of this is that

$$\text{ACE} := \frac{\text{cov}(Y, X)}{\text{var}(X)} = \mathbb{E}[\beta_{1i}]$$

is termed the *average causal effect* and will be consistently estimated by an OLS regression of Y on X alone. We therefore interpret OLS estimators as estimators of average causal effects

3.4.3 Interpretation of the Binary Regressor Model

In this set up, we have what we call potential outcomes, denoted by $Y(D)$ i.e. $Y(1)$ is the outcome of under treatment and $Y(0)$ is the outcome of under no treatment. We call this the **Neyman-Rubin Causal Model**. The causal effect of the treatment is simply $Y(1) - Y(0)$.

Definition 3.11: Average Treatment Effect

The Average Treatment Effect is

$$\text{ATE} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$$

or equivalently

$$\text{ATE} = \mathbb{E}[\beta_{1i}]$$

Definition 3.12: Average Treatment on Treated

The Average effect of the Treatment on the Treated is

$$ATT = \mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 1]$$

or equivalently

$$ATT = \mathbb{E}[\beta_{1i}|D_i = 1]$$

Remark.

■ We sometimes call the ATT the TOT

Theorem 3.13: Fundamental Equation of Causal Inference

Suppose our observed differences in group averages may be written as

$$\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0]$$

Then, we may decompose the observed differences in group averages as the sum of the average effect of treatment of the treated and selection bias

$$\begin{aligned} \mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0] &= (\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 1]) + \\ &\quad (\mathbb{E}[Y(0)|D = 1] - \mathbb{E}[Y(0)|D = 0]) \end{aligned}$$

Proof. We can simply apply linearity of expectation

$$\begin{aligned} \mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0] &= \mathbb{E}[Y(1) - Y(0) + Y(0)|D = 1] - \mathbb{E}[Y(0)|D = 0] \\ &= (\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 1]) + (\mathbb{E}[Y(0)|D = 1] - \mathbb{E}[Y(0)|D = 0]) \end{aligned}$$

and this completes the proof

□

Proposition 3.14

Assume that random assignment is successful such that $X_i \perp\!\!\!\perp u_i$. Then the following facts hold

1. $\mathbb{E}[Y(0)|D = 1] - \mathbb{E}[Y(0)|D = 0] = 0$ i.e. selection bias goes to 0
2. $ATE = \mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0]$ i.e. observed differences of group means equals Average Treatment Effect
3. $ATE = ATT$

Remark.

Sometimes we want to estimate the ATT for those that are actually treated i.e. effect of going to university on incomes for those attending university. This takes place outside an RCT and so random assignment does not apply. Hence, using OLS to estimate ATE is not very useful, because the treated group we are concerned about is not randomly assigned

Problems with RCTs. There are generally two kinds of problems in RCTs: internal and external validity problems

Definition 3.15: Internal Validity

A study is said to have internal validity if its findings for the sample are credible i.e. within-sample validity

Definition 3.16: External Validity

A study is said to have external validity if its findings can be credibly extrapolated to the population or real world policy of interest

Internal Validity

- Contamination:
 - People in the control group access the treatment anyway
 - Some of the untreated turn out to be treated
- Non-compliance:
 - Individuals who are offered a treatment refuse to take it
 - Some of the treated turn out to be untreated
- Hawthorne effect: A phenomenon in which participants alter their behaviour as a result of being part of an experiment or study
- Placebo effect: the placebo effect impacts final outcomes because of perceived changes (different from Hawthorne effect where the outcome changes due to imperceptible changes)

The first 2 re-introduce selection bias.

External Validity. If an RCT has external validity, it means that we can expect the distribution of outcomes that would occur in the population under the policy of interest would be the same as the distribution of outcomes realised by the experimental treatment group. Properly executed RCT's have high internal validity, but you need external validity too for credible policy evaluation. The internal validity of RCT's does not imply external validity, credible policy evaluation requires both.

Threats to external validity

- The sample used in an RCT may differ from the population of policy interest due to the small scale/local nature of the RCT, (e.g. a particular geographic area, institutional environment or demographic group).
- Establishing external validity requires enumerative and eliminative induction: doing a lot of RCTs and varying the circumstances under which the RCT takes place and seeing which aspects of the environment matter/don't matter to the outcome
- The assumption of individualistic treatment response may be valid within the design of the RCT but may not hold in the population where there is the potential for spillover effects
- Serious measurement problems when studies have short durations because we often want to learn long-term outcomes of treatments, but short studies reveal only immediate, surrogate outcomes. Even well-conducted RCT's may only reveal the distribution of surrogate outcomes in the study population, not the distribution of outcomes of real interest

4 Instrumental Variables

The basic motivation is that sometimes, in a model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + u$$

X_1 is endogenous. We want to be able to consistently estimate these parameters even when X_1 is endogenous. The basic idea then is to use what we call an instrument Z that determines the object of interest X_1 , but only affects Y through X_1 and is also orthogonal to unmodelled determinants of Y . Here, we have three definitions specifically for instrumental variables.

Definition 4.1: Structural Equation

We call the causal model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + u$$

the structural equation, where X_1 might be endogenous while $(X_2, \dots, X_k)$ are taken to be exogenous

Definition 4.2: First Stage Regression

We call the regression of X on $(Z_1, \dots, Z_m)$ i.e.

$$X = \pi_0 + \pi_1 Z_1 + \cdots + \pi_m Z_m + v$$

the first-stage regression, where $\mathbb{E}[v] = \mathbb{E}[Zv] = 0$ by construction

Definition 4.3: Reduced Form regression

We call the equation from substituting the first-stage regression into the structural equation the reduced form regression i.e.

$$\begin{aligned} Y &= \beta_0 + \beta_1(\pi_0 + \pi_1 Z_1 + \dots + \pi_m Z_m + v) + \dots + \beta_k X_k + u \\ &= (\beta_0 + \beta_1 \pi_0) + (\beta_1 \pi_1) Z_1 + \dots + (\beta_1 \pi_m) Z_m + \dots + \beta_k X_k + (\beta_1 v + u) \\ &=: \gamma_0 + \gamma_1 Z_1 + \dots + \gamma_m Z_m + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon \end{aligned}$$

which is indeed a regression equation (i.e. one whose parameters can be consistently estimated by OLS) since

$$\text{cov}(\varepsilon, Z) = \beta_1 \text{cov}(v, Z) + \text{cov}(u, Z) = 0$$

by first stage regression and assumed exogeneity of an instrument Z

Remark.

It is easy to see the single instrument case

4.1 One Instrument

We have a causal/structural model

$$Y = \beta_0 + \beta_1 X + u$$

where $\text{cov}(X, u) \neq 0$. We want to use another variable Z that acts as an instrument for X

Definition 4.4: Valid Instrument

An instrument Z is said to be valid just when it satisfies

Z1 Relevance: $\text{Cov}(X, Z) \neq 0$; Z is correlated with X .

Z2 Exogeneity: $\text{Cov}(Z, u) = 0$; Z is uncorrelated with any unmodelled determinants of Y .

Z3 Exclusion: If

$$Y = \beta_0 + \beta_1 X + \delta Z + u,$$

then $\delta = 0$; Z does not appear in the causal model.

Interpretation: we may take **Z2-Z3** as saying that Z cannot have a direct effect on Y – holding X constant, $\delta = 0$ – it may only have an indirect effect on Y through its effect on X / **Z1** says that the effect of Z on X must be nonzero.

Estimating β_1 . There are two ways for us to estimate β_1 using the instrument

1. **Indirect Least Squares:** The basic idea is to compare the coefficients on Z in the first stage regression and the reduced form, which gives

$$\frac{\gamma_1}{\pi_1} = \frac{\beta_1 \pi_1}{\pi_1} = \beta_1$$

This is because an increase in Z by 1 unit increases X by π_1 units and because Z has no direct effect on Y , shifts Y by $\pi \times \beta_1$ units. So, we merely have to rescale this by dividing by π_1 to get back ‘a 1 unit increase in X increases Y by β_1 units’. So we have the following

Proposition 4.5

Consider γ_1 and π_1 being coefficients in population regressions of Y on Z and X on Z respectively. Then,

$$\beta_1 = \frac{\gamma_1}{\pi_1} = \frac{\text{cov}(Y, Z)/\text{var}(Z)}{\text{cov}(X, Z)/\text{var}(Z)} = \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)}$$

where β_1 is the estimated coefficient in the causal model

Then we have the sample analogue

Corollary 4.6

The sample counterpart is the indirect least squares (ILS) estimator

$$\hat{\beta}_1 := \frac{\hat{\gamma}_1}{\hat{\pi}_1} = \frac{\hat{\text{cov}}(Y, Z)}{\hat{\text{cov}}(X, Z)}$$

and $\hat{\beta}_1 \xrightarrow{p} \beta_1$ i.e. is consistent for β_1 when Z satisfies **Z1-Z3**

2. **Two-stage least squares (2SLS):** The basic motivation here is that substituting $X = \pi_0 + \pi_1 Z =: X^* + v$ where $X^* := \pi_0 + \pi_1 Z$ into the causal model gives

$$Y = \beta_0 + \beta_1(X^* + v) + u =: \beta_0 + \beta_1 X^* + \varepsilon$$

where $\text{cov}(\varepsilon, X^*) = 0$. This is because

$$\begin{aligned} \text{cov}(X^*, \varepsilon) &= \text{cov}(\pi_0 + \pi_1 Z, u + v) \\ &= \pi_1 \text{cov}(Z, u) + \pi_1 \text{cov}(Z, v) \end{aligned}$$

where $\text{cov}(Z, u) = 0$ by exogeneity and $\text{cov}(Z, v) = 0$ by construction of the first stage regression. So,

$$\beta_1 = \frac{\text{cov}(Y, X^*)}{\text{var}(X^*)}$$

This yields

Proposition 4.7

Consider a 2SLS population regression. Then,

$$\beta_1 = \frac{\text{cov}(Y, X^*)}{\text{var}(X^*)} = \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)}$$

This follows since

$$\frac{\text{cov}(Y, X^*)}{\text{var}(X^*)} = \frac{\pi_1 \text{cov}(Y, Z)}{\pi_1^2 \text{var}(Z)} = \frac{\text{cov}(Y, Z)}{\pi_1 \text{var}(Z)} = \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)}$$

To estimate β_1 , we estimate X^* first via a first stage OLS regression of X on Z which gives

$$\hat{X}_i^* = \hat{\pi}_0 + \hat{\pi}_1 Z_i$$

which gives

$$\hat{\beta}_1 = \frac{\text{cov}(Y, \hat{X})}{\text{var}(\hat{X})}$$

Theorem 4.8: ILS and 2SLS coincide

Let β_1^{ILS} and β_1^{2SLS} be the coefficients of X from the respective regressions of Y on X with an instrument Z satisfying **Z1-Z3**. Then,

$$\beta_1^{\text{ILS}} = \beta_1^{\text{2SLS}}$$

4.2 General Case

In this setting, the structural equation is

$$Y = \beta_0 + \beta_1 X + \beta_2 W_1 + \cdots + \beta_{r+1} W_r + u$$

where X is endogenous but $\mathcal{W} := (W_1, \dots, W_r)$ are exogenous variables i.e. $\text{cov}(W_I, u) = 0$ for all $I \in \{1, \dots, r\}$. Also $\mathbb{E}[u] = 0$.

We also assume that there are $m \geq 1$ instruments $Z, \dots, Z_m$ available.

Definition 4.9: Valid Instrument (Multiple Instruments)

Instruments $Z_1, \dots, Z_m$ are said to be valid just when it satisfies

Z1 Relevance: in the first stage regression,

$$X = \pi_0 + \pi_1 Z_1 + \dots + \pi_m Z_m + \pi_{m+1} W_1 + \dots + \pi_{m+r} W_r + v$$

where $\mathbb{E}[u] = \mathbb{E}[Z_I v] = \mathbb{E}[W_J v] = 0$ by construction of the regression equation, at least one of the coefficients on the instruments must be nonzero i.e.

$$\exists I \in \{1, \dots, m\} \text{ s.t. } \text{cov}(X, \tilde{Z}_I) \neq 0$$

Z2 Exogeneity: $\text{cov}(Z_I, u) = 0$ for all $I \in \{1, \dots, m\}$ i.e. Z is uncorrelated with any unmodelled determinants of Y

Z3 Exclusion: If

$$Y = \beta_0 + \beta_1 X + (\delta_1 Z_1 + \dots + \delta_m Z_m) + \beta_2 W_1 + \dots + \beta_{r+1} W_r + u,$$

then $\delta_1 = \dots = \delta_m = 0$; $Z_1, \dots, Z_m$ does not appear in the causal model.

Remark.

We include the exogenous controls in the first stage regression as well as the second stage regression – otherwise, your second stage regression is no longer a genuine regression model, and you may not consistently estimate the second stage by OLS

Theorem 4.10: 2SLS Consistency

Assume **Z1-Z3** hold and there is no perfect multicollinearity amongst the instruments and the exogenous variables and that we want to estimate β_1 from

$$Y = \beta_0 + \beta_1 X + \beta_2 W_1 + \dots + \beta_{r+1} W_r + u$$

with $m \geq 1$ instruments $Z_1, \dots, Z_m$. Then, the 2SLS estimate $\hat{\beta}_1$ consistently estimates β_1 i.e. $\hat{\beta}_1 \xrightarrow{p} \beta_1$

Proof. The first step is to show that under **Z1-Z3**, β_1 can be expressed as a coefficient in a certain population linear regression model. Consider the first stage regression

$$X = \pi_0 + \pi_1 Z_1 + \dots + \pi_m Z_m + \pi_{m+1} W_1 + \dots + \pi_{m+r} W_r + v =: X^* + v$$

where $\text{cov}(X^*, v) = 0$, since $\mathbb{E}[u] = \mathbb{E}[Z_I v] = \mathbb{E}[W_J v] = 0$ by construction. Substituting X^*

into the structural equation yields

$$\begin{aligned} Y &= \beta_0 + \beta_1(X^* + v) + \beta_2W_1 + \dots + \beta_{r+1}W_r + u \\ &= \beta_0 + \beta_1X^* + \beta_2W_1 + \dots + \beta_{r+1}W_r + (\beta_1v + u) \\ &=: \beta_0 + \beta_1X^* + \beta_2W_1 + \dots + \beta_{r+1}W_r + \varepsilon \end{aligned}$$

to show that this is a regression equation, we want to show that

- $\mathbb{E}[\varepsilon] = 0$ which holds trivially because $\mathbb{E}[v] = \mathbb{E}[u] = 0$ by construction
- $\text{cov}(W_I, \varepsilon) = \beta_1 \text{cov}(W_I, v) + \text{cov}(W_I, u) = 0$ for $I \in \{1, \dots, r\}$. By assumption of exogeneity, $\text{cov}(W_I, u) = 0$. Since we included $\mathcal{W} := (W_1, \dots, W_r)$ in the first-stage regression, $\text{cov}(W_I, v) = 0$ by construction
- $\text{cov}(Z_I, \varepsilon) = \beta_1 \text{cov}(Z_I, v) + \text{cov}(Z_I, u) = 0$ for $I \in \{1, \dots, m\}$. This follows immediately by the exogeneity of Z_I (**Z2**) and by construction of the first stage regression

The second step is to show that there is no perfect multicollinearity. We know that the instruments and exogenous variables do not exhibit perfect multicollinearity within themselves. Furthermore, by **Z1** we know that at least one $\pi_I \neq 0$ so it is not the case that $\pi_1 = \dots = \pi_m = 0$ and X^* is simply a linear combination of the W_I 's, and so we can recover β_1 .

Thus under **Z1-Z3**, we can

1. obtain the predicted values X^* from the first stage population regression
2. recover β_1 as the coefficient on X^* in a population regression of Y on $(X^*, \mathcal{W})$

Hence, β_1 can be consistently estimated by performing a sample analogue of the above procedure which yields the 2SLS estimator

1. Perform an OLS regression of X on the instruments and exogenous variables to obtain the predicted values

$$\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_{1i} + \dots + \hat{\pi}_m \hat{Z}_{mi} + \hat{\pi}_{m+1} W_{1i} + \dots + \hat{\pi}_{m+r} W_{ri}$$

2. Compute $\hat{\beta}_1$ as the coefficient on $\hat{X}$ in an OLS regression of Y on $(\hat{X}, \mathcal{W})$

□

Remark.

- All the estimates for all the variables will be consistent for the 2SLS

Remark.

ILS can also consistently estimate β_1 with a single instrument and it will be identical to the 2SLS estimator but with more instruments, we can see that each of the ratios $\delta_I/\pi_I = \beta_1 \pi_I/\pi_I = \beta_1$ and each of the estimates $\hat{\delta}_I/\hat{\pi}_I$ provides a consistent estimator of β_1 – but in general, each of these estimators will be distinct. We prefer 2SLS to produce a single efficient estimate of β_1

Theorem 4.11: 2SLS General Case Estimator

Consider a 2SLS population regression to recover β_1 and a 2SLS estimator $\hat{\beta}_1$, with m controls and a single instrument. Then,

$$\beta_1 = \frac{\text{cov}(Y, \tilde{Z})}{\text{cov}(X, \tilde{Z})} \quad \hat{\beta}_1 = \frac{\text{c}\hat{\text{ov}}(Y, \tilde{Z})}{\text{c}\hat{\text{ov}}(X, \tilde{Z})}$$

Proof. Notice that $\tilde{X}^* = X^* - \sum_{i=1}^m \alpha_i W_i = \pi_0 + \pi_1 Z - \sum_{i=1}^m \alpha_i W_i = \pi_0 + \pi_1 (Z - \sum_{i=1}^m b_i W_i) = \pi_0 + \pi_1 \tilde{Z}$, where $b_i = \frac{\alpha_i}{\pi_1}$. It remains to show that $\tilde{Z}$ is orthogonal to each W_i . Without loss, let us just consider one W_j . Then,

$$\text{cov}(W_j, \tilde{Z}) = \text{cov}(W_j, \frac{1}{\pi_1} X^* - \frac{\pi_0}{\pi_1}) = 0$$

since by construction, $\tilde{X}^*$ is orthogonal to the regressor W_j . The rest of the proof is exactly the same as the single case $\square$

4.3 Inference in 2SLS

4.3.1 Large sample distribution

Theorem 4.12: Asymptotic Normality of 2SLS estimator

Consider the model

$$Y = \beta_0 + \beta_1 X + \beta_2 W_1 + \cdots + \beta_{r+1} W_r + u$$

where X is potentially endogenous and suppose m instruments are available. Suppose that

1. Z_I is a valid instrument for $I \in \{1, \dots, m\}$
2. W_I is exogenous i.e. $\text{cov}(W_I, u) = 0$ for $I \in \{1, \dots, k\}$
3. Y_i, X_i, Z_{Ii}, W_{Ii} are i.i.d
4. $\mathbb{E}[Y^4], \mathbb{E}[X^4], \mathbb{E}[Z_I^4], \mathbb{E}[W_I^4] < \infty$ i.e. fourth moments are finite
5. There is no perfect multicollinearity among any subset of $(W_1, \dots, W_r)$ or $(Z_1, \dots, Z_m)$

Then,

$$n^{1/2}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} \mathcal{N}(0, \omega_{\beta_1, \text{IV}}^2)$$

where $\omega_{\beta_1, \text{IV}}^2$ can be computed by statistical software as a heteroskedastic robust variance

Remark.

This theorem provides the basis for t tests and confidence intervals that can be constructed in a manner exactly analogous to their OLS counterparts

4.3.2 Precision in large samples

Proposition 4.13

Consider a simple 2SLS model

$$\begin{aligned} Y &= \beta_0 + \beta_1 X + u \\ X &= \pi_0 + \pi_1 Z_1 + \cdots + \pi_m Z_m + v \end{aligned}$$

and let $X^* := \pi_0 + \pi_1 Z_1 + \cdots + \pi_m Z_m$ denote the predicted values obtained from the first stage regression. Then the asymptotic variance of the 2SLS estimator is

$$\omega_{\beta_1, IV}^2 = \frac{\mathbb{E}[(X^* - \mu_{X^*})^2 u^2]}{[\mathbb{E}(X^* - \mu_{X^*})^2]^2}$$

where $\mu_{X^*} = \mathbb{E}[X^*]$

Interpretation: we can interpret this when u is homoskedastic i.e. $\mathbb{E}[u|\mathcal{Z}] = 0$ and $\mathbb{E}[u^2|\mathcal{Z}] = \mathbb{E}[u^2] = \sigma_u^2$ where $\mathcal{Z} := (Z_1, \dots, Z_m)$. This is because

$$\begin{aligned} \mathbb{E}[(X^* - \mu_{X^*})^2 u^2] &= \mathbb{E}[\mathbb{E}[(X^* - \mu_{X^*})^2 u^2 | \mathcal{Z}]] \\ &= \mathbb{E}[(X^* - \mu_{X^*})^2 \mathbb{E}[u^2 | \mathcal{Z}]] \\ &= \sigma_u^2 \mathbb{E}[(X^* - \mu_{X^*})^2] \\ &= \sigma_u^2 \text{var}(X^*) \end{aligned}$$

since X^* is a function only of the instrumental variable. This yields

$$\omega_{\beta_1, IV}^2 = \frac{\sigma_u^2 \text{var}(X^*)}{[\mathbb{E}(X^* - \mu_{X^*})^2]^2} = \frac{\text{var}(u)}{\text{var}(X^*)}$$

Contrast this with the asymptotic variance of the OLS estimator under homoskedasticity

$$\omega_{\beta_1, OLS}^2 = \frac{\text{var}(u)}{\text{var}(X)}$$

This means that

- A larger error variance, in the structural equation, worsens the precision of the 2SLS estimator just like for the OLS estimator
- More variability in X may not improve the precision of the 2SLS estimator whereas it would improve the precision of the OLS estimator.
- Instead, what matters is the variance of that part of X that is explained by the instruments: the better the job that these do at predicting X , the larger will be

$\text{var}(X^*)$ and hence the greater the precision of 2SLS

Corollary 4.14

If X is exogenous, then the OLS is the more efficient estimator than the 2SLS estimator

Proof. Since

$$\text{var}(X) = \text{var}(X^* + v) = \text{var}(X^*) + \text{var}(v) \geq \text{var}(X^*)$$

it follows that the asymptotic variance under homoskedasticity is such that

$$\omega_{\beta_I, \text{OLS}}^2 = \frac{\text{var}(u)}{\text{var}(X)} \leq \omega_{\beta_1, \text{IV}}^2 = \frac{\text{var}(u)}{\text{var}(X^*)}$$

□

Hence, if X is exogenous, the OLS estimator should be preferred to 2SLS

4.4 Testing Instrument Validity

4.4.1 Relevance and Weak instruments

Initial Step: Consider the first stage regression

$$X = \pi_0 + \pi_1 Z_1 + \cdots + \pi_m Z_m + \pi_{m+1} W_1 + \cdots + \pi_{m+r} W_r + v$$

We may test **Z1** via an F test of

$$H_0 : \pi_I = 0, \forall I \in \{1, \dots, m\}$$

$$H_1 : \pi_I \neq 0 \text{ for at least one } I \in \{1, \dots, m\}$$

where there are m restrictions in the test. A rejection of H_0 may be interpreted as evidence in favour of the relevance condition.

Weak Instruments: However, the F test is not necessarily reliable due to ‘weak instruments’ where π_I just barely satisfies Z1 i.e. is very close to zero. Consider the simplified setting of one instrument. Then, the asymptotic variance is

$$\omega_{\beta_1, \text{IV}}^2 = \frac{\text{var}(u)}{\text{var}(X^*)} = \frac{\text{var}(u)}{\pi_1^2 \text{var}(Z)}$$

where the second equality follows by there being one instrument Z . Then, if π_1 is close to zero, $\omega_{\beta_1, \text{IV}}^2$ will be very large and will fail to provide a reliable guide to the distribution of $\hat{\beta}_1$ even in large samples. So, it is only possible to draw reliable inferences based on the 2SLS estimator (and associated SEs) when the relevance condition holds with a sufficiently wide margin that the possibility of ‘weak instruments’ can be test out.

Testing for weak instruments: We can modify the F test to use critical values that are larger than those drawn from the $F_{m,\infty}$ distribution. Early literature compares the F statistic against a critical value of 10

Remark.

■ This value can be significantly improved upon but it is beyond the scope of the course

4.4.2 Exogeneity: overidentifying restrictions

We basically want to test **Z2** i.e. $\text{cov}(Z_I, u) = 0$ for all $I \in \{1, \dots, m\}$. We want to compare the sample analogue $\text{cov}(Z_I, \hat{u})$ where

$$\hat{u} := Y - \hat{\beta}_0 - \hat{\beta}_1 X + \hat{\beta}_2 W_1 \cdots + \cdots + \hat{\beta}_{r+1} W_r$$

If u is homoskedastic, in the sense that $\mathbb{E}[u|\mathcal{Z}, \mathcal{W}] = 0$ and

$$\mathbb{E}[u^2|\mathcal{Z}, \mathcal{W}] = \mathbb{E}[u^2]$$

then, a test of **Z2** can be conducted via an F test of

$$\begin{aligned} H_0 : \delta_I &= 0, \forall I \in \{1, \dots, m\} \\ H_1 : \delta_I &\neq 0 \text{ for at least one } I \in \{1, \dots, m\} \end{aligned}$$

in the regression

$$u = \delta_0 + \delta_1 Z_1 + \cdots + \delta_m Z_m + \delta_{m+1} W_1 + \cdots + \delta_{m+r} W_r + \eta$$

Assuming homoskedasticity, the test can thus be conducted as follows:

1. Estimate the structural equation is

$$Y = \beta_0 + \beta_1 X + \beta_2 W_1 + \cdots + \beta_{r+1} W_r + u$$

by 2SLS, and obtain the residuals $\hat{u}$ as per

$$\hat{u} := Y - \hat{\beta}_0 - \hat{\beta}_1 X + \hat{\beta}_2 W_1 \cdots + \cdots + \hat{\beta}_{r+1} W_r$$

2. Perform an OLS regression of $\hat{u}$ on $\mathcal{Z}$ and $\mathcal{W}$ and construct the usual F statistic for a test of H_0 . Adjust this as per

$$F^* := \frac{m}{m-1} F$$

and compare F^* to critical values obtained from an $F_{m-1,\infty}$ distribution

Remark.

We adjust the F statistic because the residuals $\hat{u}$ have been generated by using the instruments to estimate the structural equation and we effectively ‘use up’ one of the m restrictions entailed by the exogeneity condition. When we have only one instrument there are no restrictions left to test because the residuals generated by 2SLS have $\text{cov}(Z, \hat{u})$ by construction. Hence, tests of the exogeneity conditions are tests of ‘overidentifying restrictions’, understood as tests of those restrictions implied by **Z1** that are not actually exploited when computing the 2SLS estimates

Another way to understand how this test works is that it is implicitly comparing the estimates that would be obtained by 2SLS using each of the m instruments individually (so if $m = 1$ there is nothing to compare). If these are similar, then the null of exogeneity is unlikely to be rejected. Also, we cannot conclude from a failure to reject the null of exogeneity that the instruments are exogenous since it is possible that they may be correlated with u but in such a way that yields similar estimates of the parameters of the structural equation. Hence, **a rejection of the null of exogeneity can be interpreted unambiguously**

Remark.

The more general heteroskedasticity-robust form of this test has a limiting χ^2_{m-1} distribution

4.5 Misc

4.5.1 Instrumental Variables for Imperfect Compliance

In many cases for RCT, conditional random assignment is impossible to realise because receipt of treatment necessarily involves some element of individual choice e.g. inducement such as payment contingent on graduation can be assigned independently on student’s characteristics but decision to complete degree is up to the students as some who are offered payment will nonetheless drop out hence $\text{cov}(D, u) \neq 0$. BUT since the inducement or offer of treatment Z can be randomly assigned and clearly induces graduation i.e. $\text{cov}(D, Z) \neq 0$ (**Z1**) and $\text{cov}(Z, u) = 0$ (**Z2**) and clearly does not affect the outcome directly (**Z3**).

4.5.2 Heterogenous causal effects and LATE

Suppose we have the causal model

$$Y_i = \beta_0 + \beta_{1i}X_i + u_i$$

in which the effect β_{1i} of the treatment X_i is allowed to vary across individual. We then allow the response to the instrument Z_i to vary across individuals

$$X_i = \pi_0 + \pi_{1i}Z_i + v_i$$

Suppose that Z_i is randomly assigned such that it is independent of individual characteristics, and in particular that

$$Z_i \perp\!\!\!\perp (\beta_{1i}, \pi_{1i}, u_i, v_i)$$

This is the analogue of the usual exogeneity condition and we suppose that relevance holds i.e. $\text{cov}(X_i, Z_i) \neq 0$. Consider a 2SLS regression of Y on X , using Z as an instrument for X . We can write this as

$$\hat{\beta}_{\text{IV}} := \frac{\hat{\text{cov}}(Y, Z)}{\hat{\text{cov}}(X, Z)} \xrightarrow{p} \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)} =: \beta_{\text{IV}}$$

so the 2SLS estimator is consistent for β_{IV} . We may give the following interpretation for β_{IV}

Theorem 4.15: Local Average Treatment Effect

Consider a 2SLS regression of Y on X using Z as an instrument for X that yields β_{IV} . Then,

$$\beta_{\text{IV}} = \mathbb{E} \left\{ \beta_{1i} \cdot \frac{\pi_{1i}}{\mathbb{E}[\pi_{1i}]} \right\} =: \text{LATE}$$

where the β_{IV} is the local average treatment effect which is the weighted average of the underlying β_{1i} 's. In particular, we interpret the LATE as one in which the treatment effect of person i is weighted by his responsiveness π_{1i} to the offer of treatment (i.e. the instrument).

Proof. By $Z_i \perp\!\!\!\perp (\beta_{1i}, \pi_{1i}, u_i, v_i)$ we have

$$\mathbb{E}[X_i|Z_i] = \pi_0 + (\mathbb{E}[\pi_{1i}])Z_i$$

substituting $X_i = \pi_0 + \pi_{1i}Z_i + v_i$ into $Y_i = \beta_0 + \beta_{1i}X_i + u_i$, we obtain the reduced form equation

$$Y_i = \beta_0 + \beta_{1i}(\pi_0 + \pi_{1i}Z_i + v_i) + u_i = \beta_0 + \beta_{1i}(\pi_0 + v_i) + (\beta_{1i}\pi_{1i})Z_i + u_i$$

where by independence

$$\begin{aligned} \mathbb{E}[Y_i|Z_i] &= \mathbb{E}[\beta_0 + \beta_{1i}(\pi_0 + v_i)|Z_i] + \mathbb{E}[\beta_{1i}\pi_{1i}|Z_i] \cdot Z_i + \mathbb{E}[u_i|Z_i] \\ &= (\beta_0 + \pi_0\mathbb{E}[\beta_{1i}] + \mathbb{E}[\beta_{1i}v_i]) + \mathbb{E}[\beta_{1i}\pi_{1i}]Z_i \end{aligned}$$

the first and third equations show that these conditional expectations are linear in Z_i and must correspond to the population linear regressions of X on Z and Y on Z respectively. Hence

$$\frac{\text{cov}(X_i, Z_i)}{\text{var}(Z_i)} = \mathbb{E}[\pi_{1i}] \quad \frac{\text{cov}(Y_i, Z_i)}{\text{var}(Z_i)} = \mathbb{E}[\beta_{1i}\pi_{1i}]$$

By the relevance condition, the first term in the preceding is nonzero. Hence by

$$\hat{\beta}_{\text{IV}} := \frac{\hat{\text{cov}}(Y, Z)}{\hat{\text{cov}}(X, Z)} \xrightarrow{p} \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)} =: \beta_{\text{IV}}$$

we have

$$\beta_{\text{IV}} = \frac{\text{cov}(Y_i, Z_i)/\text{var}(Z_i)}{\text{cov}(X_i, Z_i)/\text{var}(Z_i)} = \frac{\mathbb{E}[\beta_{1i}\pi_{1i}]}{\mathbb{E}[\pi_{1i}]} = \mathbb{E} \left\{ \beta_{1i} \cdot \frac{\pi_{1i}}{\mathbb{E}[\pi_{1i}]} \right\}$$

□

Corollary 4.16

When $\beta_{1i} = \beta_1$ or $\pi_{1i} = \pi_1$ for all i or where β_{1i} and π_{1i} are independent, then $\text{LATE} = \text{ATE}$

Unlike the $\text{ATE} = \mathbb{E}[\beta_{1i}]$, the LATE has the possibly undesirable property that it depends on the instrument Z

Example.

Imagine two independent studies of the effect of completing a university degree ($D = 1$) on earnings. Both studies draw their samples from the same underlying population – say, students in the same cohort – but use different incentive schemes, Z_1 and Z_2 , to encourage selected students to complete their degree. In that case, we have two distinct treatment equations

$$D_i = \pi_0 + \pi_{1i}Z_{1i} + v_i$$

$$D_i = \alpha_0 + \alpha_{1i}Z_{2i} + v_i$$

which describe the response of the individuals to the inducement to graduate in the first and second studies respectively. Since in general the distributions of π_{1i} and α_{1i} will differ (because they describe responses to different inducements), it is likely that

$$\mathbb{E} \left\{ \beta_{1i} \cdot \frac{\pi_{1i}}{\mathbb{E}[\pi_{1i}]} \right\} \neq \mathbb{E} \left\{ \beta_{1i} \cdot \frac{\alpha_{1i}}{\mathbb{E}[\alpha_{1i}]} \right\}$$

so that even in large samples, the 2SLS estimates emerging from both studies may be distinct

In other words, β_{IV} is still consistent but for different weight average causal effects depending on the instrument.

More generally, **Theorem 4.14** provides a heuristic for interpreting 2SLS as a weighted average of individual-level causal effects, weighted by responsiveness to the instrument. In other words, the LATE measures the causal effect of the treatment on individuals who are compliant with the incentive.

5 Time Series: Stationary Processes

The key difference between time series data and cross-sectional data is that time series data is of the same entity, while cross-sectional data is of distinct entities.

5.1 Stationary Time Series

Definition 5.1: Weak Stationarity

Call $\{Y_t\}$ weakly stationary if

- its mean $\mathbb{E}[Y_t]$
- its variance $\text{var}(Y_t)$
- its autocovariance $\text{cov}(Y_t, Y_{t-h})$ for $h \in \mathbb{Z}$ do not depend on t

Remark.

Apparently, one must also show that $\text{cov}(Y_t, Y_{t+h})$ is invariant with t

Definition 5.2: Strong/Strict Stationarity

Call $\{Y_t\}$ strictly (or strongly) stationary if, for every $k \geq 0$, and every $s, t \geq 1$, we have

$$(Y_t, Y_{t+1}, \dots, Y_{t+k}) =_d (Y_s, Y_{s+1}, \dots, Y_{s+k})$$

where $=_d$ denotes ‘has the same distribution as’

Remark.

See that a process $\{\varepsilon_t\}$ being i.i.d. implies that it is strictly stationary

The next concept is rarely used

Definition 5.3: Joint Stationarity

We say that $\{Y_t\}$ and $\{X_t\}$ are jointly strictly stationary if

$$(X_t, Y_t, X_{t+1}, Y_{t+1}, \dots, X_{t+k}, Y_{t+k}) =_d (X_s, Y_s, X_{s+1}, Y_{s+1}, \dots, X_{s+k}, Y_{s+k})$$

for every $k \geq 0$ and $s, t \geq 1$

Proposition 5.4

If $\{Y_t\}$ and $\{X_t\}$ are individually stationary, then $\{X_t\}$ and $\{Y_t\}$ are jointly stationary

Remark.

Joint stationarity does not imply individual stationarity

Proposition 5.5

If $\{Y_t\}$ is strictly stationary, then $\{Y_t\}$ is weakly stationary

Proof. The proposition is trivial. If $\{Y_t\}$ is strictly stationary, then for pair of vector of random variables (Y_t, Y_{t+h}) and (Y_s, Y_{s+h}) , they must have the same distribution. Therefore, they must have the same covariance i.e. $\text{cov}(Y_t, Y_{t+h}) = \text{cov}(Y_s, Y_{s+h})$. Similarly, notice that this implies that for some fixed t and any $s \geq 0$, the two facts hold

$$\begin{aligned}\mathbb{E}[Y_t] &= \mathbb{E}[Y_s] \quad \forall s \in \mathbb{Z}^+ \\ \text{var}[Y_t] &= \text{var}[Y_s] \quad \forall s \in \mathbb{Z}^+\end{aligned}$$

□

The basic motivation for stationarity is that the sample estimator of means, variances and autocovariances have well-defined population counterparts that they are consistent for i.e. in large samples the sample mean converges to the population mean. (the relevant LLN for time series is not covered).

The sample variances and autocovariances differ from their cross-sectional counterparts in being standardised by T rather than what cross-sectional counterparts are usually standardised by (like $T - 1$ in the sample variance case due to Bessel's correction).

Definition 5.6: Sample Variance and Autocovariance

The sample variance and autocovariance can be computed as

$$\begin{aligned}\hat{\gamma}_h &= \hat{\text{cov}}(Y_t, Y_{t-h}) := \frac{1}{T} \sum_{t=h+1}^T (Y_t - \bar{Y}_T)(Y_{t-h} - \bar{Y}_T) \\ \hat{\gamma}_0 &= \hat{\text{var}}(Y_t) = \frac{1}{T} \sum_{t=1}^T (Y_t - \bar{Y}_T)^2\end{aligned}$$

Remark.

The reason why Bessel's correction is not applied to time series is because the i.i.d. assumption is not expected to hold for time series (and actually does not hold generally)

Definition 5.7: Autocorrelation

The h th autocorrelation (or serial correlation) of a time series is

$$\rho_h := \frac{\text{cov}(Y_t, Y_{t-h})}{\text{var}(Y_t)} = \frac{\text{cov}(Y_t, Y_{t-h})}{[\text{var}(Y_t)\text{var}(Y_{t-h})]^{1/2}}$$

where the second equality follows by stationarity

We estimate the autocorrelation function as

$$\hat{\rho}_h := \frac{\text{cov}(Y_t, Y_{t-h})}{\text{var}(Y_t)} = \frac{\hat{y}_h}{\hat{y}_0}$$

One property that is important is the *persistence* of a time series. A formal definition is out of the course so there is a heuristic definition:

- the speed with which $\{Y_t\}$ reverts to its mean; or
- the extent of serial correlation in the time series

In other words, a highly persistent time series will have a serial correlation for quite high values of h or equivalently, will revert to its mean quite slowly.

There is also an Autocorrelation function (ACF) of $\{Y_t\}$ $h \mapsto \rho_h$ (for $h \geq 0$). For stationary series, ρ_h generally decays to 0 as h increases. In other words, the extent of the correlation lessens as h increases. This provides a measure of persistence. In practice, ρ_h is unknown so we estimate the ACF using the sample ACF i.e. $h \mapsto \hat{\rho}_h$

5.2 Autoregressive Models

Definition 5.8: The $AR(p)$ model

Consider a sequence $\{Y_t\}$ and $p \geq 1$. Then, an $AR(p)$ model can be constructed as

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t \quad (13)$$

where $\{u_t\}$ is the shock or innovation sequence that is not forecastable on the basis of past values of $\{Y_t\}$ and hence

$$\mathbb{E}[u_t | \mathcal{Y}_t] = 0$$

where $\mathcal{Y}_t = (Y_t, Y_{t-1}, \dots, Y_s)$ where Y_s is the initialisation of the process (often Y_0)

Remark.

The simple $AR(1)$ model that we often consider then is just

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

Claim: $\{u_t\}$ is serially uncorrelated

Assume that $\{u_t\}$ is a shock sequence in the $AR(p)$ model and hence $\mathbb{E}[u_t | \mathcal{Y}_t] = 0$ holds. Then, $\{u_t\}$ is serially uncorrelated i.e.

$$\text{cov}(u_t, u_{t-h}) = 0, \quad \forall h \neq 0$$

Proof for Claim.

By definition of covariance, we may write

$$\text{cov}(u_t, u_{t-h}) = \mathbb{E}[u_t u_{t-h}] - \mathbb{E}[u_t] \mathbb{E}[u_{t-h}]$$

By assumption $\mathbb{E}[u_t] = 0$ (see that $\mathbb{E}[\mathbb{E}[u_t | \mathcal{Y}_t]] = \mathbb{E}[0] = 0$), then

$$\text{cov}(u_t, u_{t-h}) = \mathbb{E}[u_t u_{t-h}]$$

By LIE, notice that

$$\begin{aligned} \text{cov}(u_t, u_{t-h}) &= \mathbb{E}[u_t u_{t-h}] \\ &= \mathbb{E}[\mathbb{E}[u_t u_{t-h} | \mathcal{Y}_t]] \\ &=_{(1)} \mathbb{E}[u_{t-h} \mathbb{E}[u_t | \mathcal{Y}_t]] \\ &=_{(2)} \mathbb{E}[u_{t-h} \cdot 0] \\ &= 0 \end{aligned}$$

(1) follows from the fact that we know u_{t-h} based on $\mathcal{Y}_t$ and (2) follows by assumption

Intuitively, this follows because past values of $\{Y_t\}$ carry no information about the conditional mean of $\{u_t\}$. However, it does not preclude the conditional variance of $\{u_t\}$ depending on past values of $\{Y_t\}$.

Remark.

Mean independence of u_t does not preclude the conditional variance of $\{u_t\}$ depending on past values of $\{Y_t\}$

If Y_t follows the $AR(p)$ model, then we say that $\{Y_t\}$ is an autoregressive or $AR(p)$ process, or that it has been generated by an $AR(p)$ model, even though this model is intended only to be descriptive. The causal model that ‘actually’ generates $\{Y_t\}$ may be different from an $AR(p)$ model – however, the $AR(p)$ model may provide an excellent description of how $\{Y_t\}$ is correlated with its lags.

Necessary and sufficient conditions for an $AR(1)$ process to be stationary

Theorem 5.9: Necessary Condition for stationary $AR(1)$

Consider an $AR(1)$ model

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

Then, if $AR(1)$ is stationary, $\beta_1 \in (-1, 1)$

Proof. Suppose the $AR(1)$ model is stationary. Then, we consider the mean and variance

Mean: We have

$$\mathbb{E}[Y_t] = \mathbb{E}[\beta_0 + \beta_1 Y_{t-1} + u_t] = \beta_0 + \beta_1 \mathbb{E}[Y_{t-1}]$$

By stationarity, $\mu_Y = \mathbb{E}[Y_t] = \mathbb{E}[Y_{t-1}]$. Then,

$$\mu_Y = \frac{\beta_0}{1 - \beta_1}$$

Then, $\beta_1 \neq 1$. Otherwise, μ_Y is undefined.

Variance: We may write

$$\begin{aligned} \text{var}(Y_t) &= \text{var}(\beta_0 + \beta_1 Y_{t-1} + u_t) \\ &= \beta_1^2 \text{var}(Y_{t-1}) + \text{var}(u_t) + 2\beta_1 \text{cov}(Y_{t-1}, u_t) \\ &= \beta_1^2 \text{var}(Y_{t-1}) + \text{var}(u_t) \end{aligned}$$

We know this since

$$\begin{aligned} \text{cov}(Y_{t-1}, u_t) &= \mathbb{E}[Y_{t-1} u_t] - \mathbb{E}[Y_{t-1}] \mathbb{E}[u_t] \\ &= \mathbb{E}[\mathbb{E}[Y_{t-1} u_t | \mathcal{Y}_{t-1}]] \\ &= \mathbb{E}[Y_{t-1} \mathbb{E}[u_t | \mathcal{Y}_{t-1}]] \\ &= 0 \end{aligned}$$

Since $\{u_t\}$ is assumed to be stationary, then define $\text{var}(u_t) := \sigma_u^2$ which is time invariant, and since we assume $\{Y_t\}$ to be stationary as well, we have $\sigma_Y^2 = \text{var}(Y_t) = \text{var}(Y_{t-1})$. So,

$$\sigma_Y^2 = \beta_1^2 \sigma_Y^2 + \sigma_u^2 \implies \sigma_Y^2 = \frac{\sigma_u^2}{1 - \beta_1^2}$$

Given that $\text{var}(Y_t) \geq 0$, then $|\beta_1| < 1$ since $|\beta_1| \geq 1$ will contradict this property of variance or be undefined. $\square$

Theorem 5.10: Sufficiency for Stationary

Consider an $AR(1)$ model

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

Then, Y_t is stationary if the following conditions are satisfied

1. $\beta_1 \in (-1, 1)$
2. $\mathbb{E}[Y_0] = \frac{\beta_0}{1 - \beta_1}$
3. $\text{var}(Y_0) = \frac{\sigma_u^2}{1 - \beta_1^2}$

In other words, the initial value Y_0 must have the same mean and variance as the rest of the sequence

Proof. Assume that the conditions hold. Then, we show that

$$\mathbb{E}[Y_1] = \beta_0 + \beta_1 \mathbb{E}[Y_0] = \beta_0 + \beta_1 \frac{\beta_0}{1 - \beta_1} = \frac{\beta_0(1 - \beta_1) + \beta_1 \beta_0}{1 - \beta_1} = \frac{\beta_0}{1 - \beta_1} = \mathbb{E}[Y_0]$$

and

$$\begin{aligned}
\text{var}(Y_1) &= \beta_1^2 \text{var}(Y_0) + \sigma_u^2 \\
&= \beta_1^2 \frac{\sigma_u^2}{1 - \beta_1^2} + \sigma_u^2 \\
&= \frac{\beta_1^2 \sigma_u^2 + (1 - \beta_1^2) \sigma_u^2}{1 - \beta_1^2} \\
&= \frac{\sigma_u^2}{1 - \beta_1^2} \\
&= \text{var}(Y_0)
\end{aligned}$$

It is easy to see that, if we proceed by induction, if

$$\begin{aligned}
\mathbb{E}[Y_t] &= \frac{\beta_0}{1 - \beta_1} = \mathbb{E}[Y_0] \\
\text{var}(Y_t) &= \frac{\sigma_u^2}{1 - \beta_1^2} = \text{var}(Y_0)
\end{aligned}$$

then,

$$\begin{aligned}
\mathbb{E}[Y_{t+1}] &= \frac{\beta_0}{1 - \beta_1} = \mathbb{E}[Y_0] \\
\text{var}(Y_{t+1}) &= \frac{\sigma_u^2}{1 - \beta_1^2} = \text{var}(Y_0)
\end{aligned}$$

Hence, for all t ,

$$\begin{aligned}
\mathbb{E}[Y_t] &= \frac{\beta_0}{1 - \beta_1} = \mathbb{E}[Y_0] \\
\text{var}(Y_t) &= \frac{\sigma_u^2}{1 - \beta_1^2} = \text{var}(Y_0)
\end{aligned}$$

For the h th autocovariance, notice that

$$\begin{aligned}
\text{cov}(Y_t, Y_{t-h}) &= \text{cov}(\beta_0 + \beta_1 Y_{t-1} + u_t, Y_{t-h}) \\
&= \beta_1 \text{cov}(Y_{t-1}, Y_{t-h}) + \text{cov}(u_t, Y_{t-h}) \\
&= \beta_1 \text{cov}(Y_{t-1}, Y_{t-h})
\end{aligned}$$

since $\text{cov}(u_t, Y_{t-h}) = 0$ by assumption. Proceeding recursively,

$$\begin{aligned}
\text{cov}(Y_t, Y_{t-h}) &= \beta_1 \text{cov}(Y_{t-1}, Y_{t-h}) \\
&= \beta_1^2 \text{cov}(Y_{t-2}, Y_{t-h}) \\
&= \dots = \beta_1^h \text{cov}(Y_{t-h}, Y_{t-h}) \\
&= \beta_1^h \text{var}(Y_{t-h}) = \beta_1^h \frac{\sigma_u^2}{1 - \beta_1^2}
\end{aligned}$$

Hence, since the mean, variance and h th autocorrelation does not depend on t , $\{Y_t\}$ is stationary. $\square$

Remark.

For $AR(p)$ it is $\sum_{i=1}^k \beta_i \in (-1, 1)$

5.3 Forecasting

The main object of time series is to forecast values of Y_t , usually on past lags of $\{Y_t\}$.

Definition 5.11: Mean-Squared Forecast Error Minimising Forecast Problem

Consider the problem of trying to optimally forecast Y_t based on $\mathcal{Y}_{t-1} := (Y_{t-1}, Y_{t-2}, \dots)$. Then our optimal mean-squared (forecast) error minimising forecast is the function $m^*(\mathcal{Y}_{t-1})$ s.t.

$$m^*(\mathcal{Y}_{t-1}) = \underset{m(\cdot)}{\operatorname{argmin}} \mathbb{E}[Y_t - m(\mathcal{Y}_{t-1})]^2$$

Theorem 5.12: Conditional Expectation Solves MSFE Problem

Consider the MSFE problem. Then,

$$m^*(\mathcal{Y}_{t-1}) = \mathbb{E}[Y_t | \mathcal{Y}_{t-1}]$$

Proof. This is the exact same proof as for the non-time series case. We consider the MSFE problem

$$\begin{aligned} \mathbb{E}[Y_t - m(\mathcal{Y}_{t-1})]^2 &= \mathbb{E}[(Y_t - \mathbb{E}[Y_t | \mathcal{Y}_{t-1}]) - (m(\mathcal{Y}_{t-1}) - \mathbb{E}[Y_t | \mathcal{Y}_{t-1}])]^2 \\ &= \mathbb{E}[\varepsilon + g(\mathcal{Y}_{t-1})]^2, \quad \varepsilon = Y_t - \mathbb{E}[Y_t | \mathcal{Y}_{t-1}], g(\mathcal{Y}_{t-1}) = -(m(\mathcal{Y}_{t-1}) - \mathbb{E}[Y_t | \mathcal{Y}_{t-1}]) \end{aligned}$$

Notice that ε is uncorrelated with $\mathcal{Y}_{t-1}$ as $\mathbb{E}[\varepsilon | \mathcal{Y}_{t-1}] = \mathbb{E}[Y_t - \mathbb{E}[Y_t | \mathcal{Y}_{t-1}] | \mathcal{Y}_{t-1}] = \mathbb{E}[Y_t | \mathcal{Y}_{t-1}] - \mathbb{E}[Y_t | \mathcal{Y}_{t-1}] = 0$. Therefore,

$$\begin{aligned} \mathbb{E}[\varepsilon + g(\mathcal{Y}_{t-1})]^2 &= \mathbb{E}\varepsilon^2 + 2\mathbb{E}[\varepsilon g(\mathcal{Y}_{t-1})] + \mathbb{E}g(\mathcal{Y}_{t-1})^2 \\ &= \mathbb{E}\varepsilon^2 + \mathbb{E}g(\mathcal{Y}_{t-1})^2 \end{aligned}$$

which is precisely minimised if $g(\mathcal{Y}_{t-1}) = 0$ i.e. when $m^*(\mathcal{Y}_{t-1}) = \mathbb{E}[Y_t | \mathcal{Y}_{t-1}]$ □

Definition 5.13: Optimal h -step ahead forecast

The optimal forecast of $Y_{T+h|T}$ on the basis of $\{Y_t\}$ is denoted

$$Y_{T+h|T} := \mathbb{E}[Y_{T+h} | \mathcal{Y}_T]$$

Now, the optimal forecast does not depend on whether the model that generates $\{Y_t\}$ is an AR model or not. However, the $AR(p)$ model does give a useful rule for optimally forecasting

Y_{T+1} (or Y_{T+h} more generally).

Theorem 5.14: $AR(p)$ rule for optimal forecast

Consider the problem of optimally forecasting Y_{T+h} and $\{Y_t\}$ is assumed to be generated by an $AR(p)$ model such that

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t \quad \mathbb{E}[u_t | \mathcal{Y}_{t-1}] = 0$$

the optimal rule for forecasting Y_{T+h} is

$$Y_{T+h|T} = \mathbb{E}[Y_{T+h} | \mathcal{Y}_T] = \beta_0 + \sum_{i=h}^p \beta_i Y_{T+h-i} + \mathbb{E} \left[\sum_{j=1}^{h-1} \beta_j Y_{T+h-j} | \mathcal{Y}_T \right]$$

where $p \geq h$

Proof. We may write

$$\begin{aligned} Y_{T+h|T} &= \mathbb{E}[Y_{T+h} | \mathcal{Y}_T] \\ &= \mathbb{E} \left[\beta_0 + \sum_{i=1}^p \beta_i Y_{T+h-i} + u_t | \mathcal{Y}_T \right] \\ &= \beta_0 + \mathbb{E} \left[\sum_{i=1}^{h-1} \beta_i Y_{T+h-i} + \sum_{i=h}^p \beta_i Y_{T+h-i} | \mathcal{Y}_T \right] \\ &= \beta_0 + \sum_{i=h}^p \beta_i Y_{T+h-i} + \mathbb{E} \left[\sum_{j=1}^{h-1} \beta_j Y_{T+h-j} | \mathcal{Y}_T \right] \end{aligned}$$

since $((Y_T, Y_{T-1} \dots))$ is contained in $\mathcal{Y}_T$ □

Heuristically, we can decompose the rule to “stuff we know up till p lags” and the expected value of stuff not in the information set, based on current informaton.

Remark.

There is no closed form expression for the h -step forecast since we don't know what $\mathbb{E}[\sum_{i=1}^{h-1} \beta_i Y_{T+h-i} | \mathcal{Y}_T]$ is. However, there is a special case for $h = 1$ where $Y_{T+h|T} = \mathbb{E}[Y_{T+h} | \mathcal{Y}_T] = \beta_0 + \sum_{i=1}^p \beta_i Y_{T+h-i}$.

For the $AR(1)$ case, we get the expression

$$Y_{T+h|T} = \beta_0 + \beta_1 Y_{T-h-1|T}$$

with the sample analogue

$$\hat{Y}_{T+h|T} = \hat{\beta}_0 + \hat{\beta}_1 Y_{T-h-1|T}$$

Next, we digress to consider the behaviour of the optimal forecast as h grows.

Proposition 5.15

Consider a time series Y_t follows an $AR(1)$ model. Further assume Y_t is stationary. Then, we have

$$\lim_{h \rightarrow \infty} Y_{T+h|T} = \mu$$

Interpretation: at longer horizon forecasts, if $\{Y_t\}$ is stationary, the optimal forecast is simply the mean.

Proof. Since $\{Y_t\}$ is stationary, then $\mu = \beta_0/(1 - \beta_1)$. So,

$$Y_{T+h|T} - \mu = \beta_1(Y_{T+h-1|T} - \mu)$$

and recursing backwards to $h = 0$, we have

$$Y_{T+h|T} - \mu = \beta_1^h(Y_{T|T} - \mu) = \beta_1^h(Y_T - \mu)$$

Hence, if $\{Y_t\}$ is stationary, $|\beta_1| < 1$ and so, taking the limit as $h \rightarrow \infty$, this yields

$$\lim_{h \rightarrow \infty} Y_{T+h|T} = \mu + \lim_{h \rightarrow \infty} \beta_1^h(Y_T - \mu) = \mu$$

□

This follows for the sample analogue as well, provided that $|\hat{\beta}_1| < 1$, recursing similarly implies $\lim_{h \rightarrow \infty} \hat{Y}_{T+h|T} = \hat{\mu} = \hat{\beta}_0/(1 - \hat{\beta}_1)$ where the latter is approx. but not exactly equal to $\bar{Y}$

Corollary 5.16

Consider the problem of optimally forecasting Y_{T+1} and $\{Y_t\}$ is assumed to be generated by an $AR(p)$ model such that

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t \quad \mathbb{E}[u_t | \mathcal{Y}_{t-1}] = 0$$

the optimal rule for forecasting Y_{T+1} is

$$Y_{T+1|T} = \mathbb{E}[Y_{T+1} | \mathcal{Y}_T] = \beta_0 + \sum_{i=1}^p \beta_i Y_{T+h-i}$$

Notice that from our assumptions on the $AR(p)$ model that the two conditions follow

$$\mathbb{E}[u_t] = 0 \quad \text{cov}(u_t, Y_{t-i}) = 0, \quad \forall i \in \{1, \dots, p\}$$

So, **OR** is satisfied. Hence, even though we often don't know the values of $\{B_i\}_{i=0}^p$, we have the following result

Proposition 5.17

Consider an $AR(p)$ model

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-1} + u_t$$

where $\mathbb{E}[u_t | \mathcal{Y}_{t-1}] = 0$. Then, $\{\beta_i\}_{i=0}^p$ coincide with $\{\hat{\beta}_i\}_{i=0}^p$ i.e. the parameters of a population linear regression of Y_t on its past p lags.

Furthermore, the population linear regression of Y_t on its past p lags may be consistently estimated by an OLS regression.

Remark.

If $\{Y_t\}$ is also stationaty, then these estimators are also asymptotically normal and the usual t and F hypothesis tests can be conducted as per usual.

Theorem 5.18: OLS $AR(p)$ estimate of optimal 1-step forecast

Consider the problem of estimating the optimal forecast $Y_{T+1|T}$ on the basis of data $\{Y_t\}_{t=1}^T$. Letting $\{\hat{\beta}_i\}_{i=0}^p$ denote OLS estimates, we have the following estimate of the optimal forecast

$$\hat{Y}_{T+1|T} := \hat{\beta}_0 + \sum_{i=1}^p \hat{\beta}_i Y_{T+1-i}$$

The error associated with $\hat{Y}_{T+1|T}$ is thus

$$\hat{e}_{T+1|T} := Y_{T+1} - \hat{Y}_{T+1|T}$$

We may decompose

$$\begin{aligned} \hat{e}_{T+1|T} &:= Y_{T+1} - \hat{Y}_{T+1|T} = (Y_{T+1} - Y_{T+1|T}) + (Y_{T+1|T} - \hat{Y}_{T+1|T}) \\ &=: e_{T+1|T} + (Y_{T+1|T} - \hat{Y}_{T+1|T}) \end{aligned}$$

into two components

1. The error made by the optimal forecast itself:

$$\begin{aligned} e_{T+1|T} &= Y_{T+1} - Y_{T+1|T} = \left[\beta_0 + \sum_{i=1}^p \beta_i Y_{T+1-i} + u_{T+1} \right] + \left[\beta_0 + \sum_{i=1}^p \beta_i Y_{T+1-i} \right] \\ &= u_{T+1} \end{aligned}$$

2. and the additional error due to having to estimate the optimal forecast

$$(Y_{T+1|T} - \hat{Y}_{T+1|T}) = (\beta_0 - \hat{\beta}_0) + \sum_{i=1}^p (\beta_i - \hat{\beta}_i) Y_{T+1-i}$$

Lemma 5.19: Orthogonality of errors

The error of the optimal forecast itself, $e_{T+1|T}$, and the additional error introduced by estimating the optimal forecast, $(Y_{T+1|T} - \hat{Y}_{T+1|T})$ are orthogonal i.e.

$$\mathbb{E}[e_{T+1|T}] = \text{Cov}(e_{T+1|T}, (Y_{T+1|T} - \hat{Y}_{T+1|T})) = 0$$

Proof. Notice that

$$(Y_{T+1|T} - \hat{Y}_{T+1|T}) = (\beta_0 - \hat{\beta}_0) + \sum_{i=1}^p (\beta_i - \hat{\beta}_i) Y_{T+1-i}$$

implies $(Y_{T+1|T} - \hat{Y}_{T+1|T})$ is a function only of $\mathcal{Y}_T$ since it depends only on the observations $\{Y_{T+1-p}, \dots, Y_{T-1}, Y_T\}$ and $\{\hat{\beta}_i\}_{i=0}^p$ which are a function of the sample $\{Y_t\}_{t=1}^T$. By assumption,

$$\mathbb{E}[e_{T+1|T} | \mathcal{Y}_T] = \mathbb{E}[u_{T+1} | \mathcal{Y}_T] = 0$$

And we know that if $\mathbb{E}[u_{T+1} | \mathcal{Y}_T] = 0$, then for any function $m(\mathcal{Y}_T)$, $\mathbb{E}[u_{T+1} | m(\mathcal{Y}_T)] = 0$. Let $m(\mathcal{Y}_T) = (Y_{T+1|T} - \hat{Y}_{T+1|T})$, then we have

$$\mathbb{E}[e_{T+1|T}] = \mathbb{E}[u_{T+1} | m(\mathcal{Y}_T)] = 0$$

which implies $\text{Cov}(e_{T+1|T}, (Y_{T+1|T} - \hat{Y}_{T+1|T})) = 0$ □

Definition 5.20: Mean Squared Forecast Error

The mean squared forecast error associated with $\hat{Y}_{T+1|T}$ is defined as

$$\text{MSFE}(\hat{Y}_{T+1|T}) := \mathbb{E}[\hat{e}_{T+1|T}^2]$$

Theorem 5.21: Decomposition of MSFE

The MSFE of $\hat{Y}_{T+1|T}$ can be decomposed into

$$\text{MSFE}(\hat{Y}_{T+1|T}) = \sigma_u^2 + \mathbb{E}(Y_{T+1|T} - \hat{Y}_{T+1|T})^2$$

Proof. By definition, we may write

$$\begin{aligned} \text{MSFE}(\hat{Y}_{T+1|T}) &:= \mathbb{E}[\hat{e}_{T+1|T}^2] = \mathbb{E}[e_{T+1|T}^2] + \mathbb{E}[Y_{T+1|T} - \hat{Y}_{T+1|T}]^2 \\ &= \mathbb{E}[u_{T+1}^2] + \mathbb{E}[Y_{T+1|T} - \hat{Y}_{T+1|T}]^2 \\ &= \sigma_u^2 + \mathbb{E}[Y_{T+1|T} - \hat{Y}_{T+1|T}]^2 \end{aligned}$$

The first equality follows because the term $2\mathbb{E}[e_{T+1|T}(Y_{T+1|T} - \hat{Y}_{T+1|T})] = 0$ in $\mathbb{E}[\hat{e}_{T+1|T}^2] = \mathbb{E}[e_{T+1|T}^2] + 2\mathbb{E}[e_{T+1|T}(Y_{T+1|T} - \hat{Y}_{T+1|T})] + \mathbb{E}[Y_{T+1|T} - \hat{Y}_{T+1|T}]^2$ due to orthogonality by the lemma $\square$

Remark.

Clearly then σ_u^2 provides a biased estimate of the MSFE i.e. consistently underestimating the MSFE

OLS estimation of $AR(p)$ models

Definition 5.22: OLS estimation of $AR(p)$ models

Suppose we have data $\{Y_t\}_{t=1}^T$. Then, we may estimate the $AR(p)$ model via OLS by solving

$$(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p) = \underset{b_0, \dots, b_p}{\operatorname{argmin}} \sum_{t=p}^T (Y_t - b_0 - b_1 Y_{t-1} - \dots - b_p Y_{t-p})^2$$

i.e. minimising the sum of $T - p$ squared errors

Remark.

This is because there are only $T - p$ periods which we observe the dependent variable Y_t and the rest of the regressors $Y_{t-1}, \dots, Y_{t-p}$

Without loss, we may assume for convenience that we observe sufficient number of periods before $t = 1$ (i.e. $Y_{-1}, \dots, Y_{1-p}$) such that for any p lags chosen, the first value we estimate is Y_1 and we always have T observations available to estimate the $AR(p)$ model. So T may be taken as the effective size of the sample

5.4 Forecast Evaluation

Pseudo Out-Of-Sample forecast errors. Instead of OLS residuals, we want a set of out-of-sample forecast errors to estimate the MSFE($\hat{Y}_{T+1|T}$). We cannot recover $\hat{e}_{T+1|T} := Y_{T+1} - \hat{Y}_{T+1|T}$ since Y_{T+1} lies outside the sample, we can obtain a sequence of pseudo OOS errors that have similar statistical properties to $\hat{e}_{T+1|T}$ by running the forecast procedure as if we have only observed up to $s < T$ periods and finding

$$\hat{e}_{s+1|s} := Y_{s+1} - \hat{Y}_{s+1|s}$$

The general algorithm is as follows:

1. For each s in $(T - P, T - 1)$
 - (a) use the subsample $\{Y_t\}_{t=1}^s$ to estimate $\{\hat{\beta}_{i,1:s}\}$ in the $AR(p)$ model to produce the forecast

$$\hat{Y}_{s+1|s} := \hat{\beta}_{0,1:s} + \sum_{i=1}^p \hat{\beta}_{i,1:s} Y_{s+1-i}$$

(b) Evaluate pseudo OOS forecast errors

$$\hat{e}_{s+1|s} := Y_{s+1} - \hat{Y}_{s+1|s}$$

2. Using the final P periods of the sample $\{Y_t\}_{t=1}^T$, we have a sequence $\{\hat{e}_{s+1|s}\}_{s=T-P}^{T-1}$ of pseudo OOS forecast errors, which we can use to estimate the MSFE via

$$\text{MSFE}(\hat{Y}_{T+1|T}) := \frac{1}{P} \sum_{s=T-P}^{T-1} \hat{e}_{s+1|s}^2$$

where in practical applications, P is set to be an integer between $0.1T$ and $0.2T$

This extends straightforwardly to $h \geq 2$ where we would estimate MSFE on the basis of the h -step ahead pseudo OOS forecast errors $\hat{e}_{s+h|s} := Y_{s+h} - \hat{Y}_{s+h|s}$ as per

$$\text{MSFE}(\hat{Y}_{T+h|T}) := \frac{1}{P} \sum_{s=T-P-(h-1)}^{T-h} \hat{e}_{s+h|s}^2$$

Model Selection.

Theorem 5.23: Bias-Variance Decomposition

Consider a model $f(\cdot)$ and a given value x_0 . Let y_0 be the dependent variable. Then,

$$\mathbb{E}(y_0 - \hat{f}(x_0))^2 = \text{var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{var}(\varepsilon)$$

where $\mathbb{E}(y_0 - \hat{f}(x_0))^2$ is the MSE at x_0 .

In other words, we can decompose the MSE into three fundamental quantities: the variance of the estimated model $\hat{f}(x_0)$, the squared bias of $\hat{f}(x_0)$ and the variance of the error term ε . If we want to reduce MSE, we want low variance and low bias.

However, there is a **Variance-Bias tradeoff**. As our model becomes more flexible i.e. more regressors, we have a lower bias but a greater variance; in contrast, when our model becomes less flexible i.e. fewer regressors, we have a higher bias but a lower variance. This is because the model becomes biased towards the simplifying assumption we choose

The intuition for time series in terms of p lags is

- larger values of p allows for a more flexible model, with richer dynamics that may better match those of $\{Y_t\}$ and more free parameters with which to match the true conditional expectation $\mathbb{E}[Y_{T+1}|\mathcal{Y}_T]$, reducing the bias with which this is approximated by the model
- smaller values of p reduce the number of parameters that need to be estimated with a small finite amount of data, which typically reduces the variability of those estimates, and thus the variance of the implied estimate of $\mathbb{E}[Y_{T+1}|\mathcal{Y}_T]$

So, even if $\{Y_t\}$ follows an AR model with a relatively high lag order, the optimal feasible forecast may be produced by an AR model of a smaller order, which is better situated due

to a lower variance.

To determine the lag order we want to use, we want to rank models according to their estimated MSFEs, choosing the model for which this measure is minimised. To avoid the noise associated with estimates of MSFE, we use **information criteria** that optimise the bias-variance tradeoff.

Definition 5.24: Akaike Information Criteria (AIC)

Consider a model with m parameters. Then the AIC is

$$\text{AIC}_m := \log \frac{\text{SSR}_m}{T} + m \frac{2}{T}$$

where SSR_m denotes the sum of squared residuals and $m = p + 1$. We choose

$$m^* = \underset{m}{\operatorname{argmin}} \text{AIC}_m$$

Definition 5.25: Bayesian Information Criteria (BIC)

Consider a model with m parameters. Then the BIC is

$$\text{BIC}_m := \log \frac{\text{SSR}_m}{T} + m \frac{\log T}{T}$$

where SSR_m denotes the sum of squared residuals and $m = p + 1$. We choose

$$m^* = \underset{m}{\operatorname{argmin}} \text{BIC}_m$$

Remark.

Since minimising SSR_m always favours models with more parameters, we have a penalising term that punishes the criterion for each additional parameter added. So, minimising either yields a model with an intermediate number of parameters

Some extra facts: AIC is known to be conservative and will have a tendency to overfit i.e. choose a larger p in large samples, while BIC is known to be consistent i.e. choosing with probability one the number of lags which $\{Y_t\}$ is in fact generated by. In small samples, both will tend to pick smaller lag orders i.e. tend to underfit.

Connection between $\bar{R}_m^2$ and Information Criteria

Claim

Selecting the model that maximises $\bar{R}_m^2$ tends to yield a model larger than minimising AIC or BIC and hence worse forecasts

Proof for Claim.

This is less a proof than an explanation. We know that

$$\bar{R}_m^2 = 1 - \frac{\text{SSR}_m/(T-m)}{\text{TSS}/(T-1)} = 1 - \frac{\text{SER}_m^2}{\text{TSS}/(T-1)}$$

where $\text{TSS} = \sum_{t=1}^T (Y_t - \bar{Y}_T)^2$ is the total sum of squares. SER_m , the standard error of regression, is the only term that depends on the choice of the model, so maximising $\bar{R}_m^2$ is equivalent to minimising the SER or since $\log(\cdot)$ is a monotonic transformation, minimising

$$\begin{aligned} \log \text{SER}_m^2 &= \log \frac{\text{SSR}_m}{T-m} \\ &= \log \frac{\text{SSR}_m}{T} \frac{T}{T-m} \\ &= \log \frac{\text{SSR}_m}{T} - \log(1 - m/T) \\ &\simeq \log \frac{\text{SSR}_m}{T} + m \frac{1}{T} \end{aligned}$$

The penalty term is smaller than it is for both AIC and BIC and so the claim follows ■

5.5 Granger Causality and Autoregressive Distributed Lag models

Definition 5.26: Autoregressive distributed lag

Let $ADL(p, q)$ denote a model that adds q lags of $\{X_t\}$ to an $AR(p)$ model for $\{Y_t\}$ such that

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{i=1}^q \delta_i X_{t-i} + u_t \quad (14)$$

where $\{u_t\}$ is unforecastable on the basis of past lags of $\{Y_t\}$ and $\{X_t\}$ i.e.

$$\mathbb{E}[u_t | \mathcal{Y}_{t-1}, \mathcal{X}_{t-1}] = 0$$

So, u_t is orthogonal to the regressors and hence **OR** holds: the parameters of the model coincide with the population linear regression parameters and may be consistently estimated by OLS.

Remark.

Provided that $\{Y_t\}$ and $\{X_t\}$ are stationary, t and F tests may be conducted in exactly the usual way.

ADL 1-step ahead forecasts

We may give an optimal 1-step ahead forecast with an $ADL(p, q)$ model via

$$Y_{T+1|T} = \mathbb{E}[Y_{T+1} | \mathcal{Y}_T, \mathcal{X}_T] = \beta_0 + \sum_{i=1}^p \beta_i Y_{T+1-i} + \sum_{i=1}^q \delta_i X_{T+1-i}$$

of which we may use OLS estimates of the model parameters to yield

$$\hat{Y}_{T+1|T} = \hat{\beta}_0 + \sum_{i=1}^p \hat{\beta}_i Y_{T+1-i} + \sum_{i=1}^q \hat{\delta}_i X_{T+1-i}$$

What we want to know is if the introduction of $\{X_t\}$ to the model is actually helpful for forecasting $\{Y_t\}$ or we can just use an $AR(p)$ model instead.

Definition 5.27: Granger Causality

We say that $\{X_t\}$

- Granger causes $\{Y_t\}$ if

$$\mathbb{E}[Y_{T+1} - \mathbb{E}[Y_{T+1}|\mathcal{Y}_T, \mathcal{X}_T]]^2 < \mathbb{E}[Y_{T+1} - \mathbb{E}[Y_{T+1}|\mathcal{Y}_T]]^2$$

- does not Granger cause $\{Y_t\}$ if

$$\mathbb{E}[Y_{T+1} - \mathbb{E}[Y_{T+1}|\mathcal{Y}_T, \mathcal{X}_T]]^2 = \mathbb{E}[Y_{T+1} - \mathbb{E}[Y_{T+1}|\mathcal{Y}_T]]^2$$

i.e. if the mean squared error made by the optimal forecasts on the basis of $(\mathcal{Y}_T, \mathcal{X}_T)$ is the same as that made on the basis of $\mathcal{Y}_T$ alone

The intuition is to find out whether lags of $\{X_t\}$ carry useful information about $\{Y_t\}$ in addition to that carried by its own lags. Since the conditional expectation minimises the mean squared prediction error i.e.

$$\mathbb{E}[Y_{T+1}|\mathcal{Y}_T, \mathcal{X}_T] = \underset{m(\cdot)}{\operatorname{argmin}} \mathbb{E}[Y_{T+1} - m(\mathcal{Y}_T, \mathcal{X}_T)]^2$$

, then if

$$\mathbb{E}[Y_{T+1} - \mathbb{E}[Y_{T+1}|\mathcal{Y}_T, \mathcal{X}_T]]^2 = \mathbb{E}[Y_{T+1} - \mathbb{E}[Y_{T+1}|\mathcal{Y}_T]]^2$$

holds, it must be the case that $\mathbb{E}[Y_{T+1}|\mathcal{Y}_T]$ is also the solution to the minimisation problem and thus

$$\mathbb{E}[Y_{T+1}|\mathcal{Y}_T, \mathcal{X}_T] = \mathbb{E}[Y_{T+1}|\mathcal{Y}_T]$$

An equivalent way of formulating the Granger causality test is to start with an $ADL(p, q)$ model for $\{Y_t\}$ in which we have

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{i=1}^q \delta_i X_{t-i} + u_t$$

where $\mathbb{E}[u_t|\mathcal{Y}_{t-1}, \mathcal{X}_{t-1}] = 0$. Then, we test the null hypothesis

$$H_0 : \delta_1 = \delta_2 = \dots = \delta_q = 0$$

Provided that the two series are stationary, we can carry out the test via the F test of multiple restrictions, with a rejection of H_0 corresponding to a finding that $\{X_t\}$ Granger

causes $\{Y_t\}$

6 Time Series: Non-stationary processes

This section will be significantly less formal, with a focus on *knowing the procedures* and when to apply them

6.1 Breaks and parameter instability

The basic problem here is that the parameters of the model might change abruptly after some date τ , which we call a **structural break**. This is usually informed by eyeballing or economic knowledge of what's happening. This makes the series **non-stationary**

Example.

One example of this is a time series for inflation $\{\pi_t\}$ with the replacement of the Chair of the Fed with a drastically new policy at time τ

We accommodate this parameter instability via a breakpoint dummy variable

$$D_t(\tau) := \mathbf{1}\{\tau \leq t\} = \begin{cases} 1 & \text{if } t \leq \tau \\ 0 & \text{if } t > \tau \end{cases}$$

and the following model(s)

- $AR(p)$:

$$\begin{aligned} Y_t &= \beta_0 + \sum_{i=1}^p (\beta_i + \gamma_i D_t(\tau)) Y_{t-i} \\ &= \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{i=1}^p \gamma_i D_t(\tau) Y_{t-i} \end{aligned}$$

- $ADL(p, q)$

$$\begin{aligned} Y_t &= \beta_0 + \sum_{i=1}^p (\beta_i + \gamma_i D_t(\tau)) Y_{t-i} + \sum_{i=1}^q (\delta_i + \theta_i D_t(\tau)) X_{t-i} \\ &= \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{i=1}^p \gamma_i D_t(\tau) Y_{t-i} + \sum_{i=1}^q \delta_i X_{t-i} + \sum_{i=1}^q \theta_i D_t(\tau) X_{t-i} \end{aligned}$$

The basic intuition is that the dummy variable will remove the additional coefficient γ (or θ) if it is before the structural break date τ and will ‘top-up’ the remaining coefficient values (or remove) to the post-break values when $t = \tau$ i.e. when the dummy takes the value 1

Remark.

All parameters change on the same date

Chow breakpoint test: Testing for structural break To determine whether the data

is consistent with the presence of a break at date τ , we test the null of no break i.e.

$$H_0 : \bigwedge_{i=1}^p \gamma_i = 0$$

for the $AR(p)$ case, against the alternative of a break affecting at least one of the parameters i.e.

$$H_1 : \bigvee_{i=1}^p \gamma_i \neq 0$$

This is carried out via the usual F test supposing that $\{Y_t\}$ is stationary (or $\{X_t\}$ and $\{Y_t\}$ are jointly stationary) under the null, the test statistic will have the $F_{q,\infty}$ limiting distribution where q is the number of restrictions.

Remark.

Rejection of the null is evidence that at least one of the parameters has been affected by a break date τ but does not indicate whether only a certain subset of the parameters has been affected. If this were of interest, we could investigate by testing hypotheses that restrict only a subset of the parameters

The detection of a break would normally result in different models being estimated before and after the breakpoint by either using the breakpoint dummies or splitting the sample period and estimating separately on two subsamples, dealing with non-stationarity of $\{Y_t\}$.

Quandt Likelihood Ratio Test.

Motivation: The Chow breakpoint test assumes that the break date τ is known, but we usually only know the rough timing so we usually consider some interval $\{\tau_0, \tau_0 + 1, \dots, \tau_1\}$. If we wanted to run the Chow tests of H_0 for each $\tau \in \{\tau_0, \tau_0 + 1, \dots, \tau_1\}$ and check if we obtain a rejection of H_0 , this entails running the test $\tau_1 - \tau_0 + 1$ times. The problem with this is that the overall probability of a type 1 error (false detection of a non-existent break) will be non-trivially higher than the significance level of any of these individual Chow tests.

So, we use the Quandt likelihood ratio tests which is basically computing the Chow breakpoint statistic $F(\tau)$ for H_0 associated with each $\tau \in \{\tau_0, \tau_0 + 1, \dots, \tau_1\}$ and taking the maximum i.e.

$$QLR := \max_{\tau \in \{\tau_0, \dots, \tau_1\}} F(\tau)$$

to be our test statistic for the null of no break over the interval $\{\tau_0, \tau_0 + 1, \dots, \tau_1\}$.

The critical values for the QLR test depend on

1. The number of parameters q that are restricted by the null hypothesis of no break
2. The number of slope coefficients k in the unrestricted model (which allows for a break)
3. The location of the end points τ_0/T and τ_1/T of the interval to be searched for the break, expressed relative to the length of the sample

Remark.

Since the QLR statistic takes the maximum of several F statistics, these critical values are higher than corresponding $F_{q,\infty}$ critical values with the difference increasing in the magnitude of $\tau_1 - \tau_0$

The QLR test also provides an estimate of the break date, which is the τ^* that solves

$$\tau^* = \underset{\tau \in \{\tau_0, \dots, \tau_1\}}{\operatorname{argmax}} F(\tau)$$

6.2 Unit Roots and Stochastic Trends

Motivation: A second kind of nonstationarity is due to time-trends or random wandering (i.e. random walks).

Differencing: To resolve stationarity, we usually difference $\{Y_t\}$ i.e. have $\{\Delta Y_t\}$ where $\Delta Y_t := Y_t - Y_{t-1}$. In particular, we rely on the following proposition

Proposition 6.1

For all $p \geq 1$, we have for any series $\{Y_t\}$

$$\Delta Y_t \text{ is } AR(p-1) \iff Y_t \text{ is } AR(p) \text{ with } \sum_{i=1}^p \beta_i = 1$$

Proof. We prove both the forwards and backwards direction:

($\Rightarrow$) We want to show that any $AR(p)$ model $Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t$ can be written as

$$\Delta Y_t = \beta_0 + \delta Y_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t$$

where $\delta := \sum_{i=1}^p \beta_i - 1$ and $\gamma_i := -\sum_{j=i+1}^p \beta_j$ for $i \in \{0, \dots, p-1\}$ with $\gamma_p := 0$.

Consider a Y_t model that follows an $AR(p)$ process $Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t$. Define $\delta := \sum_{i=1}^p \beta_i - 1$ and $\gamma_i := -\sum_{j=i+1}^p \beta_j$ for $i \in \{0, \dots, p-1\}$ with $\gamma_p := 0$. Then we have

$$\gamma_i - \gamma_{i-1} = -\gamma_{j=i+1}^p \beta_j + \sum_{j=1}^p \beta_j = \beta_i$$

for $i \in \{1, \dots, p-1\}$ and also when $i = p$. Hence

$$\sum_{i=1}^p \beta_i Y_{t-i} = \sum_{i=1}^p (\gamma_i - \gamma_{i-1}) Y_{t-i}$$

where using that $\gamma_p = 0$,

$$\begin{aligned}
\sum_{i=1}^p (\gamma_i - \gamma_{i-1}) Y_{t-i} &= -\gamma_0 Y_{t-1} + \gamma_1 Y_{t-1} - \gamma_1 Y_{t-2} + \gamma_2 Y_{t-2} + \cdots + \gamma_{p-1} Y_{t-p+1} - \gamma_{p-1} Y_{t-p} + \gamma_p Y_{t-p} \\
&= -\gamma_0 Y_{t-1} + \gamma_1 (Y_{t-1} - Y_{t-2}) + \cdots + \gamma_{p-1} (Y_{t-p+1} - Y_{t-p}) \\
&= -\gamma_0 + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i}
\end{aligned}$$

So, we may rewrite the equation $Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t$ as

$$Y_t = \beta_0 - \gamma_0 Y_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t$$

Subtracting Y_{t-1} from both sides and recognising that $\delta = \sum_{i=1}^p \beta_i - 1 = -\gamma_0 - 1$ we obtain

$$\begin{aligned}
\Delta Y_{t-1} &= \beta_0 + (-\gamma_0 - 1) Y_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t \\
&= \beta_0 + \gamma Y_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t
\end{aligned}$$

($\Leftarrow$) We start with an $AR(p-1)$ model

$$\Delta Y_t = \beta_0 + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t$$

and define $\gamma_0 := -1$ and $\gamma_o := 0$ (notice that the model only specifies values of $\{\gamma_i\}_{i=1}^{p-1}$). Adding Y_{t-1} to both sides we yield

$$\begin{aligned}
Y_t &= \beta_0 + Y_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t \\
&= \beta_0 - \gamma_0 Y_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta Y_{t-i} + u_t
\end{aligned}$$

Defining $\beta_i := \gamma_i - \gamma_{i-1}$ for $i \in \{1, \dots, p\}$ and running the reverse argument for the forwards implication yields

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t$$

where

$$\sum_{i=1}^p \beta_i = \sum_{i=1}^p (\gamma_i - \gamma_{i-1}) = -\gamma_0 + \gamma_p = 1$$

This completes the proof □

Remark.

Note that the $AR(p-1)$ model describing $\{\Delta Y_t\}$ does not need to be stationary. This is something we need to check. So the above proposition is merely an algebraic identity that differencing an $AR(p)$ model reduces the lag order by 1

6.2.1 AR models and Unit Roots

Definition 6.2: The unit root $AR(p)$ model

A time series $\{Y_t\}$ follows a unit root $AR(p)$ model when

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t$$

with $\sum_{i=1}^p \beta_i = 1$

Assuming $\{\Delta Y_t\}$ is stationary, we want to decompose the unit root $AR(p)$ process into tractable components.

Proposition 6.3

Suppose $\{\Delta Y_t\}$ is a stationary $AR(p-1)$ process. Then, we have

$$Y_t := Y_0 + \mu t + V_t$$

where $\mu := \mathbb{E}[\Delta Y_t]$, $V_t = \sum_{s=1}^t v_s$ and $v_t := \Delta Y_t - \mu$.

Proof. First, observe that v_t is a mean zero i.e. $\mathbb{E}[v_t] = 0$ stationary process. Then, given stationarity of $\{Y_t\}$, we have $\mu := \mathbb{E}[\Delta Y_t]$ and $v_t = \Delta Y_t - \mu$ (second is by construction). This yields

$$\begin{aligned} Y_t &= Y_0 + \sum_{s=1}^t \Delta Y_s \\ &= Y_0 + \mu t + \sum_{s=1}^t (\Delta Y_s - \mu) \\ &= Y_0 + \mu t + \sum_{s=1}^t v_s \\ &=: Y_0 + \mu t + V_t \end{aligned}$$

□

This means that we can decompose the unit root processes $AR(p)$ into three components:

- Y_0 : the initial value (or initialisation) of the process
- μt : a linear deterministic trend (or drift)

- $V_t := \sum_{s=1}^t v_s$: the partial sum of a stationary process i.e. a stochastic trend

Definition 6.4: Stochastic Trend

A process W_t is a stochastic trend if $W_t = \sum_{s=1}^t w_s$, where

1. $\{w_t\}$ is a stationary process, with $\gamma_{w,h} := \text{cov}(w_t, w_{t-h})$
2. $\text{Irvar}(w_t) := \sum_{h=-\infty}^{\infty} \gamma_{w,h} = \gamma_{w,0} + 2 \sum_{h=1}^{\infty} \gamma_{w,h} > 0$

Remark.

Irvar is the long run variance i.e. the sum of covariances of w_t at all positive and negative lags. A detailed understanding is beyond the course. It is to exclude the possibility that there is so much negative serial correlation between the w_t 's that they effectively cancel each other out when added together. This ensures that the variance of W_t grows as t grows, so that W_t shares the random wandering behaviour characteristic of a random walk.

Remark.

Notice that a random walk $U_t = \sum_{s=1}^t u_s$ is a special case of a stochastic trend: since $\{u_t\}$ is stationary and since it is serially uncorrelated, $\text{Irvar}(u_t) = \gamma_{u,0} = \sigma_u^2 > 0$

Some discussion on the stochastic process condition: When might condition (2) fail?

- If w_t follows a stationary $AR(p)$ model, this condition is guaranteed to be satisfied
- $\text{Irvar}(w_t) = 0$ can systematically occur (and really only in this case) when w_t itself is the difference of a stationary process.

Proposition 6.5

Suppose w_t is the difference of a stationary process. Then,

$$\text{Irvar}(w_t) = 0$$

This is most easily seen when $w_t = \Delta u_t = u_t - u_{t-1}$ for u_t a serially uncorrelated process, since then

$$\begin{aligned} \gamma_{w,0} &= \text{var}(u_t - u_{t-1}) = 2\sigma_u^2, \text{ and} \\ \gamma_{w,1} &= \text{cov}(u_t - u_{t-1}, u_{t-1} - u_{t-2}) = -\sigma_u^2 \end{aligned}$$

and $\gamma_{w,h} = 0$ for all $h \geq 2$. Hence

$$\text{Irvar}(w_t) = \gamma_{w,0} + 2 \sum_{h=1}^{\infty} \gamma_{w,h} = 2\sigma_u^2 + 2(-\sigma_u^2) = 0$$

Since $v_t = \Delta Y_t - \mu$ is itself a stationary $AR(p)$ process, we must have $\text{Irvar}(v_t) > 0$ and thus V_t is a stochastic trend. By implication, Y_t will be governed by this stochastic trend and the

deterministic trend μt . But this means that we no longer need to assume that $\Delta Y_t = \mu + v_t$ is serially uncorrelated (and only stationary). This is because we sometimes assume that v_t is serially uncorrelated as in the unit root $AR(1)$ process (and hence ΔY_t) but in this broader set up we need not do so, so it may have an autocorrelation structure.

Special case of unit root $AR(1)$ process

Definition 6.6: The unit root $AR(1)$ model

We say that $\{Y_t\}$ follows a unit root $AR(1)$ process such that

$$\begin{aligned} Y_t &= \beta_0 + \beta_1 Y_{t-1} + u_t \\ &= \beta_0 + Y_{t-1} + u_t \end{aligned}$$

with $\mathbb{E}[u|\mathcal{Y}_{t-1}] = 0$

We may similarly decompose the unit root $AR(1)$ into three tractable components.

Claim: Decomposition of unit root $AR(1)$ model

Consider that $\{Y_t\}$ follows a unit root $AR(1)$ process i.e. $Y_t = \beta_0 + Y_{t-1} + u_t$. Then, we may write

$$Y_t = Y_0 + \beta_0 t + \sum_{s=1}^t u_s = Y_0 + \beta_0 t + U_t$$

where $U_t = \sum_{s=1}^t u_s$

Proof for Claim.

We prove this via recursive substitution, where

$$\begin{aligned} Y_t &= \beta_0 + Y_{t-1} + u_t \\ &= \beta_0 + (\beta_0 + Y_{t-2} + u_{t-1}) + u_t \\ &= 2\beta_0 + Y_{t-2} + (u_{t-1} + u_t) \\ &= 3\beta_0 + Y_{t-3} + (u_{t-2} + u_{t-1} + u_t) \end{aligned}$$

Proceeding recursively until the initial period, we obtain

$$Y_t = Y_0 + \beta_0 t + \sum_{s=1}^t u_s = Y_0 + \beta_0 t + U_t$$

This shows that the unit root $AR(1)$ process may be decomposed into

1. Y_0 : the initial value of the process
2. $\beta_0 t$: the linear deterministic trend (or drift)

3. $U_t := \sum_{s=1}^t u_s$: the cumulation (or partial sum) of a sequence of stationary uncorrelated random variables i.e. a random walk

Definition 6.7: Random Walk

We say that $U_t = \sum_{s=1}^t u_s$ is a random walk when it is the cumulation of serially uncorrelated, stationary random variables $\{u_t\}_{s=1}^t$

Remark.

$\{u_t\}$ being i.i.d. is sufficient for U_t to be a random walk because it implies that $\{u_t\}$ is stationary and serially uncorrelated

Proposition 6.8

A random walk is not stationary

Proof. Consider $U_t = \sum_{s=1}^t u_s$. Then even though it has a time invariant mean

$$\mathbb{E}[U_t] = \sum_{s=1}^t \mathbb{E}[u_s] = 0$$

it does not have any tendency to revert back towards that mean since

$$\text{var}(U_t) = \text{var}\left(\sum_{s=1}^t u_s\right) = \sum_{s=1}^t \text{var}(u_s) = \sigma_u^2 t$$

where the second equality follows by uncorrelatedness. □

Hence, the variance of U_t grows at rate t and its distribution spreads out over the whole of the real line.

Intuition

- The trend component implies that in the absence of any shocks, Y_t would grow by β_0 per period
- The random walk component implies that time plots of U_t will exhibit ‘random wandering’ that is characteristic of a random walk, with no definitive tendency to revert to any fixed point
- Together, the three components imply that depending on the sign of β_0 , Y_t will exhibit
 1. a tendency to grow (or decline if negative) over time
 2. randomly wandering around the linear trend $\beta_0 t$

The dominating component depends on relative magnitudes of β_0 and σ_u

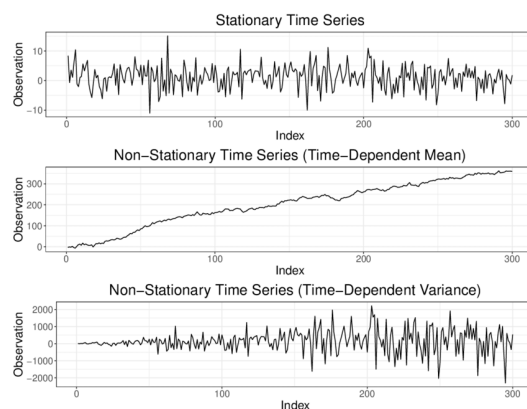
So, the unit root $AR(1)$ process can generate a trending and random wandering behaviour (like GDP).

However, the model has the unfortunate implication that are unlikely to be satisfied – e.g. subtracting Y_{t-1} from both sides imply that

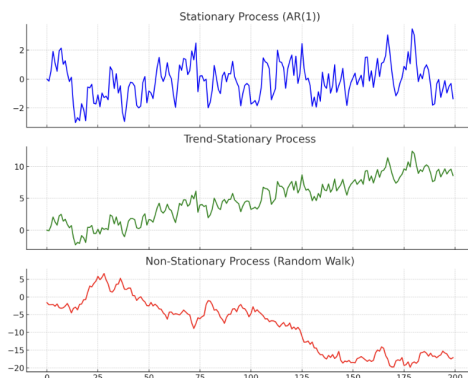
$$\Delta Y_t = \beta_0 + u_t$$

i.e. that the first difference of Y_t should be unforecastable which is rarely the case. So, the unit root $AR(p)$ model generalises the unit root $AR(1)$ to allow Y_t to exhibit trends and random wandering while permitting ΔY_t to be serially correlated.

Time plots of time series



Note that the middle picture may describe a trend-stationary process if it just reverts around the time-dependent mean. Alternatively, ChatGPT produced these nice charts at my request



Note that

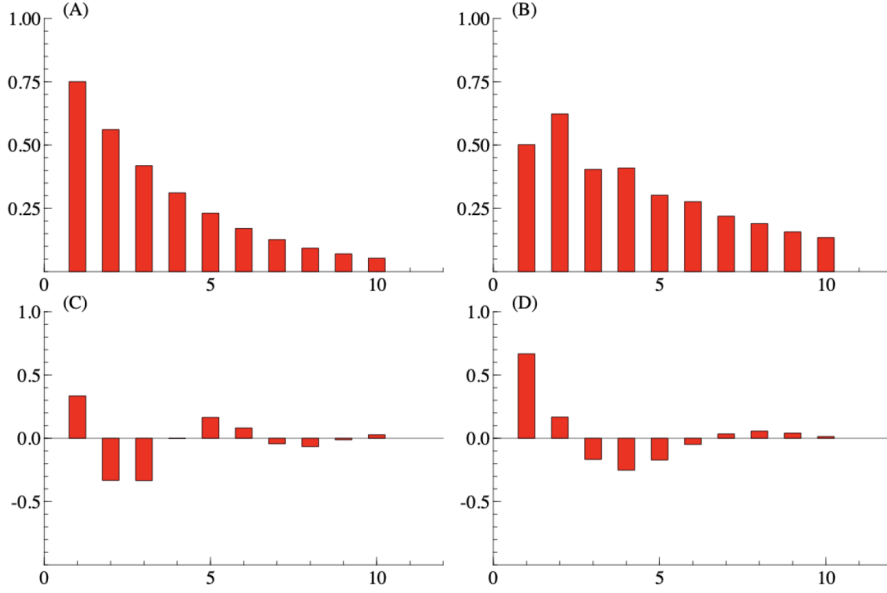
- Stationary process (blue): an $AR(1)$ that fluctuates around a constant mean with constant variance.
- Trend-stationary process (green): a deterministic upward trend plus stationary noise; deviations are mean-reverting around the trend.
- Non-stationary process (red): a random walk that drifts without bound; variance grows over time and there's no stable long-run mean.

Correlogram

A correlogram has the x -axis for k lags and y axis for $\rho_k = \frac{\text{cov}(Y_t, Y_{t+k})}{\text{var}(Y_t)}$. For instance, consider an AR(2) function

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + u_t$$

with $\{u_t\}$ being i.i.d with mean 0 and $\text{var}(u_t) = \sigma^2$. Now, we have



We want to match the following values of β_1, β_2 to each plot.

	β_1	β_2
(i)	0.25	0.50
(ii)	0.50	-0.50
(iii)	0.75	0.00
(iv)	1.00	-0.50

(A) Matches (iii) ($\beta_1 = 0.75, \beta_2 = 0$):

- This is essentially an AR(1) process.
- ACF is strictly positive and decays geometrically at a moderate rate.
- Shape: monotone decay, no oscillation.

(B) Matches (i) ($\beta_1 = 0.25, \beta_2 = 0.50$):

- Both lags enter positively, so the series is more persistent.
- The lag-2 term reinforces the lag-1 effect, slowing down the decay.
- ACF remains positive for many lags.

(C) Matches (ii) ($\beta_1 = 0.50, \beta_2 = -0.50$):

- Negative β_2 introduces alternating correlation across lags.
- β_1 is only moderate, so the oscillations appear quickly.
- ACF alternates sign and dies out relatively fast.

(D) Matches (iv) ($\beta_1 = 1.00, \beta_2 = -0.50$):

- Strong positive β_1 keeps the first few autocorrelations positive.
- But negative β_2 eventually pulls the ACF below zero.
- Result: a slowly decaying, oscillatory pattern (initially positive, then alternating).

6.2.2 Testing for Unit Roots

Motivation for testing for unit roots: The central question is why do we not use the consistency of the OLS to estimate the unit root in Y_t i.e. such that in our $AR(p)$ model we estimate $\sum_{i=1}^p \hat{\beta}_i \xrightarrow{P} \sum_{i=1}^p \beta_i = 1$.

- The OLS is consistent irrespective of whether $\{Y_t\}$ is stationary or follows a unit root $AR(p)$ process
- However, standard inferences are only guaranteed to be applicable (if using OLS) when the process is stationary i.e. if we estimate an $AR(1)$ model and Y_t has a unit root so that $\beta_1 = 1$, $\hat{\beta}_1 \xrightarrow{P} \beta_1 = 1$ (will be consistent) but standard inferences cannot be made on β_1 since the CLT no longer applies to ensure the asymptotic normality of $\hat{\beta}_1$ and hence the associated t statistic
- The OLS estimator suffers from non-trivial biases that become pronounced in the vicinity of a unit root.

In the context of the unit root $AR(1)$ model, $\hat{\beta}_1$ has a non-normal distribution which is centred to the left of 1 so the bulk of estimates of β_1 will lie well below 1. So, we may be better off imposing β_1 if we have evidence that $\beta_1 = 1$ and using forecasts based on that basis than using OLS

- More generally, since the biases above are inherited by the estimators of the parameters of the $AR(p)$ model, if we have reason to believe that Y_t follows a unit root $AR(p)$ process, we should

1. impose that unit root by estimating the implied $AR(p-1)$ model for ΔY_t
2. use the estimated model to compute the h or one step ahead forecast ($\widehat{\Delta Y}_{t+1|T}$) for ΔY_{T+1}
3. recover the implied forecast for Y_t as per

$$\hat{Y}_{T+1|T} = Y_T + (\widehat{\Delta Y}_{t+1|T})$$

Alternatively, if Y_t follows a stationary $AR(p)$ model, we should estimate an $AR(p)$ model for it directly – estimating an AR model for ΔY_t is inappropriate since it incorrectly imposes a unit root of the implied model Y_t

Ultimately, we want to be able to discriminate between when $\{Y_t\}$ is a unit root $AR(p)$ process and $\{Y_t\}$ is a stationary $AR(p)$ process.

The Augmented Dickey-Fuller test: there are two kinds of tests for $AR(p)$ process

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + u_t$$

For both tests, we have hypotheses

$$H_0 : \sum_{i=1}^p \beta_i = 1$$

$$H_1 : \sum_{i=1}^p \beta_i < 1$$

1. **Constant only:** We have the following ADF form

$$\Delta Y_t = \beta_0 + \delta Y_{t-1} + \sum_{i=1}^{p_\Delta} \gamma_i Y_{t-i} + u_t$$

where $p_\Delta := p - 1$ and $\delta = \sum_{i=1}^p \beta_i - 1$. The null can be rephrased as

$$H_0 : \delta = 0$$

$$H_1 : \delta < 0$$

- What the hypotheses represent: null represents the hypothesis that Y_t is a unit root while the alternative represents the hypothesis that Y_t is stationary
- Advantage: test is more powerful since the critical values are less negative, hence it is easier to reject the null
- For an $AR(1)$ process, the Dickey-Fuller form is

$$\Delta Y_t = \beta_0 + \delta Y_{t-1} + u_t$$

- Large sample inference: The test statistic under the null is $t(0) = \hat{\delta}/\text{se}(\hat{\delta})$ such that

$$t(0) \xrightarrow{d} \text{DF}_{\text{cn}}$$

i.e. the t statistic has approximately a Dickey-Fuller (constant only) distribution

- Decision rule: We reject the null if $t(0) < c_\alpha$ drawn from DF_{cn} at α significance level.

2. **Trend and Constant:** We have the following ADF form

$$\Delta Y_t = \beta_0 + \alpha t + \delta Y_{t-1} + \sum_{i=1}^{p_\Delta} \gamma_i Y_{t-i} + u_t$$

where $p_\Delta := p - 1$ and $\delta = \sum_{i=1}^p \beta_i - 1$. The null can be rephrased as

$$H_0 : \delta = 0$$

$$H_1 : \delta < 0$$

- What the hypotheses represent: null represents the hypothesis that Y_t is a unit root while the alternative represents the hypothesis that Y_t is trend stationary
- Advantage: If ΔY_t is trend stationary but one only estimates ΔY_t with constant, $\hat{\delta}/\text{se}(\hat{\delta})$ will be biased towards 0 so we would be even less likely to reject the null and conclude that Y_t has a unit root
- For an $AR(1)$ process, the Dickey-Fuller form is

$$\Delta Y_t = \beta_0 + \alpha t + \delta Y_{t-1} + u_t$$

- Large sample inference: The test statistic under the null is $t(0) = \hat{\delta}/\text{se}(\hat{\delta})$ such that

$$t(0) \xrightarrow{d} DF_{\text{tr}}$$

i.e. the t statistic has approximately a Dickey-Fuller (trend and constant) distribution

- Decision rule: We reject the null if $t(0) < c_\alpha$ drawn from DF_{cn} at α significance level.

What to write in exam:

1. Write down the ADF model we want to estimate (with trend or constantly only)
2. State the null and alternative hypothesis

$$H_0 : \delta = 0$$

$$H_1 : \delta < 0$$

3. Say, under H_0 , $t(0) \xrightarrow{d} DF_{\text{tr}}$ or DF_{cn} depending
4. Use decision rule: we reject (do not reject the null) if $t(0) < c_\alpha$ ($t(0) \geq c_\alpha$) drawn from DF_{tr} or DF_{cn} distribution at α significance level
5. Explicitly say what rejection implies i.e. if we fail to reject the null, then (in the case of with trend) $\{Y_t\}$ is a unit root $AR(p)$ process

Orders of Integration

If we do not reject the null of the ADF test (for constant and trend), then Y_t has a unit root, but this does not imply that ΔY_t is stationary. We need to run the test again on ΔY_t to determine whether it is stationary or if it has a unit root. We do this until we find $\Delta^d Y_t$ such that we reject the null and conclude that it is stationary i.e. the algorithm is:

Order of Integration Algorithm: for d in $\mathbb{N}$,

1. test for a unit root in $\Delta^d Y_t$ (use ADF test with constant only or constant and trend depending on the data)
2. if reject: conclude $Y_t \sim I(d)$ and break

For those that don't know how for loops work, else, we increase d by 1 and return to the first step.

Definition 6.9: Order of Integration

The order of integration d of a process $\{Y_t\}$ is defined as the smallest $d \in \mathbb{Z}$ such that $\Delta^d Y_t$ is stationary. We denote this $Y_t \sim I(d)$. If $Y_t \sim I(d)$ with $d \geq 1$, it is said to be an integrated process.

For a stationary process $\{Y_t\}$, no differencing is required so $Y_t \sim I(0)$. For a process that has a unit root but whose first differences are stationary, $Y_t \sim I(1)$. We sometimes say that d is the number of unit roots for a process $Y_t \sim I(d)$

Remark.

$Y_t \sim I(0)$ is consistent with Y_t being any stationary process, not just an $AR(p)$ process e.g.

$$Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

Example.

Write down an ADF regression for a variable X_t that includes a constant and linear trend. Explain how this model can be used to test the null hypothesis that $X_t \sim I(1)$ against $X_t \sim I(0)$, clearly stating the null and alternative hypotheses.

Solution:

We write the ADF model

$$\Delta X_t = \beta_0 + \delta t + \delta X_{t-1} + \sum_{i=1}^p \gamma_i \Delta X_{t-i} + u_t$$

where $\Delta X_t := X_t - X_{t-1}$, $\mathbb{E}(u_t) = 0$, and

$$\text{Cov}(u_t, t) = \text{Cov}(u_t, X_{t-1}) = \text{Cov}(u_t, \Delta X_{t-i}) = 0 \quad \text{for } i \in \{1, \dots, p\} \text{ by construction.}$$

Estimate the parameters of the regression by OLS. Our hypotheses are

$$H_0 : \delta = 0 \quad (\text{unit root}) \quad H_1 : \delta < 0 \quad (\text{trend stationary})$$

Our test statistic is

$$t(\delta) = \frac{\hat{\delta}}{\text{se}(\hat{\delta})}$$

Under the null, $t(\delta) \rightsquigarrow \text{DF}_{\text{tr}}$ (Dickey-Fuller distribution with trend). Reject the null if $t(\delta) < \text{critical value } c_\alpha$, where α is the level of significance and c_α is such that:

$$\alpha = \mathbb{P}(\text{DF}_{\text{tr}} < c_\alpha)$$

Conclusion:

- If we **fail to reject** the null, $\{X_t\}$ is a unit root autoregressive process, and the first difference ΔX_t can be described by an $\text{AR}(p)$ model. Assuming this model is stationary, then $X_t \sim I(1)$.
- If we **reject** the null, $\{X_t\}$ is a (trend) stationary process, hence $X_t \sim I(0)$.

6.3 Cointegration

6.3.1 Spurious Regression

Motivation: Consider two time series $\{Y_t\}$ and $\{X_t\}$ – we want to ask how are $\{Y_t\}$ and $\{X_t\}$ related in the descriptive sense.

Proposition 6.10

Suppose $\{Y_t\}$ and $\{X_t\}$ are (jointly) stationary and independent, where $Y_t = \beta_0 + \beta_1 X_t + u_t$. Then,

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 = \frac{\text{cov}(X_t, Y_t)}{\text{var}(X_t)} = 0$$

To conduct inferences in this case, we need to use *heteroskedasticity and autocorrelation-*

consistent (HAC) standard errors. This is because it is very unlikely that X_t captures all the serial correlation in Y_t so $\{u_t\}$ will be serially correlated which violates our standard assumptions. With this adjustment for the serial correlation in u_t , standard inferences based on the t statistic can be drawn about whether $\beta_1 = 0$.

Proposition 6.11

Consider $\{Y_t\}$ and $\{X_t\}$ that are independent where $Y_t = \beta_0 + \beta_1 X_t + u_t$ but $Y_t \sim I(1)$ and $X_t \sim I(1)$. Then,

1. $\hat{\beta}_1$ will not be consistent for zero (and for any value) but remains a random variable no matter how large T becomes
2. the t statistic $t(0) = \hat{\beta}_1 / \text{se}(\hat{\beta}_1)$ associated with $H_0 : \beta_1 = 0$ will not settle down to any distribution but diverge (in magnitude) as the sample size grows, so whatever critical value c is chosen, the probability $\mathbb{P}\{|t(0)| > c\}$ of a rejection of H_0 converges to one so we will falsely reject true nulls
3. R^2 does not converge to 0 but remains random as $T \rightarrow \infty$; it will be very high with a non-negligible probability

This is bad because we will think that X_t and Y_t are related even though they are independent – so we call this a *Spurious Relationship*

Definition 6.12: Spurious Regression

The tendency to find statistically significant regression relationships between $I(1)$ series even when they are entirely independent

Intuition: Spurious regression arises because $I(1)$ series tend to exhibit large swings of increase and decrease with much regularity so we may match a movement in one $I(1)$ series with another to call that a statistically significant relationship.

Cointegration and Long-run Equilibria Motivation: It is not generally true that the results of a regression of one $I(1)$ series on another is spurious – such ‘matching’ swings can be explained by cointegration.

Definition 6.13: Cointegration

X_t and Y_t are said to be cointegrated if $X_t, Y_t \sim I(1)$ and there is a θ such that

$$\xi_t := Y_t - \theta X_t \sim I(0)$$

i.e. if there is a linear combination of the series that is stationary. θ is called the cointegrating coefficient

Proposition 6.14

The cointegrating coefficient θ is uniquely defined such that for any two series $X_t, Y_t \sim I(1)$, if

$$\begin{aligned}\xi_t &= Y_t - \theta X_t \sim I(0) \\ \xi_t &= Y_t - \theta^* X_t \sim I(0)\end{aligned}$$

for some θ^* , then $\theta^* = \theta$

Some facts:

- X_t and Y_t wander around randomly (and may also have deterministic trends) but move together in such a way that the equilibrium error $\xi_t := Y_t - \theta X_t$ is mean reverting
- ξ_t is typically serially correlated i.e. not independent of $\xi_{t-1}, \xi_{t-2}, \dots$

Intuition: The interpretation of the cointegrating relationship is that it describes the long run equilibrium towards which the series will always tend to revert, even though the levels of the series themselves are not ‘anchored’ to any particular value.

Remark.

We may generalise the definition of cointegration to multiple series i.e. X_t, Y_t, Z_t are cointegrated if $X_t, Y_t, Z_t \sim I(1)$ and there exists a θ_X and θ_Z such that

$$Y_t - \theta_X X_t - \theta_Z Z_t \sim I(0)$$

Proposition 6.15

The sum of an $I(1)$ and $I(0)$ series is always $I(1)$

6.3.2 Testing for cointegration

Goal: How can we use $\hat{\xi}_t = Y_t - \hat{\theta}X_t$ to determine whether a regression involving $I(1)$ variables is potentially spurious or whether it estimates a cointegrating relationship.

Procedure for testing for cointegration:

1. Perform ADF tests to verify that $X_t, Y_t \sim I(1)$: if either (or both) are $I(0)$, then there is no possibility of spurious regression, nor of cointegration either
2. If (we think) we know that cointegrating coefficient θ (e.g. because an economic theory makes a precise prediction about θ)
 - (a) compute $\xi_t := Y_t - \theta X_t$ and conduct an ADF test for a unit root in ξ_t (a null of no cointegration since cointegration means $\xi_t \sim I(0)$)
 - (b) if we reject the null of a unit root, conclude that $\xi \sim I(0)$: so X_t and Y_t are cointegrated with cointegrating coefficient θ

A finding of $\xi_t \sim I(1)$ does not imply that X_t and Y_t are not cointegrated, only that they are not cointegrated with coefficient θ

3. If we do not know the cointegrating coefficient θ

- (a) estimate θ by OLS and compute the estimated equilibrium error $\hat{\xi}_t := Y_t - \hat{\theta}X_t$ (or equivalently the OLS residuals)
- (b) perform an ADF test on $\hat{\xi}_t$ (with a null of a unit root and alternative of stationarity), using adjusted Engle-Granger critical values
- (c) if we reject the null that $\hat{\xi}_t \sim I(1)$, we can conclude that X_t and Y_t are cointegrated; otherwise they are not

Remark.

We only need the adjustment of critical values when we do not know θ because $\hat{\xi}_t$ is the result of a regression of Y_t on X_t while we compute ξ_t directly when θ is known

The possibility of cointegration provides a reason for running regressions with $I(1)$ series, since differencing the series will exclude any long run relationships between the series that we find via θ .

Inference: We cannot use the OLS standard errors to construct t statistics and CIs because $X_t \sim I(1)$ distribution of $\hat{\theta}$ is not approximately normal – must use alternatives that are not covered.

6.3.3 Mathematical Illustration of Cointegration

The goal here is to illustrate the possible that two series could be $I(1)$ while also having a linear combination of the two series which are stationary.

Suppose $\{v_t\}, \{w_{yt}\}, \{w_{xt}\}$ are jointly stationary and that X_t and Y_t are given by

$$Y_t = \mu t + \sum_{s=1}^t v_s + w_{yt} \quad X_t = \frac{1}{\theta} \left[\mu t + \sum_{s=1}^t v_s \right] + w_{xt}$$

Both Y_t and X_t are $I(1)$, since

$$\begin{aligned} \Delta Y_t &= Y_t - Y_{t-1} = \left[\mu t + \sum_{s=1}^t v_s + w_{yt} \right] - \left[\mu(t-1) + \sum_{s=1}^{t-1} v_s + w_{y,t-1} \right] \\ &= \mu + v_t + w_{yt} - w_{y,t-1} \sim I(0) \end{aligned}$$

The case for X_t follows analogously. Alternatively, defining $S_t := \mu t + \sum_{s=1}^t v_s \sim I(1)$, we recognise that

$$Y_t = S_t + w_{yt} = I(1) + I(0) = I(1)$$

Similarly,

$$X_t = \frac{1}{\theta} S_t + w_{xt} \sim I(1)$$

The source of the nonstationarity in both series is their common deterministic and stochastic trend component. Hence, for any c

$$Y_t - cX_t = [1 - \frac{c}{\theta}]S_t + (w_{yt} - cw_{xt})$$

Irrespective of the value c , $w_{yt} - cw_{xt} \sim I(0)$. Therefore, so long as S_t is present on the r.h.s, $Y_t - cX_t \sim I(1)$. The only exception is when $c = \theta$ so that

$$Y_t - \theta X_t = [1 - \frac{\theta}{\theta}]S_t + (w_{yt} - \theta w_{xt}) = w_{yt} - \theta w_{xt} \sim I(0)$$

which is precisely when they are cointegrated with coefficient θ . This also shows why θ is unique.

Recall that the OLS estimators $(\hat{\alpha}, \hat{\theta})$ are the (a, c) that minimises the sum of squared errors i.e.

$$\sum_{t=1}^T (Y_t - a - cX_t)^2 = \sum_{t=1}^T [(1 - \frac{c}{\theta})S_t + (w_{yt} - cw_{xt}) - a]^2$$

Since $S_t \sim I(1)$, it has a large and growing variance (i.e. it has the properties of a stochastic trend). So in attempting to make the r.h.s as small as possible, OLS will choose $\hat{\theta}$ that is close to θ and $\hat{\theta} \xrightarrow{p} \theta$. Thus if Y_t and X_t are cointegrated, OLS consistently estimates the cointegrating coefficient θ .

In a large sample, $\hat{\theta} \simeq \theta$, and so the estimated equilibrium error (or OLS residuals)

$$\hat{\xi}_t = Y_t - \hat{\theta}X_t \simeq Y_t - \theta X_t \sim I(0)$$

behaves like an $I(0)$ process (this is also true of the OLS residuals). By contrast, suppose that $X_t, Y_t \sim I(1)$ but are not cointegrated where we take $\mu = 0$ such that

$$Y_t = \sum_{s=1}^t v_{ys} + w_{yt} \quad X_t = \mu t + \sum_{s=1}^t v_{xs} + w_{xt}$$

where $\{v_{ys}\}$ and $\{v_{xs}\}$ are jointly stationary processes. Both X_t and Y_t have stochastic trends but lack a common stochastic trend. So, every linear combination of Y_t and X_t will still be $I(1)$ i.e.

$$Y_t - cX_t = \sum_{s=1}^t (v_{ys} - cv_{xs}) + w_{yt} - cw_{xt} \sim I(1)$$

since there is no value of c that estimates $\sum_{s=1}^t (v_{ys} - cv_{xs})$; so X_t, Y_t are not cointegrated. Hence, if we perform a regression of Y on X_t as before and compute the estimated equilibrium error (or OLS residuals), we find that

$$\hat{\xi}_t := Y_t - \hat{\theta}X_t \sim I(1)$$

because whatever $\hat{\theta}$ happens to be, it must always give a linear combination of Y_t and X_t that is $I(1)$

7 Appendix I: Probability Theory

7.1 Axioms of Probability

Definition 7.1: Probability Space

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where

1. Ω is a set, called the *sample space*,
2. $\mathcal{F}$ is a collection of subsets of Ω , called *events*, satisfying axioms **F₁–F₃** below,
3. $\mathbb{P}$ is a *probability measure*, which is a function $\mathbb{P}: \mathcal{F} \rightarrow \mathbb{R}$ satisfying axioms **P₁–P₃**.

Definition 7.2: Komolgorov Axioms

$\mathcal{F}$ is a collection of subsets of Ω , with:

F₁: $\Omega \in \mathcal{F}$.

F₂: If $A \in \mathcal{F}$, then also $A^c \in \mathcal{F}$.

F₃: If $\{A_i, i \in I\}$ is a finite or countably infinite collection of members of $\mathcal{F}$, then $\bigcup_{i \in I} A_i \in \mathcal{F}$.

$\mathbb{P}$ is a function from $\mathcal{F}$ to $\mathbb{R}$, with:

P₁: For all $A \in \mathcal{F}$, $\mathbb{P}(A) \geq 0$.

P₂: $\mathbb{P}(\Omega) = 1$.

P₃: If $\{A_i, i \in I\}$ is a finite or countably infinite collection of members of $\mathcal{F}$, and $A_i \cap A_j = \emptyset$ for $i \neq j$, then $\mathbb{P}(\bigcup_{i \in I} A_i) = \sum_{i \in I} \mathbb{P}(A_i)$.

Proposition 7.3

The following results follow immediately from the axioms

1. Probabilities are bounded between 0 and 1: $0 \leq \mathbb{P}(A) \leq 1$ for all $A \subseteq \Omega$
2. Empty event has probability zero: $\mathbb{P}(\emptyset) = 0$
3. Complement rule: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
4. Relative complement rule: if $A \subseteq B$, then $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$
5. Union rule: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Definition 7.4: Partition

A family of events $\{B_1, \dots\}$ is a partition if

1. $\Omega = \bigcup_{i \geq 1} B_i$
2. $B_i \cap B_j = \emptyset$ whenever $i \neq j$

Definition 7.5: Conditional Probability

For any $A, B \subseteq \Omega$,

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Theorem 7.6: Law of Total Probability

Suppose $\{B_1, \dots\}$ is a partition of Ω by sets from $\mathcal{F}$ such that $\mathbb{P}(B_i) > 0$ for all $i \geq 1$. Then, for any $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i \geq 1} \mathbb{P}(A|B_i)\mathbb{P}(B_i)$$

Theorem 7.7: Bayes' Theorem

Suppose that $\{B_1, \dots\}$ is a partition of Ω by sets from $\mathcal{F}$ such that $\mathbb{P}(B_i) > 0$ for all $i \geq 1$. Then, for any $A \in \mathcal{F}$ such that $\mathbb{P}(A) > 0$,

$$\mathbb{P}(B_k|A) = \frac{\mathbb{P}(A|B_k)\mathbb{P}(B_k)}{\sum_{i \geq 1} \mathbb{P}(A|B_i)\mathbb{P}(B_i)}$$

Definition 7.8: Independence

Consider two events $A, B \subseteq \Omega$. We say that A and B are independent just when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B)$$

or equivalently,

$$\mathbb{P}(A|B) = \mathbb{P}(A)$$

We denote this $A \perp\!\!\!\perp B$

Interpretation: We interpret independence as B giving us no information about A

7.2 Distributions

Discrete Distributions

Definition 7.9: Discrete Random Variables

A random variable X is said to be discrete if there is a finite list of values $a_1, \dots, a_n$ or an infinite list of values $a_1, a_2, \dots$ such that $\mathbb{P}(X = a_j \text{ for some } j) = 1$. If X is a discrete r.v., then the finite or countably infinite set of values x such that $\mathbb{P}(X = x) > 0$ is called the support of X .

Definition 7.10: Probability mass function

The probability mass function (PMF) of a discrete r.v. X is the function $p_X(x) = \mathbb{P}(X = x)$. Note that this is positive if x is in the support of X and 0 otherwise.

Theorem 7.11: Valid PMFs

Let X be a discrete r.v. with support $x_1, x_2, \dots$. The PMF p_X of X must satisfy the following criteria:

1. $p_X(x) > 0$ if $x = x_j$ for some j and $p_X(x) = 0$ otherwise
2. Sums to 1: $\sum_{j=1}^{\infty} p_X(x_j) = 1$

Definition 7.12: Bernoulli distribution

An r.v. X is said to have the Bernoulli distribution with parameter p if $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$ where $0 < p < 1$. We write this as $X \sim \text{Bern}(p)$.

Story (Bernoulli Trial): An experiment that can result in either a “success” or “failure” (but not both) can be called a Bernoulli trial. A Bernoulli random variable can be thought of as the indicator of success in a Bernoulli trial: it equals 1 if the success occurs and 0 if the failure occurs in the trial.

Definition 7.13: Binomial Distribution

Suppose that n independent Bernoulli trials are performed, each with the same success probability p . Let X be the number of successes. The distribution of X is called the Binomial distribution with parameters n and p . We write $X \sim \text{Bin}(n, p)$, where $n \in \mathbb{Z}^+$ and $0 < p < 1$.

Remark.

■ This acts as its story

Theorem 7.14: Binomial PMF

If $X \sim \text{Bin}(n, p)$, then the PMF of X is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

for $k = 0, 1, \dots, n$ and $\mathbb{P}(X = k) = 0$ otherwise

Corollary 7.15

Let $X \sim \text{Bin}(n, p)$ and $q = 1 - p$. Then $n - X \sim \text{Bin}(n, q)$

Proposition 7.16

If $X \sim \text{Bin}(n, p)$, $Y \sim \text{Bin}(m, p)$, and X is independent of Y , then $X + Y \sim \text{Bin}(n + m, p)$

Definition 7.17: Cumulative Distribution Function

The cumulative distribution function (CDF) of an r.v. X is the function F_X given by $F_X(x) = \mathbb{P}(X \leq x)$

Theorem 7.18: Valid CDFs

Any CDF F has the following properties:

- Increasing: If $x_1 \leq x_2$, then $F(x_1) \leq F(x_2)$
- Right-continuous: the CDF is continuous except possibly for having some jumps. Whenever there is a jump, the CDF is continuous from the right. That is, for any a , we have

$$F(a) = \lim_{x \rightarrow a^+} F(x)$$

- Convergence to 0 and 1 in the limits:

$$\lim_{x \rightarrow -\infty} F(x) = 0 \text{ and } \lim_{x \rightarrow \infty} F(x) = 1$$

Theorem 7.19: PMF of $g(X)$

Let X be a discrete r.v. and $g : \mathbb{R} \rightarrow \mathbb{R}$. Then, the support of $g(X)$ is the set of all y such that $g(x) = y$ for at least one x in the support of X and the PMF of $g(X)$ is

$$\mathbb{P}(g(X) = y) = \sum_{x: g(x)=y} \mathbb{P}(X = x)$$

Definition 7.20: Geometric Distribution

Consider a sequence of independent Bernoulli trials, each with the same success probability $p \in (0, 1)$, with trials performed until a success occurs. Let X be the number of failures before the first successful trial. Then X has the Geometric distribution with parameter p . We denote this $X \sim \text{Geom}(p)$

Theorem 7.21: Geometric PMF

If $X \sim \text{Geom}(p)$, then the PMF of X is

$$\mathbb{P}(X = k) = q^k p$$

for $k = 0, 1, \dots$, where $q = 1 - p$

Continuous Distributions

Definition 7.22: Continuous random variable

An r.v. has a continuous distribution if its CDF is differentiable. We also allow there to be endpoints (or finitely many points) where the CDF is continuous but not differentiable, as long as the CDF is differentiable everywhere else. A continuous random variable is a random variable with a continuous distribution

Definition 7.23: Probability Density Function

For a continuous r.v. X with CDF F , the probability density function (PDF) of X is the derivative f of the CDF, given by $f(x) = F'(x)$. The support of X and of its distribution is the set of all x where $f(x) > 0$

Proposition 7.24

Let X be a continuous r.v. with PDF f . Then, the CDF of X is given by

$$F(x) = \int_{-\infty}^x f(t) dt$$

Theorem 7.25: Valid PDFs

The PDF f of a continuous r.v. must satisfy the following two criteria:

- Nonnegative: $f(x) \geq 0$
- Integrates to 1: $\int_{-\infty}^{\infty} f(x) dx = 1$

Definition 7.26: Uniform Distribution

A continuous r.v. U is said to have the Uniform distribution on the interval (a, b) if its PDF is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

We denote this as $U \sim \text{Unif}(a, b)$

Theorem 7.27: Universality of the Uniform

Let F be a CDF which is a continuous function and strictly increasing on the support of the distribution. This ensures that the inverse function F^{-1} exists, as a function from $(0, 1)$ to $\mathbb{R}$. Then we have the following

1. Let $U \sim \text{Unif}(0, 1)$ and $X = F^{-1}(U)$. Then, X is an r.v. with CDF F
2. Let X be an r.v. with CDF F . Then, $F(X) \sim \text{Unif}(0, 1)$

Definition 7.28: Standard Normal Distribution

A continuous r.v. Z is said to have the standard Normal Distribution if its PDF φ is given by

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

We denote this $Z \sim \mathcal{N}(0, 1)$

We also denote the standard Normal CDF as Φ where

$$\Phi(z) = \int_{-\infty}^z \varphi(t) dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Definition 7.29: Normal Distribution

If $Z \sim \mathcal{N}(0, 1)$, then

$$X = \mu + \sigma Z$$

is said to have the normal distribution with mean μ and variance σ^2 . We denote this $X \sim \mathcal{N}(\mu, \sigma^2)$

Theorem 7.30: Properties of the Normal Distribution

We have the following (we use the standard normal WLOG)

1. Symmetry of PDF: φ satisfies $\varphi(z) = \varphi(-z)$ i.e. φ is an even function
2. Symmetry of tail areas: the area under the PDF curve to the left of $-z$ equals the area to the right of z i.e.

$$\Phi(z) = 1 - \Phi(-z)$$

for all z

3. Symmetry of Z and $-Z$: If $Z \sim \mathcal{N}(0, 1)$, then $-Z \sim \mathcal{N}(0, 1)$

Multivariate Distributions

We go through only the continuous counterparts

Definition 7.31: Joint CDF and PDF

The joint CDF for bivariate events is

$$F(x, y) = \mathbb{P}(X \leq x, Y \leq y)$$

or, through double integrals,

$$F(x, y) = \int \int f(x, y) dx dy$$

Of course it follows that

$$F(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

And the joint PDF is

$$f(x, y) = \mathbb{P}(X = x, Y = y) \quad \text{discrete}$$

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} \quad \text{continuous}$$

Remark.

It is easy to see the discrete counterpart for the CDF

Definition 7.32: Marginal PDF

The marginal PDF of x is

$$f(x) = f(x, \text{whatever the value of } y)$$

or more formally,

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad \text{for all } x$$

The PDF would be in terms of x , so you essentially get the value of x for all y values. Same for marginal PDF of y .

Definition 7.33: Conditional PDF

Consider r.v.s Y and X with marginal PDFs $f_Y(y)$ and $f_X(x)$ respectively and joint PDF $f_{YX}(x, y)$. Then, we have

$$f_{Y|X=x_0}(y|X=x_0) := \frac{f_{YX}(y, x_0)}{f_X(x_0)}$$

when conditioning on a value of $X = x_0$ and

$$f_{Y|X}(y|x) := \frac{f_{YX}(y, x)}{f_X(x)}$$

when conditioning on a r.v. X

7.3 Moments

Definition 7.34: Expectation

The expected value of a r.v. X is

$$\mathbb{E}(X) = \begin{cases} \sum_{j=1}^{\infty} x_j \mathbb{P}(X = x_j) & \text{if discrete} \\ \int_{-\infty}^{\infty} x f(x) & \text{if continuous} \end{cases}$$

We replace the sum with a finite sum if the support is finite

Theorem 7.35: Properties of Expectation

For any r.v.s X, Y and any constant $c \in \mathbb{R}$, we have

- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- $\mathbb{E}[cX] = c\mathbb{E}[X]$

Theorem 7.36: LOTUS

If X is a r.v. and g is a function from $\mathbb{R}$ to $\mathbb{R}$, then

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x)\mathbb{P}(X = x) & \text{if discrete} \\ \int_{-\infty}^{\infty} g(x)f(x) & \text{if continuous} \end{cases}$$

Definition 7.37: Variance and Standard Deviation

The variance of an r.v. X is

$$\text{var}(X) = \mathbb{E}[X - \mathbb{E}(X)]^2 = \mathbb{E}[X^2] - [\mathbb{E}(X)]^2$$

The square root of the variance is called the standard deviation:

$$\text{sd}(X) = \sqrt{\text{var}(X)}$$

Theorem 7.38: Expectation and Variance of Discrete Distributions

We have the following:

1. $X \sim \text{Bern}(p)$:

- Expectation: $\mathbb{E}[X] = p$
- Variance: $\text{var}(X) = p(1 - p)$

2. $X \sim \text{Bin}(n, p)$:

- Expectation: $\mathbb{E}[X] = np$
- Variance: $\text{var}(X) = np(1 - p)$

3. $X \sim \text{Geom}(p)$:

- Expectation: $\mathbb{E}[X] = \frac{1}{p}$
- Variance: $\text{var}(X) = \frac{1-p}{p^2}$

Theorem 7.39: Expectation and Variance of Continuous Distributions

We have the following

1. $U \sim \text{Unif}(a, b)$:

- Expectation: $\mathbb{E}[X] = \frac{a+b}{2}$
- Variance: $\text{var}(X) = \frac{(b-a)^2}{12}$

2. $X \sim \mathcal{N}(\mu, \sigma^2)$:

- Expectation: $\mathbb{E}[X] = \mu$
- Variance: $\text{var}(X) = \sigma^2$

Theorem 7.40: Jensen's Inequality

Suppose X is a r.v. and φ is a convex function. Then,

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$$

If φ is a concave function, then

$$\mathbb{E}[\varphi(X)] \leq \varphi(\mathbb{E}[X])$$

Definition 7.41: Covariance

The covariance between X and Y is

$$\text{cov}(X, Y) = \mathbb{E}((X - \mathbb{E}[X])(Y - \mathbb{E}[Y]))$$

or equivalently,

$$\text{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

Corollary 7.42

If X and Y are independent, then $\text{cov}(X, Y) = 0$

Theorem 7.43: Properties of Variance

We have the following properties

1. $\text{var}(aX + b) = a^2 \text{var}(X)$
2. $\text{var}(aX + bY) = a^2 \text{var}(X) + b^2 \text{var}(Y) - 2ab \text{cov}(X, Y)$
3. If X and Y are independent, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

Theorem 7.44: Properties of Covariance

We have the following properties

1. $\text{cov}(X, X) = \text{var}(X)$
2. $\text{cov}(X, Y) = \text{cov}(Y, X)$
3. $\text{cov}(X, c) = 0$ for $c \in \mathbb{R}$
4. $\text{cov}(X + Y, Z) = \text{cov}(X, Z) + \text{cov}(Y, Z)$
5. $\text{cov}(X + Y, Z + W) = \text{cov}(X, Z) + \text{cov}(X, W) + \text{cov}(Y, Z) + \text{cov}(Y, W)$
6. For n r.v.s $X_1, \dots, X_n$,

$$\text{var}(X_1 + \dots + X_n) = \text{var}(X_1) + \dots + \text{var}(X_n) + 2 \sum_{i < j} \text{cov}(X_i, X_j)$$

Definition 7.45: Correlation

The correlation between X and Y is

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$$

Remark.

Correlation is a unitless measurement i.e. scaling and shifting X or Y has no effect

Corollary 7.46

$$\text{cov}(X, Y) = 0 \text{ iff } \text{corr}(X, Y) = 0$$

Proposition 7.47

For any r.v.s X, Y

$$-1 \leq \text{corr}(X, Y) \leq 1$$

Theorem 7.48: Variance-Bias decomposition (for fixed parameter)

The mean-squared error of $\hat{\theta}_n$ $\text{mse}(\hat{\theta}_n) := \mathbb{E}(\hat{\theta}_n - \theta)^2$ can be decomposed into

$$\text{mse}(\hat{\theta}_n) = [\text{bias}(\hat{\theta}_n)]^2 + \text{var}(\hat{\theta}_n)$$

where $\text{bias}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n) - \theta$

Remark.

- This is different from above where we have a third quantity ε that is unexplainable –
- here we are decomposing the mse of a fixed quantity rather than predicting a r.v. Y .
- The latter will have a part unexplainable by X which is ε

Conditional Moments

Definition 7.49: Conditional expectation given an event

Let $X = x_0$ be an event with positive probability. Then

$$\mathbb{E}[Y|X = x_0] = \begin{cases} \sum_y y \mathbb{P}(Y = y|X = x_0) & \text{if discrete} \\ \int_{-\infty}^{\infty} y f(y|X = x_0) dy & \text{if continuous} \end{cases}$$

Remark.

- This is a fixed quantity

Definition 7.50: Conditional expectation given an r.v.

Suppose that X, Y are r.v.s. Then,

$$\mathbb{E}[Y|X] = \begin{cases} \sum_y y \mathbb{P}(Y = y|X = x) & \text{if discrete} \\ \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy & \text{if continuous} \end{cases}$$

Remark.

- Notice that since we sum or integrate wrt y only, $\mathbb{E}[Y|X]$ is always a function of x only

Theorem 7.51: Properties of Conditional Expectation

Suppose X and Y are r.v.s. Then,

- If X and Y are independent, $\mathbb{E}[Y|X] = \mathbb{E}[Y]$
- For any function h , $\mathbb{E}[h(X)Y|X] = h(X)\mathbb{E}[Y|X]$
- For r.v.s Y_1, Y_2 , $\mathbb{E}[Y_1 + Y_2|X] = \mathbb{E}[Y_1|X] + \mathbb{E}[Y_2|X]$ and $\mathbb{E}[cY|X] = c\mathbb{E}[Y|X]$
- For any function h ,

$$\text{corr}(Y - \mathbb{E}[Y|X], h(X)) = 0$$

or equivalently

$$\mathbb{E}((Y - \mathbb{E}[Y|X])h(X)) = 0$$

Theorem 7.52: Law of Iterated Expectation

For any r.v.s X and Y ,

$$\mathbb{E}(\mathbb{E}[X|Y]) = \mathbb{E}[X]$$

Proposition 7.53

For any r.v.s X, Y, Z we have

$$\mathbb{E}(\mathbb{E}[Y|X, Z]|Z) = \mathbb{E}[Y|Z]$$

Definition 7.54: Conditional Variance

The conditional variance of Y given X is

$$\text{var}(Y|X) = \mathbb{E}((Y - \mathbb{E}[Y|X])^2|X)$$

or equivalently,

$$\text{var}(Y|X) = \mathbb{E}[Y^2|X] - (\mathbb{E}[Y|X])^2$$

Theorem 7.55: Law of total Variance

For any r.v.s X and Y ,

$$\text{var}(Y) = \mathbb{E}[\text{var}(Y|X)] + \text{var}(\mathbb{E}[Y|X])$$

7.4 Some Limit Theorems**Definition 7.56: Convergence in probability**

Let $\{\xi_n\}$ denote a sequence of random variables. We say that ξ_n converges in probability to a constant c if for every small $\epsilon > 0$,

$$\mathbb{P}(|\xi_n - c| \leq \epsilon) \rightarrow 1$$

as $n \rightarrow \infty$. We denote this $\xi_n \rightarrow c$

Theorem 7.57: Weak Law of Large Numbers

Suppose $\{X_i\}_{i=1}^n$ is i.i.d with $\mathbb{E}[X_i] = \mu$. Then,

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

Proof. Trivial. Follows immediately from Chebyshev's inequality and the variance of $\bar{X}_n$

□

Definition 7.58: Convergence in distribution

ξ_n converges in distribution to a continuous distribution F , denoted $\xi \xrightarrow{d} F$, if for every x ,

$$F_{\xi_n}(x) = \mathbb{P}(\xi_n \leq x) \rightarrow F(x)$$

as $n \rightarrow \infty$. We denote this $\xi \xrightarrow{d} \xi$

Theorem 7.59: Central Limit Theorem

Suppose $\{X_i\}_{i=1}^n$ is i.i.d with $\mathbb{E}[X_i] = \mu$ and $\text{var}(X_i) = \sigma^2$. Then,

$$Z_n := \frac{\bar{X}_n - \mu}{\sigma/n^{1/2}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Proof. (Outline) Proven by showing that the characteristic function of any distribution converges to the characteristic function of the normal distribution in the limit $\square$

8 Appendix II: Statistical Inference and Basic Estimation Theory

8.1 Estimation

We go through some essential definitions

Definition 8.1: Unbiased

An estimator $\hat{\theta}_n$ is unbiased for a population parameter θ just when $\mathbb{E}[\hat{\theta}_n] = \theta$

Remark.

Unbiasedness is a finite sample property of an estimator since it does not depend on sample size n

Definition 8.2: Consistency

A sequence of estimators $\{\hat{\theta}_n\}$ is consistent for θ if $\hat{\theta}_n \xrightarrow{p} \theta$

Remark.

We also have a term called *efficiency* which we heuristically take to just mean an estimator $\hat{\theta}_n$ is more efficient just when it has a smaller standard error $\text{sd}(\hat{\theta}_n)$

8.2 Estimators and Sample Distribution

Definition 8.3: Sample Mean, Variance and Covariance

Consider i.i.d random variables $\{(X_i, Y_i)\}_{i=1}^n$. Then we have the following estimators

- Sample mean: $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$
- Sample variance: $\text{vâr}(X) = s_{n,X}^2 := \frac{n}{n-1} \hat{\sigma}_{n,X}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$
- Sample covariance: $\text{côv}(X, Y) = s_{n,XY} := \frac{n}{n-1} \hat{\sigma}_{n,XY}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)$

Remark.

We normalise $\hat{\sigma}$ by $n/(n-1)$ due to Besel's correction; otherwise the estimators would be biased since we can derive the n -th observation via taking the difference between the sample mean and the $n-1$ observations before that. Notice that the sample variance is a special case of the sample covariance where $X = Y$

Remark.

The sample estimators retain the usual properties that the population moments have (i.e. linearity, additivity etc)

Proposition 8.4

The sample mean, variance and covariance are unbiased

Proof. Lazy to prove

□

Proposition 8.5

Suppose our sample $\{X_i\}_{i=1}^n$ is i.i.d with finite moments, and that $\mathbb{E}[X_i] = \mu$ and $\text{var}(X_i) = \sigma^2$ Then,

$$\bar{X}_n \xrightarrow{d} \mathcal{N}(\mu, \sigma^2/n)$$

8.3 Hypothesis testing

Here, we go through applications of t statistic as in year 1.

Sample Means

Test statistic for each sample type

1. Single sample of size N from population with mean μ and variance σ^2 . Our hypotheses are

$$H_0 : \mu = \mu_0; \quad H_1 : \mu \neq \mu_0$$

Our test statistic is

$$z = \frac{\bar{X} - \mu_0}{\sqrt{\frac{s^2}{N}}}$$

- Two independent samples of size N_x and N_y from populations with means μ_x and μ_y and variances σ_x^2 and σ_y^2 . Our hypotheses are

$$H_0 : \mu_x - \mu_y = \delta_0; H_1 : \mu_x - \mu_y \neq \delta_0$$

Our test statistic is

$$z = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{s_x^2}{N_x} + \frac{s_y^2}{N_y}}}$$

- A paired sample of size N of two variables with population means μ_x and μ_y and variances σ_x^2 and σ_y^2 . Our hypotheses are

$$H_0 : \mu_x - \mu_y = \delta_0; H_1 : \mu_x - \mu_y \neq \delta_0$$

Our test statistic is

$$z = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{s_D^2}{N}}}$$

Sample Proportions

Approach closely mirrors the mean

- The first is that we work out the variance of the sampling distribution of a proportion from the proportion itself
- New: In a hypothesis test, everything is done under the null. This therefore includes the calculation of the standard error of the test statistic

Test statistic for each sample type

- Single sample of size N from population with mean p . Our hypotheses are

$$H_0 : p = p_0; H_1 : p \neq p_0$$

Our test statistic is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{N}}}$$

- Two independent samples of size N_x and N_y from populations with means p_x and p_y . Our hypotheses are

$$H_0 : p_x - p_y = \delta_0; H_1 : p_x - p_y \neq \delta_0$$

Our test statistic is

$$z = \frac{(\hat{p}_x - \hat{p}_y) - (p_x - p_y)}{\sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{N_x} + \frac{\hat{\pi}(1-\hat{\pi})}{N_y}}}$$

- A paired sample of size N of two variables with population means p_x and p_y . Our hypotheses are

$$H_0 : p_x - p_y = \delta_0; H_1 : p_x - p_y \neq \delta_0$$

Our test statistic is

$$z = \frac{(\hat{p}_x - \hat{p}_y) - (p_x - p_y)}{\sqrt{\frac{s_D^2}{N}}}$$

Tests regarding the Difference in Two Independent Proportions

As usual, the test statistic is

$$z = \frac{(\hat{p}_x - \hat{p}_y) - (p_x - p_y)}{SE(\hat{p}_x - \hat{p}_y)}$$

The Standard Error of the sampling distribution: under the null hypothesis, the population proportions are the same in both samples. That is $p_x = p_y = \pi$ where π is the common value.

Derivation of π . We can estimate it by applying the Law of Iterated Expectations:

$$\hat{\pi} = \left(\frac{N_x}{N_x + N_y} \right) \hat{p}_x + \left(\frac{N_y}{N_x + N_y} \right) \hat{p}_y$$

Hence

$$Var(\hat{p}_x) = \frac{\hat{\pi}(1 - \hat{\pi})}{N_x} \quad Var(\hat{p}_y) = \frac{\hat{\pi}(1 - \hat{\pi})}{N_y}$$

Then, since due to independence, $Var(\hat{p}_x - \hat{p}_y) = Var(\hat{p}_x) + Var(\hat{p}_y)$, we yield

$$SE(\hat{p}_x - \hat{p}_y) = \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{N_x} + \frac{\hat{\pi}(1 - \hat{\pi})}{N_y}}$$

For paired data, you just have to go find and calculate the sample variance of the difference directly as per usual